Conceptual Topology Mapping Benchmark: Waypoint Elicitation Reveals Structured, Asymmetric, Model-Specific Navigation in LLM Conceptual Space

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Abstract

Static geometric structure in LLM representations is increasingly well-characterized: linear directions encode concepts, hierarchies form polytopes, and translation symmetry in data statistics determines manifold shape. But no work has tested whether this geometry supports reliable *behavioral traversal* — whether models can consistently navigate between concepts, not merely represent them. The distinction between map and route is logically independent, and static structure alone does not entail navigational consistency.

We introduce the Conceptual Topology Mapping Benchmark, a waypoint-elicitation paradigm that probes navigational geometry across approximately 21,540 runs, 12 models from 11 independent training pipelines, and 11 experimental phases. Models are given concept pairs and asked to produce intermediate waypoints; the resulting paths are evaluated against metric axioms, compositional structure, and causal perturbation, yielding six robust empirical claims plus one qualified.

Models exhibit distinct "conceptual gaits" — navigational signatures with consistency spanning a 2.9x range (0.258 to 0.747 across 12 models) — on a shared underlying geometry. Navigation is fundamentally asymmetric (mean directional asymmetry 0.811), consistent with quasi-metric rather than metric structure. Bridge concepts function as structural bottlenecks, not mere associations, with frequencies exceeding 0.97 under robustness testing. These properties generalize across all 12 models and resist protocol variation. Single-variable explanations consistently fail: seven follow-up hypotheses — spanning mechanistic, model-deficit, methodological, and structural types — were falsified and one resurrected with additional data, revealing a mechanism ceiling where qualitative structure is characterizable but quantitative dynamics remain opaque to current tools. Twenty-nine documented dead ends provide transparent accounting of explanatory limits. The benchmark establishes that conceptual navigation has geometric regularity that can be reliably detected but cannot yet be mechanistically decomposed.

1 Introduction

1.1 The Navigation Gap

LLM embedding spaces have rich geometric structure. Linear directions encode concepts (Park et al., 2024). Hierarchies form polytopes (Park et al., 2025). Hyperbolic geometry captures tree-like relations (Nickel and Kiela, 2017). Most fundamentally, Karkada and others (2025) proved that this geometry arises inevitably from translation symmetry in data statistics — the manifold shape is determined by co-occurrence structure, not architectural choices.

The static geometry of LLM representations is increasingly well-characterized, and theoretically grounded. What remains uncharacterized is the *dynamic* behavior on that geometry.

But geometry is not navigation.

A map is not a route.

Static analysis tells us what landmarks look like; it does not tell us whether models can reliably walk between them. Embedding probes reveal that "warm" sits between "hot" and "cold" in representational space, but they do not test whether a model traversing from hot to cold will actually pass through warm — or whether it will take a different route entirely, through "tepid" and "chilly" and "brisk." The distinction matters: two agents can share an identical map and prefer entirely different paths.

The gap is not merely philosophical. Static geometry answers questions about structure — what is near what, which directions encode which features, what the curvature looks like. Navigation answers questions about behavior — which routes models actually take, whether those routes are consistent across repeated traversals, whether they differ by direction, and whether they compose when chained through intermediate concepts. A model might have perfect geometric representations and still navigate erratically, or conversely, navigate with rigid consistency through a geometry that embedding analysis would not predict. The static and dynamic questions are logically independent, and neither entails the other.

This paper asks the dynamic question directly: **Do LLMs have consistent, measurable geometric structure in how they *navigate* between concepts?**

Not what their representations look like, but how they behaviorally traverse conceptual space when asked to produce intermediate waypoints between two endpoints.

The answer, across approximately 21,540 runs, 12 models from 11 independent training pipelines, and 11 experimental phases, is yes — with substantial qualifications about what that structure can and cannot explain. Navigation is structured (gait consistency spanning a 2.9x range across models), asymmetric (mean directional asymmetry 0.811), organized around bottleneck landmarks (bridge frequencies exceeding 0.97 under robustness testing), and consistent with quasimetric topology (triangle inequality holding at approximately 91%). But 7 of 8 follow-up hypotheses were falsified (one later resurrected), and 29 documented dead ends attest to the gap between characterizing structure and explaining it.

Theoretical complement. Karkada and others (2025) proved *why* static geometry exists: translation symmetry in co-occurrence statistics determines manifold shape. We test *whether that geometry supports behavioral navigation* — whether the geometric structure that arises inevitably from data statistics produces consistent, structured, asymmetric paths when models are asked to walk between concepts. Their contribution explains the map; ours probes the routes. The two contributions are complementary: static geometry is necessary for navigation to be structured, but not sufficient — the dynamics could still be noisy, inconsistent, or model-independent. They are none of these.

1.2 Why Navigation Matters

Four independent lines of research motivate the navigation question.

Cognitive science predicts navigable conceptual spaces. Gärdenförs (2000) theorized that concepts live in geometric spaces with measurable distance — spaces that support not just representation but traversal. Constantinescu et al. (2016) provided neural evidence: grid cells in the human entorhinal cortex, originally characterized for spatial navigation, fire during *conceptual* navigation, suggesting that spatial mechanisms have been repurposed for abstract thought. If human

conceptual spaces are navigable, the question of whether LLM conceptual spaces share this property is natural and testable.

Our benchmark provides a large-scale computational probe: 12 models, 100+ concept pairs, and over 21,000 runs testing the navigability prediction directly.

Representational convergence does not imply navigational convergence. The Platonic Representation Hypothesis (Huh et al., 2024) claims that models trained on similar data converge toward shared representations, supported by measures like CKA alignment. But representational alignment is not navigational alignment. Two agents can agree perfectly on a map — identical embeddings, identical similarity structures — and still prefer different routes.

Our benchmark tests the behavioral complement: do models that share representations also share navigational behavior? The answer requires distinguishing three levels. Navigational *structure* converges — all 12 models exhibit gait consistency, directional asymmetry, and bridge bottleneck topology. Navigational *content* converges among frontier-scale models — the same bridge concepts ("spectrum," "dog," "warm") appear across providers, with aggregate bridge frequency differences whose confidence intervals include zero. But navigational *dynamics* diversify: the 2.9x gait range from GPT (0.258) to Mistral (0.747) does not collapse with scale.

Same map, different routes. The Platonic Representation Hypothesis receives partial confirmation with a clear boundary: structure and content converge, but gaits remain model-specific.

The reversal curse reveals directional structure. Berglund et al. (2024) demonstrated that autoregressive models cannot reverse learned associations: training on "A is B" does not teach "B is A." This has been framed as a failure mode. We reframe it as a structural signal.

Tversky (1977) established decades earlier that human similarity judgments are inherently asymmetric — North Korea is judged more similar to China than China is to North Korea. Our benchmark finds pervasive directional asymmetry in LLM navigation (mean 0.811 in the Phase 2 cohort, all 12 tested models exceeding the 0.60 threshold), consistent with quasimetric topology. Forward and reverse paths between the same two concepts share less than 19% of their waypoints on average. The asymmetry is not noise — it tracks semantic properties of the concept pairs and model-specific navigational styles.

The connection across these three findings is thematic: all concern directional asymmetry in conceptual structure. We do not claim that the reversal curse mechanistically causes our observed asymmetry; rather, both phenomena reflect a shared property of directed conceptual spaces in which the path from A to B traverses fundamentally different intermediate territory than the path from B to A.

Multi-hop reasoning failures call for richer diagnostics. The compositionality gap (Press et al., 2023) and the two-hop curse (Two-Hop Curse Authors, 2025) document that models struggle with latent multi-step reasoning. These findings use binary pass/fail evaluation: the model either retrieves the composed fact or it does not.

Our benchmark generalizes the diagnostic. Rather than asking whether a model can compose two facts, we ask what *shape* the composition takes — which intermediate concepts appear, how consistently they appear across repeated runs, whether they compose properly (the triangle inequality holds at approximately 91%), and whether they differ by direction. This produces a geometric characterization of compositional structure rather than a scalar accuracy measure.

The characterization is informative even when the model "fails." Asymmetry and bridge fragility are structured phenomena, not random noise. A model that produces different paths in opposite directions is not failing at composition — it is revealing the directional structure of its conceptual space.

1.3 Contributions

This paper makes six contributions, spanning paradigm design, empirical findings, and methodological transparency:

1. **A novel evaluation paradigm** — waypoint elicitation as behavioral probe for conceptual geometry, distinct from static embedding analysis, factual QA, and free association. The paradigm is self-grounding: model outputs generate their own ground truth, enabling scalable evaluation without human annotation. This introduces a circularity concern addressed through external mathematical tests (metric axioms), cross-model validation (12 independent models), and causal intervention (pre-fill experiments).

2. **Six robust empirical claims** about navigational structure, replicated across multiple phases and models, plus one qualified claim (R5, control validation) that reveals a fundamental paradigm limitation (Section 9.5):

   - Models have characteristic conceptual gaits spanning a 2.9x range [robust] R1.
   - Navigation is consistent with quasimetric structure: systematically asymmetric (mean 0.811) with the triangle inequality holding at approximately 91% [robust] R2.
   - Polysemy sense differentiation is genuine, with cross-pair Jaccard of 0.011-0.062 for different senses [robust] R3.
   - Hierarchical paths are compositional with 4.9x higher transitivity than random controls [robust] R4.
   - Control validation supports the paradigm, though stapler-monsoon fails strict criteria for all 12 models [robust, qualified] R5.
   - Bridge concepts function as structural bottlenecks with frequencies exceeding 0.97 under robustness testing [robust] R6.
   - Cue-strength gradient exists: 12/16 family-model combinations show monotonic decrease [robust] R7.
   - R3, R4, and R7 are untested beyond the original 4-model cohort.

3. **A mechanism ceiling result** — 7 follow-up hypotheses fail across Phases 8–10 — spanning mechanistic, model-deficit, methodological, and effect-direction types — with one additional hypothesis initially falsified then resurrected with more data, establishing that conceptual navigation resists simple explanations (Section 7). Prediction accuracy descended from 81.3% for characterization questions to 20–24% for mechanistic ones — tracking a systematic shift in prediction type, not random degradation.

4. **Cross-architecture generality** — navigational structure and content generalize across 12 models from 11 independent training pipelines; scale differentiates (Section 8). Llama 3.1 8B retains gait and asymmetry but navigates through different landmarks (bridge frequency 0.200 vs 0.717–0.817 for frontier models). The within-family Llama comparison (Maverick 0.724 vs 8B 0.200) provides the sharpest evidence.

5. **Protocol robustness** — bridge frequency insensitive to waypoint count and temperature (exceeding 0.97 across all conditions); gait rankings largely stable (W = 0.840); model identity dominates protocol variation (eta-squared = 0.242 vs approximately 0 for protocol factors) (Section 9). Asymmetry is resolution-dependent — a measurement sensitivity finding that qualifies R2 without undermining it.

6. **29 documented dead ends** as honest accounting of the benchmark's learning curve. Failed predictions — "metaphor" at frequency 0.00 for language-to-thought, gait normalization producing 0.000 improvement, three successive Gemini deficit characterizations all falsified — contain more methodological information than several of the successful findings. The graveyard is a methodological contribution, not a failure record.

1.4 Paper Structure

The paper follows a six-act narrative structure, tracing the benchmark from initial characterization through mechanism and generality to robustness.

Section 2: The Benchmark. Defines the waypoint elicitation paradigm, derived metrics (gait consistency, path asymmetry, bridge frequency, navigational distance, triangle inequality), concept pair selection across 9 categories, the 12-model cohort accessed via OpenRouter, and the evidential strategy including self-grounding circularity mitigations.

Section 3: Related Work. Positions the benchmark against existing work on LLM geometry, conceptual spaces, compositionality, and evaluation methodology.

Section 4: Act I – Structure. Establishes that models have distinct, stable conceptual gaits spanning a 2.9x range; paths are self-consistent across time (no temporal drift) and resolution (5-waypoint paths appear as subsequences of 10-waypoint paths 67.9% of the time); both endpoints shape the trajectory via a dual-anchor effect; and polysemy sense differentiation (cross-pair Jaccard near zero for "bank," "bat," "crane") confirms that paths reflect conceptual structure rather than surface word statistics.

Section 5: Act II – Topology. Characterizes navigational geometry as consistent with quasi-metric structure: systematically asymmetric (mean 0.811), with the triangle inequality holding at approximately 91% across three independent samples, compositional hierarchy at 4.9x over random controls, bottleneck bridge topology where navigational salience depends on directional information content rather than associative strength, and non-uniform navigational traffic concentrated at preferred waypoints.

Section 6: Act III – Mechanism. Pre-filling the first waypoint causally displaces bridges (mean displacement 0.515, CI excluding zero), revealing associative primacy as the primary mechanism with secondary content modulation. The relation class of the pre-fill systematically affects bridge survival (Friedman $p = 0.034$), and the effect is bidirectional — displacement for dominant bridges, facilitation up to 8x for marginal bridges under semantically aligned pre-fills.

Section 7: Act IV – Limits. Seven mechanistic hypotheses fail across three phases; prediction accuracy descends from 81.3% to 20%; cross-model distance is fundamentally model-dependent ($r = 0.170$, gait normalization produces 0.000 improvement); the Gemini mystery survives three successive characterization attempts; and the mechanism ceiling establishes that navigation resists single-variable explanation.

Section 8: Act V – Generality. Gait and asymmetry replicate universally across 12 models from 11 independent training pipelines. Bridge content generalizes among frontier-scale models but not to Llama 3.1 8B (bridge frequency 0.200 vs 0.717–0.817). The structure/content/scale hierarchy emerges as the organizing framework for cross-architecture comparison.

Section 9: Act VI – Robustness. Model identity dominates protocol variation (eta-squared = 0.242 vs approximately 0 for protocol factors). Bridge frequency is the most robust property, exceeding 0.97 across all tested conditions. Asymmetry is resolution-dependent: 9-waypoint conditions clear the 0.60 threshold; 5-waypoint conditions fall just short. The control pair problem — 0 of 24 replacement cells passing strict criteria — is real and acknowledged directly.

Section 10: Discussion. Synthesizes the six-act narrative, states nine limitations directly (six

benchmark design, three phenomenon), connects findings to Karkada et al.'s geometric inevitability, Gardenfors's conceptual spaces, the Platonic Representation Hypothesis, and spectral semantic attractors (Wyss and others, 2025), and identifies ten future directions from multi-variable mechanism models to human comparison.

Section 11: Conclusion. Distills the central empirical contribution: LLMs navigate conceptual space with structured, asymmetric, model-specific geometry that generalizes across architectures and resists mechanistic decomposition — structure is detectable before it is explainable.

2 The Benchmark

This section defines the Conceptual Topology Mapping Benchmark precisely enough for replication. We describe the elicitation paradigm, derived metrics, concept pair selection, model cohort, experimental phases, and evidential strategy. The goal is to establish the methodological foundation on which all subsequent findings rest; summary statistics that appear in the phase overview table (Section 2.5) and limitations discussion (Section 2.6) are included for context but are developed fully in their respective results sections.

2.1 Waypoint Elicitation

Task. Given a source concept A and a destination concept B, the model is asked to produce N intermediate waypoints that form a conceptual path from A to B. The waypoints are ordered — they represent a traversal through conceptual space, not an unstructured set of associations.

Prompt design. Two prompt formats were evaluated in a holdout experiment (Phase 1, 15 pairs, 4 models):

- **Semantic format** (selected):

  > Imagine walking through conceptual space from "[A]" to "[B]". What [N] landmarks do you pass along the way? List only the landmark concepts, one per line, numbered 1 through [N]. Do not include the starting or ending concepts.

- **Direct format** (backup, used for models that parse poorly under semantic):

  > List exactly [N] intermediate concepts that form a path from "[A]" to "[B]". Respond with only the concepts, one per line, numbered 1 through [N]. Do not include the starting or ending concepts.

The semantic format was selected based on higher extraction accuracy and more interpretable waypoints on the holdout set. The direct format was retained as a fallback for models that produced unparseable outputs under the semantic prompt (notably some Phase 10A models during reliability probing).

Default parameters. Unless otherwise noted, all elicitations used 7 waypoints at temperature 0.7 with 10–20 independent runs per (model, pair, direction) combination. Phases 1–2 used 20 runs; later phases standardized to 10 runs per condition to manage API costs while maintaining adequate statistical power for bootstrap confidence intervals.

Pre-fill variant. Beginning in Phase 7A, a causal intervention variant was introduced: the first waypoint is pre-filled with a specified concept, and the model is asked to continue the remaining 6 waypoints. The pre-fill prompt format is:

> The first intermediate concept between "[A]" and "[B]" is "[pre-fill]". List exactly 6 more intermediate concepts that continue the path from "[pre-fill]" to "[B]". Respond with only the concepts, one per line, numbered 1 through 6. Do not include "[A]", "[pre-fill]", or "[B]" in your list.

This intervention tests whether navigational structure is causally sensitive to the starting trajectory, rather than merely statistically regular. (The pre-fill prompt is defined per-experiment in phase-specific scripts rather than in the shared prompt builder, as its structure differs from standard elicitation.)

Self-grounding. A distinctive feature of this benchmark is that the model's own outputs generate ground truth. There are no external labels, human annotations, or correct answers. Metrics are derived entirely from the statistical properties of the model's path outputs across repeated runs. This is both a strength (no annotation required, scalable to arbitrary model counts) and a limitation (circular — see Section 2.6).

Comparison to prior paradigms. Waypoint elicitation occupies a distinct position among probes of conceptual structure:

- *Word analogy tasks* (Mikolov et al., 2013) test single-step relational reasoning ("A is to B as C is to ?"). Our paradigm is multi-step and produces full path structure.
- *Free association norms* (Nelson et al., 2004; LLM World of Words Authors, 2025) produce local, undirected chains from a single cue. Waypoint elicitation is endpoint-conditioned: the model must produce a path from A *to* B, maintaining awareness of the destination throughout. This constraint transforms association into navigation — goal-directed traversal rather than local spreading activation.
- *Word convergence games* (Hodge, 2025) ask two players to alternate single words until they converge. The paradigm that inspired this benchmark, but convergence games test bilateral negotiation; our paradigm tests unilateral navigation.

Why this is not free association. The endpoint-conditioning is the critical distinction. Three forms of evidence support this claim: (a) paths between different endpoints sharing an intermediate concept diverge after that concept (the dual-anchor effect, Section 4.3), demonstrating that both endpoints shape the trajectory; (b) nonsense control pairs produce near-zero consistency while meaningful pairs produce structured, repeatable paths; and (c) bridge concepts are structural bottlenecks that depend on the specific pair geometry, not high-frequency associates of either endpoint (Section 5.4).

2.2 Derived Metrics

All metrics operate on canonicalized waypoint sequences (see Section 2.3 for the canonicalization pipeline). Some metrics reduce sequences to unordered sets (Jaccard similarity, bridge frequency); others depend on position within the ordered sequence (positional bridge profile, positional convergence). Unless otherwise stated, confidence intervals are 95% bootstrap percentile intervals computed from 1,000 resamples using a seeded PRNG (seed = 42) for reproducibility.

Gait consistency (intra-model Jaccard). For a given (model, pair, direction) condition with n independent runs producing waypoint sets $W_1, W_2, \ldots, W_n$, gait consistency is the mean pairwise Jaccard similarity:

$$\text{Gait} = \frac{2}{n(n-1)} \sum_{i<j} J(W_i, W_j)$$

where $J(A, B) = |A \cap B|/|A \cup B|$. Higher values indicate more deterministic navigation. Range: [0, 1].

> **Lexical limitation.** Jaccard similarity is computed on canonicalized token sets. Semantically equivalent waypoints — "melody" and "tune," "cemetery" and "graveyard" — are treated as entirely distinct. This means Jaccard *underestimates* true conceptual consistency. All consistency values, asymmetry values, and "distance" claims throughout this paper are lexical, not semantic. The magnitude of this bias is bounded by the canonicalization pipeline (which handles morphological variants such as running/run, cities/city) but is otherwise unknown. Embedding-based semantic similarity was implemented but deferred due to API cost. This is the benchmark's most pervasive methodological limitation, never fully mitigated across 11 phases.

Path asymmetry. For a pair (A, B), we collect forward runs (A→B) and reverse runs (B→A), then compute the mean cross-direction Jaccard across all forward×reverse run pairs:

$$\text{CrossJaccard} = \frac{1}{n_f \cdot n_r} \sum_{i=1}^{n_f} \sum_{j=1}^{n_r} J(W_i^{\text{fwd}}, W_j^{\text{rev}})$$

The asymmetry index is defined as 1 − CrossJaccard. Values near 0 indicate symmetric paths; values near 1 indicate complete asymmetry. Statistical significance is assessed via permutation test (1,000 permutations, null hypothesis: forward and reverse runs are drawn from the same distribution, using the correction of Phipson & Smyth, 2010 to avoid exact-zero p-values).

Bridge frequency. For a concept triple (A, B, C) where B is a candidate bridge, bridge frequency is the fraction of A→C runs in which B appears as a waypoint:

$$\text{BridgeFreq}(B \mid A \to C) = \frac{|\{r \in \text{runs}(A \to C) : B \in r\}|}{|\text{runs}(A \to C)|}$$

Bridge detection uses fuzzy matching: exact match after canonicalization, word-boundary containment (e.g., "domestic dog" matches bridge "dog"), and token-level matching (splitting on whitespace and hyphens). This catches multi-word waypoints containing the bridge but does not capture true synonyms (e.g., "canine" would not match "dog").

Navigational distance. For a set of runs in the same direction, navigational distance is defined as:

$$d(X \to Y) = 1 - \text{mean within-direction Jaccard}$$

High within-direction consistency yields low distance (easy, deterministic navigation); low consistency yields high distance (difficult, variable navigation). This is used for triangle inequality testing.

Triangle inequality. For a triple (A, B, C), the triangle inequality is tested as:

$$d(A \to C) \leq d(A \to B) + d(B \to C)$$

A small epsilon tolerance (0.001) is applied for floating-point comparison. The compliance rate is the fraction of (triple, model) combinations where the inequality holds.

> **Methodological note.** The triangle inequality uses within-direction Jaccard (consistency of repeated runs in the same direction) as the distance function, while the

asymmetry test uses cross-direction Jaccard (comparing A→B vs B→A paths). These are different operationalizations of "distance." Findings are reported as *consistent with* quasimetric structure rather than formally establishing it on a single unified distance function.

Waypoint transitivity. For a triple (A, B, C), transitivity measures overlap between the direct path A→C and the composed path A→B ∪ B→C:

$$\text{Transitivity} = J(\text{waypoints}(A \to C),\ \text{waypoints}(A \to B) \cup \text{waypoints}(B \to C))$$

Higher transitivity indicates that the direct path shares waypoints with the two-leg composed path — evidence for compositional structure.

Positional bridge profile. For each of the N waypoint positions (0-indexed), the bridge position profile records the fraction of runs where the bridge concept appears at that position. This reveals whether bridges anchor early in the path, appear at the midpoint, or distribute uniformly.

Pre-fill bridge survival. For pairs tested under the pre-fill intervention (Phases 7A, 9C, 10B), bridge survival rate is the ratio of bridge frequency under pre-fill to bridge frequency unconstrained:

$$\text{SurvivalRate} = \frac{\text{BridgeFreq}_{\text{pre-filled}}}{\text{BridgeFreq}_{\text{unconstrained}}}$$

Values below 1.0 indicate bridge displacement; values above 1.0 indicate bridge facilitation (observed for marginal bridges under semantically aligned pre-fills). Survival rate is the primary metric for comparing pre-fill conditions.

Positional displacement. The Phase 7A analysis additionally computes a position-specific displacement: bridge frequency at the bridge's unconstrained modal position minus bridge frequency at that same position under pre-fill. This captures whether pre-filling shifts the bridge to a different position in the path, not merely whether it reduces overall bridge frequency. The positional and overall metrics are related but not identical — a bridge can survive (appear somewhere in the path) while being positionally displaced.

Shannon entropy. For a given (model, pair, direction), the waypoint frequency distribution is computed by counting each unique waypoint once per run, normalizing to proportions, and computing Shannon entropy:

$$H = -\sum_i p_i \log_2 p_i$$

Lower entropy indicates concentrated, near-deterministic navigation; higher entropy indicates diverse, exploratory behavior.

Uniformity test. Departure from a uniform waypoint distribution is assessed via chi-squared goodness-of-fit test (comparing observed waypoint counts against the expectation that all waypoints are equally likely), with p-values computed using the Wilson-Hilferty normal approximation. When testing multiple pairs, Bonferroni correction is applied at the pair level (dividing the significance threshold by the number of pairs tested). Pair-level rejection is assessed per model; aggregate summaries report the fraction of pairs where at least one model rejects uniformity at the corrected threshold.

Metric	Formal Definition	Range	Interpretation	Key Limitation
Gait consistency	Mean pairwise Jaccard across runs	[0, 1]	Higher = more deterministic	Lexical only; underestimates true consistency
Path asymmetry	1 − mean cross-direction Jaccard	[0, 1]	Higher = more asymmetric	Lexical; true asymmetry likely lower
Bridge frequency	Fraction of A→C runs containing B	[0, 1]	Higher = stronger bottleneck	Fuzzy matching misses synonyms
Navigational distance	1 − mean within-direction Jaccard	[0, 1]	Higher = harder navigation	Not a true metric (asymmetric)
Triangle inequality	d(A,C) ≤ d(A,B) + d(B,C)	Boolean	Holding = metric-like structure	Different distance operationalization than asymmetry
Transitivity	Jaccard of direct vs composed paths	[0, 1]	Higher = more compositional	Triple cross-product pseudoreplication (mitigated by run-level bootstrap)
Bridge position	Per-position bridge frequency	[0, 1] per position	Reveals anchoring pattern	Resolution limited by waypoint count
Bridge survival rate	Pre-filled freq / unconstrained freq	$[0, \infty)$	< 1 = displaced; > 1 = facilitated	Requires nonzero baseline frequency
Positional displacement	Modal-position freq change under pre-fill	[−1, 1]	Positive = bridge displaced at peak	Position-specific; misses total-frequency effects
Shannon entropy	$-\Sigma\ p_i \log_2 p_i$ of waypoint frequencies	$[0, \infty)$	Lower = concentrated; higher = diverse	Sensitive to canonicalization granularity

2.3 Concept Pair Selection

Categories. Concept pairs were selected to span the major axes of semantic variation. The Phase 1 inventory comprised 36 pairs across 9 categories:

- **Anchor** (5 pairs): Pairs with known basin structure from prior word convergence experiments (e.g., Beyoncé→erosion with known basin "formation"). Used to calibrate expectations from prior work.
- **Hierarchy** (4 pairs): Hypernym–hyponym relationships at varying taxonomic depth (e.g., animal→poodle, emotion→nostalgia). Test compositional path structure.
- **Cross-domain** (4 pairs): Pairs spanning unrelated conceptual domains (e.g., music→mathematics, love→death). The benchmark's primary test cases — conceptually distant, requiring genuine navigation.
- **Polysemy** (6 pairs): Pairs where one concept is polysemous, with targets selecting different senses (e.g., bank→river vs bank→mortgage; bat→cave vs bat→baseball). Test sense differentiation.

- **Near-synonym** (4 pairs): Semantically close concepts (e.g., happy→joyful, student→pupil). Expected to produce short conceptual distances.
- **Antonym** (2 pairs): Opposite concepts (e.g., hot→cold, love→hate). Test spectrum traversal.
- **Control-identity** (2 pairs): Same concept in both positions (e.g., apple→apple). Expected to produce low-distance, high-consistency paths.
- **Control-random** (7 pairs): Arbitrarily paired concepts (e.g., origami→gravity, stapler→monsoon). Expected to produce low consistency and no systematic bridge concepts.
- **Control-nonsense** (2 pairs): Nonsense strings as endpoints (e.g., "blicket"→"daxin"). Expected to produce near-zero consistency.

These were split into a holdout set (15 pairs, used only for prompt format selection) and a reporting set (21 pairs, used for all published analyses), with even category distribution across both sets to avoid selection bias.

Bridge concept types. Across 11 phases, bridge concepts were classified into four functional types:

- **Bottleneck bridges** name the primary axis of connection between two domains. Example: "spectrum" for light→color (names the mechanism), "deposit" for loan→shore (names the action linking financial and geographic senses of "bank").
- **Gradient midpoint bridges** occupy a position on a continuous dimension between the endpoints. Example: "warm" for hot→cold, "twilight" for day→night.
- **Causal-chain bridges** name a process step connecting cause to effect. Example: "fermentation" for grape→wine, "germination" for seed→garden.
- **Too-central bridges** are informationally redundant — both endpoints already imply the candidate bridge. Example: "fire" for spark→ash, "water" for rain→ocean. These are expected to produce low bridge frequencies because they add no navigational information.

Empirical results distinguishing these types are presented in Section 5.4.

Expansion across phases. The initial 21 reporting pairs were supplemented in each subsequent phase based on findings from the previous one. By Phase 11, the benchmark encompassed over 100 unique concept pairs and triples, each selected to test specific structural hypotheses. Pair selection was not random — it followed the exploration-first methodology, targeting the most informative comparisons based on prior results.

Canonicalization pipeline. All model outputs pass through a four-stage canonicalization pipeline before metric computation:

1. **Lowercase:** Convert all characters to lowercase.
2. **Article stripping:** Remove leading articles ("the", "a", "an").
3. **Lemmatization:** Apply morphological normalization using the compromise NLP library. Verbs are converted to infinitive form ("running" → "run," "walked" → "walk"); nouns are converted to singular form ("cities" → "city," "dogs" → "dog").
4. **Phrase normalization:** Collapse whitespace, strip trailing punctuation, trim.

After canonicalization, duplicate waypoints within a single run are removed (preserving order, keeping first occurrence). This pipeline handles morphological variants but not synonyms — "melody" and "tune" remain distinct tokens after canonicalization.

Response parsing. Model responses are parsed using a multi-strategy extraction pipeline that attempts, in order: JSON array extraction, numbered list parsing, bullet list parsing, comma-separated value extraction, arrow-separated extraction (recognizing ASCII arrow patterns: `->`, `=>`,

`-->`, `==>`), and line-by-line fallback. This accommodates the wide variation in output formatting across 12 models.

Extraction count handling. After parsing and canonicalization, if the number of extracted waypoints exceeds the requested count N, the list is truncated to N. If fewer than N waypoints are extracted, the run is retained with the shortened list and flagged (`extractionCountMismatch`). This lenient policy avoids discarding partial data, but extraction mismatches can affect downstream metrics; phases report extraction accuracy as a data quality indicator.

2.4 Model Selection

Twelve models from 11 independent training pipelines were tested across the benchmark's 11 phases, all accessed via the OpenRouter API for a uniform interface. Models were added incrementally as the benchmark expanded:

Model	Provider	OpenRouter ID	Approx. Scale	Cohort	Phases
Claude Sonnet 4.6	Anthropic	`anthropic/claude-sonnet-4.6`	Frontier	Core (Phases 1–9)	1–11
GPT-5.2	OpenAI	`openai/gpt-5.2`	Frontier	Core (Phases 1–9)	1–11
Grok 4.1 Fast	xAI	`x-ai/grok-4.1-fast`	Frontier	Core (Phases 1–9)	1–9
Gemini 3 Flash	Google	`google/gemini-3-flash-preview`	Frontier	Core (Phases 1–9)	1–9
MiniMax M2.5	MiniMax	`minimax/minimax-m2.5`	Frontier	Expanded (Phase 10A)	10
Kimi K2.5	Moonshot AI	`moonshotai/kimi-k2.5`	Frontier	Expanded (Phase 10A)	10
GLM 5	Zhipu AI	`z-ai/glm-5`	Frontier	Expanded (Phase 10A)	10*
Qwen 3.5 397B-A17B	Alibaba	`qwen/qwen3.5-397b-a17b`	397B MoE	Expanded (Phase 10A)	10
Llama 3.1 8B Instruct	Meta	`meta-llama/llama-3.1-8b-instruct`	8B	Expanded (Phase 10A)	10
DeepSeek V3.2	DeepSeek	`deepseek/deepseek-v3.2`	Frontier	Final (Phase 11A)	11
Mistral Large 3	Mistral AI	`mistralai/mistral-large-2512`	Frontier	Final (Phase 11A)	11

Model	Provider	OpenRouter ID	Approx. Scale	Cohort	Phases
Cohere Command A	Cohere	`cohere/command-a`	Frontier	Final (Phase 11A)	11
Llama 4 Maverick	Meta	`meta-llama/llama-4-maverick`	Frontier	Final (Phase 11A)	11

*GLM 5 was rate-limited during Phase 10A and produced insufficient data for full analysis; it is excluded from the 12-model cohort reported throughout the paper but listed here for completeness.

Cohort structure. The core cohort (Claude, GPT, Grok, Gemini) anchored Phases 1–9; Claude and GPT continued into Phases 10–11 for comparative baselines, while Grok and Gemini were not tested beyond Phase 9. Expanded and final cohort models were added in Phases 10–11 specifically to test whether findings generalize beyond the original four models. This design means that some claims (R1, R2, R6) are supported by all 12 models, while others (R3, R4, R7) are tested only on the original 4-model cohort.

Diversity. The 12 models span 11 independent training pipelines (Llama 3.1 8B and Llama 4 Maverick share the Meta Llama family), ranging from 8B parameters (Llama 3.1 8B) to frontier-scale models. This enables both cross-architecture comparison and within-family scale comparison (Llama 8B vs Llama 4 Maverick).

Reliability probing. Before full data collection, each new model underwent a reliability probe: 3 concept pairs × 2–5 runs, testing connectivity rate, parse rate, median latency, and waypoint extraction accuracy. Models failing reliability probing were excluded (GLM 5, partially) or tested with extended timeouts (Qwen, MiniMax, and Kimi were recovered using a 300-second "patient mode" after failing the default 60-second timeout).

2.5 Experimental Phases

The benchmark followed an exploration-first methodology: each phase was designed to follow the most interesting signal from the previous phase's findings. This produced a non-linear progression through the space of possible experiments, with the direction of inquiry determined by data rather than a fixed plan. Eleven phases were completed, comprising approximately 21,540 total API runs.

Phase	Name	New Runs	Key Question	Primary Finding
1	Pilot & prompt selection	~2,480	Do models produce consistent waypoint paths?	Models have distinct gaits: Claude 0.578 vs GPT 0.258 (2.2× gap)
2	Reversals & asymmetry (5 waypoints)	~960	Is A→B navigation symmetric with B→A?	Navigation is fundamentally asymmetric (mean 0.811); symmetry axiom fails
3	Positional convergence & transitivity	~1,260	Do paths compose? Does triangle inequality hold?	Hierarchical transitivity 4.9× over random; triangle inequality 91%

Phase	Name	New Runs	Key Question	Primary Finding
4	Cross-model bridge topology	~1,600	Do models agree on bridge concepts?	Bridges are structural bottlenecks, not associations; prediction accuracy 81.3%
5	Cue-strength & dimensionality	~3,720	Is bridge frequency graded by semantic cue strength?	Gradient real (12/16 monotonic); forced crossing discovery
6	Navigational salience & position	~2,080	Where do bridge concepts appear in paths?	Bridges anchor early (position 1–2); distributions non-uniform
7	Early anchoring & navigational mechanics	~2,360	Does pre-filling a waypoint causally affect the path?	Displacement 0.515 (CI excludes zero); gradient > causal-chain bridges
8	Bridge fragility & Gemini deficit	~2,690	Can we predict bridge fragility from route competition?	All 3 hypotheses fail; prediction accuracy 24% (mechanism floor)
9	Dominance, transformation, facilitation	~3,037	Can we predict bridge survival from dominance ratios?	All 3 hypotheses fail; prediction accuracy 20%; mechanism ceiling confirmed
10	Model generality & relation classes	~1,680	Do structural findings generalize to new models?	R1/R2 replicate across 4 new models; bridge structure generalizes; scale differentiates
11	Expanded generality & robustness	~2,040	Do findings survive more models and protocol variation?	R1/R2 universal across 12 models; bridge frequency most robust property; model identity dominates protocol

Run counts are approximate due to retries, failures, and data reuse across phases. Reused runs (from earlier phases providing baseline data for later experiments) are counted once in their original phase.

2.6 Validity and Evidential Strategy

What this benchmark does not measure. It does not measure human conceptual geometry (no human participants were tested). It does not measure latent model geometry directly (we test

behavior, not internal representations or embeddings). It does not measure factual correctness (there are no correct answers — only structural properties to characterize). And, as argued above, it does not measure free association (waypoints are endpoint-conditioned).

Why waypoint elicitation measures navigation. Free association produces local, undirected chains — each word triggers the next with no global constraint. Waypoint elicitation is *constrained*: it requires producing a path conditioned on both endpoints simultaneously. The model must maintain awareness of both source and destination, producing intermediate steps that form a directed traversal. This endpoint-conditioning is what transforms association into navigation. Three lines of evidence support this distinction:

1. *Dual-anchor effect*: Paths exhibit U-shaped convergence profiles, with both the source and destination concepts exerting measurable influence on waypoint selection at their respective ends (Section 4.3). This would not occur under undirected spreading activation.
2. *Bridge specificity*: Bridge concepts are pair-specific structural bottlenecks, not high-frequency associates of either endpoint. The same concept can serve as a bridge for one pair and be absent for another pair sharing an endpoint (Section 5.4).
3. *Causal sensitivity*: Pre-filling the first waypoint with an alternative concept causally displaces bridge concepts downstream, demonstrating that navigational structure is not a fixed property of the endpoints but emerges from the trajectory itself (Section 6.1).

Self-grounding and circularity mitigation. Model outputs generate their own ground truth, which introduces a circularity concern: are we measuring genuine navigational structure, or artifacts of the generation process? Three mitigations address this:

1. *External mathematical properties*: Metric axioms (triangle inequality, symmetry/asymmetry) have definitions independent of any model. Testing whether model-generated paths satisfy these axioms grounds the benchmark in external mathematical structure.
2. *Cross-model comparison*: Twelve independent models serve as mutual validation. When all 12 models exhibit the same structural property (e.g., asymmetry > 0.60), the finding cannot be attributed to idiosyncrasies of any single model's generation process.
3. *Causal intervention*: The pre-fill experiments (Phases 7A, 9C, 10B) demonstrate that navigational structure is causally sensitive to perturbation, not merely a statistical regularity. Bridge displacement under pre-filling is structural, not an artifact of decoding.

The benchmark's evidential standard is *relative*, not absolute: we measure the magnitude gap between experimental and control conditions, not whether controls achieve a fixed baseline threshold. This is explicitly acknowledged as the primary limitation (see below).

Paradigm limitations.

1. *Lexical metrics only*: All metrics use Jaccard set similarity after canonicalization. Semantically equivalent waypoints are treated as distinct. The asymmetry value of 0.811 (Phase 2 cohort) should be interpreted as *lexical* asymmetry; true conceptual asymmetry is likely lower. A spot-check using embedding-based similarity on a subset of pairs would help bound this effect but was not conducted.

2. *Prompt sensitivity*: The benchmark's findings could be sensitive to the specific prompt wording. This is partially addressed by Phase 11C, which tested a 2×2 grid of waypoint counts (5, 9) and temperatures (0.5, 0.9) and found that model identity dominates protocol variation ($\eta^2 = 0.242$ for model vs 0.008 for waypoints and 0.002 for temperature). However, only three models were tested in the robustness analysis.

3. *API-mediated access*: All models are accessed via OpenRouter, which introduces potential confounds: temperature may be approximate for some providers, internal model state cannot be controlled, and silent model updates may occur during multi-day data collection. Partial mitigation comes from the no-temporal-drift finding (cross-batch Jaccard within 0.05 of within-batch, Phase 4B) and the robustness to temperature variation (Phase 11C).

4. *Control pair failure*: The benchmark's most serious limitation. LLMs find navigable routes between *any* concept pair at 7 waypoints, defeating strict single-pair control validation. All 12 models show some structure on the stapler–monsoon control pair (top waypoint frequency 0.650–1.000 across models). The benchmark's validity rests on relative performance gaps — experimental pairs show predicted bridges with higher consistency than controls show unpredicted bridges — not on absolute control baselines. This limitation is discussed in detail in Section 9.5.

3 Related Work

The benchmark sits at the intersection of seven distinct literatures: static geometry of LLM representations, the theoretical origin of that geometry, conceptual navigation and interpolation, consistency and directionality, multi-hop reasoning, cross-model comparison, and cognitive science foundations. Each has produced results that inform our design and findings, but none asks the question we ask: whether models can *navigate* conceptual space consistently and whether the resulting paths have measurable geometric structure. We review each literature and state precisely what gap the benchmark fills.

3.1 Static Geometry of LLM Representations

The linear representation hypothesis (Park et al., 2024) established that individual concepts correspond to linear directions in activation space. Its extension (Park et al., 2025) showed that hierarchical relationships form polytope structures — convex regions whose geometric relationships encode taxonomic nesting. Separately, hyperbolic embedding methods (Nickel and Kiela, 2017) demonstrated that tree-like hierarchies are better captured in negatively curved spaces, a finding extended to LLM representations by recent work on HELM (HELM Authors, 2025) and direct measurement of delta-hyperbolicity and ultrametricity (Delta-Hyperbolicity Authors, 2025), which quantify how tree-like a given embedding space actually is.

Topological data analysis has added finer-grained structural characterization. Explainable Mapper (Explainable Mapper Authors, 2025) identified circular manifolds for temporal concepts and linear manifolds for ordered quantities in LLM representations. Persistent Topological Features (Persistent Topological Features Authors, 2025) used persistence diagrams to characterize the homological structure of concept clusters. Wyss and others (2025) proved mathematically that LLM activation manifolds partition into discrete basins of attraction via spectral analysis of semantic attractors — a result directly relevant to the bottleneck bridge structure we observe empirically (Section 5.4). Joshi (2025) characterized intrinsic dimensionality profiles across transformer layers, finding a universal early-expand-late-compress pattern that holds across architectures.

Gap we fill. This entire literature characterizes *what shape the space has* — its curvature, dimensionality, manifold topology, and basin structure. All of it is static. We probe whether models can *navigate* that space consistently, producing paths with measurable geometric properties. The distinction is between a map and a route: static geometry gives us the map; our benchmark tests whether models can walk it.

3.2 Theoretical Origin of Geometric Structure

Karkada and others (2025) proved that translation symmetry in word co-occurrence statistics mathematically determines manifold geometry in embedding spaces. Their result is architecture-independent: the same geometric structures arise in Word2Vec, GloVe, and Gemma because all train on data distributions with shared statistical regularities. Geometry comes from data, not architecture.

This theoretical result connects directly to our empirical findings. If geometric structure is determined by data statistics that all models share, then navigational structure derived from that geometry should also generalize across architectures — which is what we observe (R1 and R2 universal across 12 models from 11 training pipelines; Section 8). The scale effect (Llama 8B as outlier in bridge frequency) is consistent with their framework: if extracting the full geometric structure requires sufficient capacity to represent translation symmetries, smaller models may have the coarse structure but lack the resolution to navigate through the same landmarks as frontier models.

Gap we fill. Karkada et al. study the static manifold shape and its mathematical origin. We test the dynamic complement: whether that geometry supports consistent behavioral navigation. Their theory is consistent with our empirical finding of cross-architecture generalization; our data provides the behavioral evidence their framework implies but does not test. The contributions are complementary — they explain why the geometry exists; we show what it supports.

3.3 Conceptual Navigation and Interpolation

The mathematical framework for navigation through latent spaces exists but has not been applied to LLM behavioral outputs. Arvanitidis et al. (2018) developed Riemannian geodesics as semantically meaningful paths in generative model latent spaces — shortest paths through curved space that respect the local geometry. The framework is directly applicable to conceptual navigation but has been used for image generation and VAE interpolation, not for probing how language models traverse between concepts.

The closest precedent for waypoint-based navigation is Relation Embedding Chains (Relation Embedding Chains Authors, 2023), which discovered intermediate concepts X such that A-to-B can be decomposed as A-to-X-to-B using knowledge graph paths. This is the nearest prior work to our waypoint elicitation paradigm, though it operates on static knowledge graph structure rather than behavioral generation. The Geometry of Knowledge (Geometry of Knowledge Authors, 2025) traversed semantic manifolds for generative diversity — the closest existing work to "conceptual traversal" — but used traversal as a tool for improving generation, not as an evaluation paradigm.

More recently, reasoning-as-path-search has entered the mainstream. RiemannInfer (RiemannInfer Authors, 2026) frames multi-step inference as trajectory optimization on a Riemannian manifold, and Geometric Reasoner (Geometric Reasoner Authors, 2026) uses geodesic metaphors for chain-of-thought planning. These works treat the path metaphor as productive for reasoning but do not test the geometric properties of the paths themselves.

Gap we fill. No prior work asks models to produce intermediate waypoints between two concepts and then tests whether the resulting paths have consistent geometric structure — satisfying metric axioms, exhibiting compositional properties, and responding to causal perturbation. The mathematical framework exists (Riemannian geodesics); the precedent for intermediate concepts exists (relation embedding chains); the path metaphor is becoming mainstream (reasoning-as-search). What is missing is the empirical test of whether behaviorally generated paths have measurable, replicable geometry. We provide that test.

3.4 Consistency, Directionality, and the Reversal Curse

Berglund et al. (2024) established the reversal curse: autoregressive models trained on "A is B" systematically fail to learn "B is A." This is an architectural limitation — the left-to-right generation order creates inherent directional bias. the directional optimization asymmetry literature (Directional Asymmetry Authors, 2025) deepened this finding by showing that inverse mappings incur higher excess loss even under symmetric training objectives, establishing that the A-to-B versus B-to-A gap is architectural, not merely a training-data artifact. Barakat (2026) extended directionality to semantic caching, demonstrating that input ordering steers inference trajectories into different attractor basins — directional hysteresis at the system level.

On the consistency side, recent work has documented pervasive self-consistency failures in frontier models (Self-Consistency Authors, 2025), with high variability even in GPT-4 across repeated evaluations on identical prompts. Trajectory variance in agentic settings (Trajectory Variance Authors, 2025) shows substantial action-sequence diversity across runs of the same agent on the same task, establishing that behavioral variability is not confined to simple generation tasks.

Our contribution. Existing work treats inconsistency and asymmetry as failure modes — problems to be solved or architectural limitations to be documented. We reframe asymmetry as data. The pervasive directional asymmetry we observe (mean 0.811 in the Phase 2 cohort, exceeding 0.60 across all 12 models tested) is not a bug but a structural property of navigational space, consistent with quasimetric topology. The pattern of asymmetry reveals structure: it tracks semantic properties of concept pairs and model-specific navigational styles, producing systematic geometric signatures rather than noise.

A defensive clarification is warranted here. Reviewers will ask why this is not simply self-consistency work with a geometry gloss. The contrast is precise: self-consistency work measures the *magnitude* of variance across samples and treats it as error to be minimized. We test whether the *structure* of that variance satisfies formal geometric axioms — metric properties, compositionality, causal sensitivity. We do not ask whether models are consistent; we ask whether the pattern of their inconsistency has geometric regularity. Variance is our data, not our failure mode.

3.5 Multi-Hop Reasoning and Compositionality

The compositionality gap (Press et al., 2023) demonstrated that models can answer individual subquestions correctly while failing the composed question — knowing both "A is B" and "B is C" but failing "A is C." the two-hop curse (Two-Hop Curse Authors, 2025) formalized this for knowledge composition: models trained on A-to-B and B-to-C facts fail A-to-C inference without explicit chain-of-thought prompting. TWOHOPFACT (TWOHOPFACT Authors, 2024) studied latent multi-hop reasoning through bridge entities, identifying the intermediate step as the critical point of failure — the model must internally activate the bridge concept to compose the two hops.

Our contribution. We generalize the compositionality question from binary pass/fail to geometric characterization. Multi-hop reasoning work asks whether the model gets the composed answer *correct*. We ask what *shape* the intermediate sequence takes — whether waypoints compose consistently (hierarchical transitivity 4.9x over random controls), whether bridge concepts function as structural bottlenecks (frequencies of 0.60-1.00 for predicted bridges), and whether the compositional structure satisfies the triangle inequality (~91% across three independent samples). Waypoints externalize latent composition, making the intermediate steps available for geometric analysis rather than accuracy scoring.

A second defensive clarification applies here. The paradigm will invite comparison to chain-of-thought elicitation. The contrast is fundamental: chain-of-thought elicits reasoning steps toward

a *correct answer* — intermediate steps are evaluated by whether they lead to a right response. We elicit navigational waypoints toward a *destination concept* — there is no correct answer, only geometric structure to characterize. Chain-of-thought is evaluated by task accuracy; our paradigm is evaluated by structural consistency, metric properties, and causal sensitivity. The two paradigms share surface similarity (both produce ordered intermediate steps) but differ in what they measure and how they are validated.

3.6 Cross-Model Comparison and the Platonic Representation Hypothesis

The Platonic Representation Hypothesis (Huh et al., 2024) predicts that model representations converge with scale, measured by CKA alignment between activation spaces. Larger models from different families produce increasingly similar internal representations — a convergence toward a shared "platonic" geometry. Multi-way alignment work (Multi-Way Alignment Authors, 2026) extended this to shared reference spaces across N models, and model stitching experiments (Model Stitching Authors, 2025) demonstrated that features transfer meaningfully but non-uniformly across architectures.

On the behavioral side, LLM behavioral fingerprinting (Behavioral Fingerprint Authors, 2025) established that models have characteristic behavioral signatures — stable patterns of response that distinguish one model from another across tasks. Nanda (2026) identified conversational attractor states: model-specific behavioral loops (Claude trending toward philosophical synthesis, GPT toward system-building) that function as the conversational analog of our quantitative gaits. Behavioral failure manifold mapping (Failure Manifold Authors, 2025) characterized model-specific topographies with qualitatively different basin structures, showing that models fail in geometrically distinct ways.

Our contribution. The PRH measures static representational alignment. We test *dynamic navigational alignment* — whether shared representations imply shared routes. The finding is a three-level hierarchy: structure generalizes (all 12 models have gait and asymmetry), content mostly generalizes among frontier-scale models (aggregate bridge frequency difference CI includes zero), but gaits diverge (the 2.9x gait range does not collapse with scale). Same map, different routes. This is a behavioral refinement of the PRH: representational convergence does not imply behavioral convergence. The models share structural properties (gait and asymmetry are universal) and largely converge on the strongest navigational landmarks, but they navigate with model-specific route preferences that remain distinctive even at frontier scale — and their navigational distances are fundamentally model-dependent (Section 7.4).

3.7 Cognitive Science Foundations

Gärdenförs (2000) proposed Conceptual Spaces as geometric similarity spaces organized along quality dimensions, with natural concepts forming convex regions and betweenness relations defining navigable paths. This theoretical framework posits that conceptual structure is inherently geometric — concepts have locations, distances between them are meaningful, and navigation between concepts follows principled geometric trajectories.

Cognitive neuroscience has found biological mechanisms consistent with this theory. Tolman (1948) proposed cognitive maps for spatial navigation. O'Keefe and Dostrovsky (1971) discovered place cells in the hippocampus. Constantinescu et al. (2016) demonstrated that grid cells in the human entorhinal cortex fire during *conceptual* navigation — not just physical movement through space but traversal through abstract conceptual dimensions. The spatial navigation machinery is repurposed for abstract thought.

Two additional cognitive science results bear directly on our findings. Tversky (1977) established that human similarity judgments are asymmetric: North Korea is judged more similar to China than China is to North Korea. This is the direct cognitive parallel to our quasimetric finding — asymmetry in conceptual distance is not an artifact of autoregressive generation but a property also observed in human cognition. Collins and Loftus (1975) proposed spreading activation through semantic networks, where activation propagates directionally from a source concept through associative links — a process model that maps onto the early-anchoring and bridge-bottleneck phenomena we observe.

More recently, LLM World of Words (LLM World of Words Authors, 2025) demonstrated that free association norms elicited from LLMs are directly comparable to human norms collected over decades, establishing that LLM conceptual organization shares at least surface-level structure with human semantic memory.

Our contribution. Gardenfors theorized that concepts live in navigable geometric spaces. Cognitive neuroscience found biological mechanisms (grid cells, place cells) that support navigation through those spaces. We provide a computational probe of this theoretical framework at scale — 12 models, 100+ concept pairs, ~21,540 runs. The partial confirmation is substantive: LLM behavior is consistent with concepts inhabiting navigable spaces with measurable, compositional distance structure (triangle inequality ~91%, hierarchical transitivity 4.9x over random). But the pervasive asymmetry (mean 0.811) challenges the standard metric assumption in conceptual spaces theory, suggesting that quasimetric structure may be a more appropriate formal framework. We do not claim that LLMs replicate human conceptual navigation — only that they exhibit analogous geometric regularities testable by similar methods. The parallel to Tversky's asymmetry and Constantinescu's grid-cell navigation is structural, not mechanistic.

The seven literatures reviewed here converge on a single gap. Static geometry is well-characterized and its theoretical origin is understood. The mathematical framework for navigation exists. Consistency and directionality are documented as phenomena. Multi-hop composition is studied as a reasoning failure mode. Cross-model convergence is measured at the representational level. Cognitive science provides both the theoretical framework and the biological precedent. What is missing — and what this benchmark provides — is an empirical test of whether LLM conceptual navigation has consistent, measurable geometric structure: not what the space looks like, but whether models can walk through it reliably, and what the shape of their walking reveals.

4 Act I — Structure: Models Have Navigational Geometry

The first three experimental phases established the empirical foundation for the benchmark: models navigate conceptual space with measurable, stable, model-specific structure. This section summarizes the five structural findings from the early phases; some were later replicated across expanded model cohorts (R1), while others remain well-supported observations from the original cohort (O1, O2, O9, O10).

4.1 Conceptual Gaits Are Distinct and Stable

The most robust finding in the benchmark is that each model navigates conceptual space with a characteristic level of consistency — a property we term its *conceptual gait*. Gait is operationalized as the mean pairwise Jaccard similarity across independent runs on the same (model, pair, direction)

condition: high gait indicates that a model produces similar waypoints each time it traverses a given conceptual path; low gait indicates that each traversal explores different intermediate territory.

The 12-model gait spectrum. Across 12 models from 11 independent training pipelines, conceptual gait spans a 2.9× range:

Model	Mean Jaccard	Cohort	Phases
Mistral Large 3	0.747	Phase 11A	11
Claude Sonnet 4.6	0.578	Original	1–11
DeepSeek V3.2	0.540	Phase 11A	11
Llama 4 Maverick	0.539	Phase 11A	11
Qwen 3.5 397B-A17B	0.508	Phase 10A	10
Cohere Command A	0.502	Phase 11A	11
MiniMax M2.5	0.419	Phase 10A	10
Moonshot Kimi K2.5	0.414	Phase 10A	10
Gemini 3 Flash	0.372	Original	1–9
Llama 3.1 8B Instruct	0.298	Phase 10A	10
Grok 4.1 Fast	0.293	Original	1–9
GPT-5.2	0.258	Original	1–11

At the extremes, Mistral produced near-deterministic paths on several pairs (Jaccard 0.936 on music→mathematics, 0.934 on hot→cold), while GPT's most consistent experimental pair (hot→cold, Jaccard 0.911 in Phase 1) was an outlier against its otherwise exploratory profile.

Temporal stability. For the four core models (Claude, GPT, Grok, Gemini), gait consistency was measured across Phases 1 through 9 — a span of multiple days and over 9,500 runs. The gait ordering (Claude > Gemini > Grok > GPT) was preserved throughout. Claude's mean Jaccard remained within [0.55, 0.60] across all phases in which it was tested; GPT's remained within [0.24, 0.28]. No systematic drift was detected.

Cross-architecture generality. Gait was initially characterized on 4 models from 4 providers (Phase 1). Phase 10A added 4 models from 4 additional providers, all falling within the previously observed range (0.298–0.508). Phase 11A added 4 more models extending the range upward (Mistral 0.747). Across 12 models from 11 independent training pipelines, every model exhibited a characteristic gait, confirming that this is a universal property of LLM conceptual navigation rather than a quirk of the original cohort.

Protocol robustness. Phase 11C tested whether gait was an artifact of the default elicitation protocol (7 waypoints, temperature 0.7) by varying both parameters in a 2×2 grid (5 and 9 waypoints × temperatures 0.5 and 0.9) for three models (Claude, GPT, DeepSeek). An ANOVA decomposition found that model identity accounted for the largest share of variance in Jaccard consistency ($\eta^2 = 0.242$, $p \approx 0.001$), while waypoint count ($\eta^2 = 0.008$) and temperature ($\eta^2 = 0.002$) were negligible. The gait rank ordering across models was largely preserved (Kendall's W = 0.840), though not perfectly — GPT and DeepSeek swapped positions in 4 of 5 conditions, with DeepSeek's baseline condition showing anomalously low Jaccard that was not replicated at other settings. These p-values are approximate (chi-squared approximation), and the robustness test covers only three models (Claude, GPT, DeepSeek); it addresses waypoint-count and temperature variation but not prompt sensitivity more broadly.

Control validation. The nonsense control pair (xkplm→qrvzt) produced mean Jaccard of 0.062 across all models — effectively zero consistency. GPT's nonsense Jaccard ranged from 0.006 to 0.015, indistinguishable from chance. This supports the interpretation that gait is content-sensitive

rather than a decoding artifact: when there is no conceptual structure to navigate, consistency vanishes. (Control validation is qualified by Phase 11B's finding that single-pair controls are insufficient for broader generality testing — see Section 2.6 — but the relative gap between experimental and control conditions remains large and consistent across the original cohort.)

Interpretation. Gait reflects a model's propensity to re-traverse the same conceptual territory across independent runs. A high-gait model like Mistral or Claude has a strongly determined internal map — given two concepts, it finds essentially the same intermediate landmarks every time. A low-gait model like GPT explores broadly, sampling different landmarks on each traversal while still arriving at the destination. Neither extreme is intrinsically better; they represent different navigational strategies. Crucially, gait is pair-dependent even within a model: Claude's Jaccard on hot→cold (0.911) far exceeds its Jaccard on random control pairs (0.291–0.343), indicating that gait interacts with the structure of the conceptual terrain, not just the model's decoding behavior.

4.2 Paths Are Self-Consistent Within Models

Before attributing structure to waypoint paths, it is necessary to establish that they are stable model properties rather than transient artifacts of API versioning or server state.

No temporal drift. Phase 4B (the targeted bridge topology experiment) collected data across multiple days. Cross-batch Jaccard — the similarity between runs collected on different days — fell within 0.05 of within-batch Jaccard for all tested (model, pair) combinations. The navigational behavior measured in the benchmark was stable over the tested collection window.

Scale consistency. Phase 1 collected paths at both 5 and 10 waypoints for all 21 reporting pairs and 4 models. The 5-waypoint paths were not independent inventions but genuine coarse-grainings of the 10-waypoint paths: on average, 70.5% of 5-waypoint concepts also appeared in the corresponding 10-waypoint paths, and 67.9% of the time, the 5-waypoint sequence appeared as a proper subsequence of the 10-waypoint sequence — same concepts in the same order, with additional waypoints interpolated between them. For example, GPT's 5-waypoint path from hot to cold (*warm, tepid, lukewarm, cool, chilly*) appeared intact within its 10-waypoint path (*scald, sear, warm, tepid, lukewarm, cool, chilly, brisk, frigid, icy*), with the longer path extending coverage at both endpoints and interpolating finer detail.

These two findings support the conclusion that waypoint elicitation measures a stable, resolution-consistent property of each model's navigational geometry, not a transient or scale-dependent artifact.

4.3 The Dual-Anchor Effect

Phase 2 established that conceptual navigation is fundamentally asymmetric: forward (A→B) and reverse (B→A) paths share less than 19% of their waypoints on average (full results in Section 5). This prompted the starting-point hypothesis — that models construct paths by greedy forward search from the starting concept, producing near-complete divergence between forward and reverse trajectories. Phase 3A tested this hypothesis by examining where in the waypoint sequence forward and reverse paths converge.

The U-shaped convergence profile. To measure positional convergence, forward position i was compared against reverse position $(N+1-i)$ — the mirror position. If forward waypoint 1 matched reverse waypoint 5, forward waypoint 2 matched reverse waypoint 4, and so on, this would indicate that both paths traversed the same intermediate concepts in opposite order. The per-position mirror-match rates across all 84 pair/model combinations were:

Position	Forward	Reverse (mirror)	Match Rate
1	wp1	wp5	0.102
2	wp2	wp4	0.057
3	wp3	wp3	0.065
4	wp4	wp2	0.085
5	wp5	wp1	0.129

The starting-point hypothesis predicted a monotonic increase from position 1 to position 5 — early waypoints should diverge most (each direction dominated by its starting concept's neighborhood), with convergence increasing as both paths approach the shared target. Instead, the data revealed a U-shape: positions 1 and 5 were both elevated relative to the middle. The overall convergence slope was weakly positive (0.0082, 95% CI [−0.0030, 0.0198]) and only 50% of pair/model combinations showed a positive slope. The starting-point hypothesis, in its simple form, was not supported.

The dual-anchor interpretation. Both endpoints exert "gravitational pull" on nearby waypoints. Position 1 (forward waypoint 1 vs. reverse waypoint 5) shows elevated overlap because both measurements sample from the neighborhood of concept A — the forward path's first step away from A and the reverse path's last step arriving at A draw from the same constrained vicinity. Position 5 shows even higher overlap for the same reason applied to concept B. The middle positions (2–4) form a valley because this is the region of maximum navigational freedom — unconstrained by either endpoint, dominated by model-specific and run-specific choices.

Category-dependent signatures. The U-shape was not uniform across relationship types. Each category produced a distinctive convergence profile:

- *Antonyms* (hot–cold) showed extreme late convergence: 0.054, 0.035, 0.215, 0.338, 0.344. The temperature axis acts as a funnel — once either direction has traversed 2–3 waypoints, both paths enter the same tightly constrained central region. This was the one category that supported the original starting-point hypothesis (slope 0.088, CI above zero).

- *Identity controls* (apple–apple) showed middle domination: 0.000, 0.044, 0.319, 0.048, 0.018. With no distance to traverse, both paths circulate through the same core associations (fruit, tree, harvest), and the middle waypoint is where they are most likely to align.

- *Random controls* showed a pure U-shape: 0.099, 0.025, 0.010, 0.031, 0.121. Both endpoints exert local pull, but no semantic bridge constrains the middle. The path between telescope and jealousy wandered freely through intermediate territory.

- *Nonsense controls* were flat near zero: 0.001, 0.003, 0.008, 0.001, 0.009. No anchoring effect, because nonsense strings have no semantic neighborhoods to constrain waypoint selection.

The category signatures in the convergence profile encode the type of semantic relationship between endpoints. Tightly constrained relationships (antonyms, hierarchies) show structured convergence; unconstrained relationships (random, nonsense) show the pure dual-anchor pattern or no pattern at all. This suggests that positional convergence profiles may help discriminate relationship types, though this observation has not been validated as a predictive tool in subsequent phases.

Model-level variation. Claude showed the only negative convergence slope (−0.005) among the four core models, meaning its U-shape tilted slightly toward the start-side anchor. This is consistent with Claude's rigid navigational style: its paths are so determined by the pair's global

topology that the target exerts less incremental pull than it does for more variable navigators. GPT (0.015) and Grok (0.016) showed positive slopes, consistent with their more exploratory styles producing stronger late-stage convergence toward the target.

4.4 Three Models Agree, Gemini Diverges

Across the 6 non-control triples analyzed in Phase 4A, Claude, GPT, and Grok aligned more closely with one another than with Gemini, which systematically diverged. The sample is small (6 triples, with two universal-zero triples that inflate pairwise correlations), but the pattern was consistent across multiple metrics.

Bridge frequency agreement. Phase 4A analyzed inter-model agreement on bridge concept usage across 6 non-control triples. Claude, GPT, and Grok showed 100% binary bridge agreement — for every triple, all three models either found the bridge or all three missed it. Gemini agreed with the other models on only 50% of triples.

Pairwise Pearson correlations on the 6-element bridge frequency vectors:

Model Pair	Pearson r	Binary Agreement
Claude–GPT	0.772	100%
Claude–Grok	0.768	100%
GPT–Grok	0.667	100%
GPT–Gemini	0.679	50%
Claude–Gemini	0.446	50%
Grok–Gemini	0.340	50%

To quantify Gemini's divergence, the mean absolute bridge frequency difference ($|\Delta\text{freq}|$) was computed for Gemini-paired model pairs versus non-Gemini pairs. Gemini-paired mean $|\Delta\text{freq}|$ was 0.336, compared to 0.200 for non-Gemini pairs — an isolation index of 0.136. The 95% CI $[-0.092, 0.378]$ includes zero, so this trend may not be reliable at the aggregate level. The qualitative pattern, however, was consistent: Gemini's bridge frequency disagreements were concentrated on specific triples where the other three models agreed.

Characteristic divergence patterns. The divergence was not random. Gemini consistently failed to route through bridge concepts that the other three models found reliably:

- *bank→ocean*: Claude, GPT, and Grok all routed through "river" (frequency 0.90–1.00); Gemini did not (0.00), following alternative non-river routes.
- *music→mathematics*: Claude (1.00), Grok (0.85), and GPT (0.15) routed through "harmony"; Gemini did not (0.00).
- *emotion→melancholy via nostalgia*: Grok found the bridge reliably (0.90); Claude and GPT found it at low frequency (0.10); Gemini did not (0.00).

The only case where all four models agreed on strong bridge presence was the hierarchical triple animal→dog→poodle (bridge frequency 0.95–1.00). (All four models also agreed on universal zero for the hot→energy→cold and beyoncé→justice→erosion triples, where no model found the designated bridge.) Gemini's topology preserved hierarchical/taxonomic structure but fragmented on cross-domain and abstract bridges.

Connection to later findings. The Gemini divergence persisted through the later core-cohort phases in which Gemini was tested (through Phase 9). Three mechanistic hypotheses for Gemini's

deficit — frame-crossing threshold (Phase 5), gradient blindness (Phase 8B), and transformation-chain blindness (Phase 9B) — were all falsified (graveyard entries G7, G21, G24). The deficit is real, stable within the tested phases, and mechanistically uncharacterized as of the end of data collection. It is discussed further in Section 7.5.

4.5 Polysemy Sense Differentiation

Polysemous concepts — words with multiple distinct meanings — provided a natural test of whether waypoint elicitation measures genuine conceptual structure or surface-level word statistics. If paths reflect word co-occurrence patterns, a polysemous source word should produce partially overlapping paths regardless of the target (since the same word token drives the associations). If paths reflect conceptual navigation, different targets should activate different senses and produce non-overlapping paths.

The result was unambiguous. Three polysemy groups were tested in Phase 1 (bank, bat, crane), each with two sense-steering target concepts. Phase 2 supplementary runs corrected an initial artifact (the holdout pairs had lacked pilot data, inflating the Phase 1 zero-overlap result). The corrected cross-pair Jaccard values — measuring the overlap between paths from the same source word to different sense-targets — were:

Group	Pair A	Pair B	Cross-Pair Jaccard
bank	bank→river	bank→mortgage	0.062
bat	bat→cave	bat→baseball	0.059
crane	crane→construction	crane→wetland	0.011

All values were near zero. The crane group (0.011) was particularly clean — essentially a single shared waypoint across all models and runs between the construction-crane and bird-crane paths. For comparison, within-pair Jaccard for these same models on meaningful pairs ranged from 0.258 (GPT) to 0.578 (Claude), and even the random control mean was 0.337.

Interpretation. Under the benchmark's lexical similarity measure, the target concept strongly determines which sense of the polysemous source is activated, and this sense activation redirects the navigational trajectory. The word "bank" paired with "river" activates a geographic/hydrological frame; the same word paired with "mortgage" activates a financial frame. The resulting paths share almost no intermediate territory. This is strong evidence that waypoint elicitation probes conceptual structure — the semantic content of the path — rather than surface statistics about the source word.

The polysemy result also connects to the dual-anchor finding (Section 4.3): even though the source concept ("bank") is identical in both conditions, the target concept exerts sufficient gravitational pull to redirect the path from waypoint 1 onward. This is incompatible with a pure forward-chaining model in which only the starting concept determines the trajectory. The target's influence is detectable from the very beginning of the path, at least when the source is polysemous and the target provides strong sense-disambiguating signal.

Extension to later phases. Phase 5B extended the polysemy finding to forced crossings — concept triples where a polysemous word serves as the only connection between two domains (e.g., loan→bank→shore). These triples produced the highest bridge frequencies in the benchmark (0.95–1.00 for three of four models), confirming that polysemous concepts can serve as obligatory navigational bottlenecks when they are the sole path between semantic domains. The full forced-crossing analysis appears in Section 5.4.

The five findings in this section — distinct gaits, self-consistent paths, dual-anchor convergence, cross-model agreement structure, and polysemy sense differentiation — collectively support the conclusion that waypoint elicitation measures real geometric structure in LLM conceptual navigation, as captured by lexical overlap under the current canonicalization pipeline (Section 2.3). The structure is stable over time, consistent across resolutions, driven by semantic content, and characteristic of each model. This foundation supports the more demanding geometric claims in the sections that follow.

5 Act II — Topology: Navigational Geometry Is Consistent with Quasimetric Structure

Section 4 established that waypoint elicitation measures real navigational structure. This section characterizes the *geometry* of that structure. The central finding is that LLM conceptual spaces exhibit properties consistent with a quasimetric space — satisfying non-negativity and triangle inequality while systematically violating symmetry. We present the evidence for each property, then show that the space exhibits compositional structure (hierarchical paths compose), selective bottleneck routing (bridge concepts are not mere associations), and non-uniform navigational traffic.

Methodological caveat. The asymmetry result (Section 5.1) and the triangle inequality result (Section 5.2) use different operationalizations of "distance." Asymmetry is measured via cross-direction Jaccard (comparing A→B paths against B→A paths), while the triangle inequality uses within-direction Jaccard (consistency of repeated runs on a single direction) to define navigational distance. The quasimetric characterization is therefore *consistent with* the data rather than formally established on a single unified distance function. We present this as convergent evidence from two complementary distance measures, not as a deductive proof.

5.1 Asymmetry Is Fundamental

Phase 2 reversed all 21 reporting pairs across 4 models (840 reverse-direction runs; ~960 total including polysemy supplementary) and computed the directional asymmetry index: 1 minus the mean Jaccard similarity between forward (A→B) and reverse (B→A) waypoint sets. A value of 0 indicates perfect symmetry (identical paths in both directions); a value of 1 indicates complete non-overlap.

The symmetry axiom fails comprehensively. The mean directional asymmetry across all 84 pair/model combinations was 0.811 (95% CI [0.772, 0.848]). Of 84 combinations, 73 (87%) showed statistically significant asymmetry by permutation test ($p < 0.05$, 1,000 resamples). The distribution was heavily right-skewed: 54 of 84 combinations (64%) exceeded 0.8, and only 5 (6%) fell below 0.5. Forward and reverse paths typically shared less than 19% of their waypoints.

Contextualizing the 0.811 mean. The Phase 2 asymmetry was computed on the original 4-model cohort at 5 waypoints, and the 0.811 mean includes control pairs — nonsense controls (0.986) and random controls (0.908) — which inflate the overall average. Experimental-category means were lower but still substantial (antonym 0.596, hierarchy 0.683, cross-domain 0.822, polysemy 0.824, anchor 0.911). The 0.811 figure should be understood as characterizing the full distribution rather than as a universal constant.

Phase 11A confirmed asymmetry across 4 additional models, all exceeding the 0.60 threshold: Mistral 0.729, DeepSeek 0.722, Cohere 0.718, Llama 4 Maverick 0.673. Phase 11C revealed a resolution-dependence qualification in the 3-model robustness subset (Claude, GPT, DeepSeek):

only the 9-waypoint conditions cleared the 0.60 threshold (0.669–0.684); the 5-waypoint conditions (0.593–0.594) and the 7-waypoint baseline (0.599) did not. The asymmetry property is real but requires sufficient path length to manifest reliably — a measurement sensitivity finding, not a protocol artifact.

Statistical caveat. The 87% significance rate was computed at $\alpha = 0.05$ across 84 independent tests with no explicit multiple-comparison correction. With Bonferroni correction (threshold $\approx$ 0.0006), the percentage would be lower, and the permutation test's resolution (1,000 resamples, minimum achievable p = 0.001) limits precision at corrected thresholds. The qualitative conclusion — that the vast majority of pair/model combinations show genuine asymmetry — is robust, but the specific 87% figure should be interpreted conservatively.

Theoretical connection. Tversky (1977) established that human similarity judgments are asymmetric: North Korea is judged more similar to China than China is to North Korea. Berglund et al. (2024) showed that autoregressive models exhibit a reversal curse — training on "A is B" does not teach "B is A." Our finding connects these: the reversal curse is not merely a failure mode but produces structured asymmetry consistent with quasimetric topology. The asymmetry patterns track semantic properties of the concept pairs (Section 4.3) and model-specific navigational styles (Section 4.1), indicating systematic structure rather than noise.

5.2 Triangle Inequality Shows Stable Replication

The triangle inequality — d(A,C) $\leq$ d(A,B) + d(B,C) — was tested across three independent samples using navigational distance defined as 1 minus mean within-direction Jaccard similarity.

Phase	N Triangles	Models	% Holding	Violations
3B	8	4	90.6% (29/32)	Gemini ×2 (music-harmony-math, bank-river-ocean), Claude ×1 marginal (hot-energy-cold)
4B	8	4	93.8% (30/32)	Claude ×1 (tree-forest-ecosystem), Grok ×1 (light-chandelier-color, a random-control triple)
7B	8	4	90.6% (29/32)	Claude ×1 (deposit-bank-shore), Gemini ×2 (seed-germination-garden, music-harmony-math)

The satisfaction rate shows a stable replication pattern around 91% across three independent triple sets. This is remarkably high given the magnitude of the asymmetry violations: a space where forward and reverse distances differ by 0.811 on average might be expected to show widespread triangle inequality failures, yet the directional distances compose properly in over 90% of cases.

The violations in Phase 3B were concentrated in Gemini: the music→harmony→mathematics triple showed a slack of −0.319 (the direct path was much longer than the sum of the two legs), and

bank→river→ocean showed −0.061. Claude's hot→energy→cold violation was marginal (−0.023). GPT and Grok each satisfied the triangle inequality in all 8 of their respective Phase 3B cases.

Quasimetric characterization. Combined with the asymmetry data, the metric-axiom profile of LLM conceptual space is: - *Identity:* not directly testable with Jaccard-based distance (identity controls show low directional divergence for some models — Claude 0.017 — but mean cross-direction asymmetry was 0.456, and within-direction distance was not measured for identity pairs) - *Non-negativity:* holds trivially - *Symmetry:* **fails comprehensively** (mean violation 0.811) - *Triangle inequality:* holds in ~91% of cases

This profile is consistent with a quasimetric space — a space satisfying all metric axioms except symmetry. Quasimetric spaces are standard mathematical objects arising in directed graphs, asymmetric norms, and computational complexity theory. The characterization provides a useful framework for interpreting the navigational geometry observed throughout the benchmark, though it rests on convergent evidence from two complementary distance operationalizations rather than a single unified metric (see methodological caveat above).

5.3 Hierarchical Paths Are Compositional

If conceptual navigation merely produces plausible-sounding word chains, there is no reason to expect that the path from A to C should systematically include the concept B that lies "between" them in a taxonomic hierarchy. Phase 3B tested this directly with 8 concept triples across 4 models.

Hierarchical triples showed 4.9× higher waypoint transitivity than random controls. Mean transitivity for hierarchical triples was 0.175 (95% CI [0.112, 0.238]); for random controls, 0.036 (95% CI [0.011, 0.070]). The confidence intervals do not overlap. Bridge concepts appeared systematically for taxonomic triples — "dog" at 15–100% frequency on the animal→poodle path — and never for random controls ("stapler" at 0% on music→mathematics, "flamingo" at 0% on hot→cold).

Polysemy-extend triples (bank→river→ocean) showed even higher bridge frequency (0.725) than hierarchical triples (0.456), reflecting the bottleneck effect of sense-mediated routing: once "bank" resolves to its geographic sense, "river" is the only natural waypoint en route to "ocean." Semantic-chain triples (music→harmony→mathematics, hot→energy→cold) showed intermediate transitivity (0.073) with strong model dependence — "harmony" appeared on 100% of Claude's music→mathematics paths but 0% of Gemini's.

The key negative result was hot→energy→cold, where "energy" never appeared as a bridge despite being associated with both endpoints. The hot→cold paths traversed the experiential temperature gradient (warm, tepid, cool, chilly); the hot→energy and energy→cold paths traversed physics terminology (thermodynamics, work, heat). Association with both endpoints is necessary but not sufficient for bridge function — the concept must lie *on the navigational path* between the endpoints, not merely in their shared associative neighborhood.

5.4 Bridge Concepts Are Bottlenecks, Not Associations

Phases 4 and 5 converged on a characterization of what makes a concept function as a navigational bridge. Associative strength alone is insufficient; structural role — whether the concept names the primary axis of connection between the endpoints — can override it (though cue strength remains a real contributing factor; see Section 5.5).

Bottleneck bridges succeed universally. "Spectrum" (the mechanism by which light produces color) appeared at 1.00 frequency across all four models on the light→color path. "Deposit" (the action connecting bank to savings) appeared at 0.95–1.00. "Sentence" (the compositional unit

between word and paragraph) appeared at 1.00. These bridges name the *mechanism* or *transformation* that connects the endpoints.

Off-axis associations fail universally. "Metaphor" (associated with both language and thought) appeared at 0.00 on the language→thought path. "Energy" appeared at 0.00 on hot→cold. These concepts are associated with both endpoints but do not name the primary axis of connection.

Process-naming outperforms object-naming. "Germination" (the process connecting seed to garden) outperformed "plant" (the object that inhabits both) in all four models: Claude 1.00 vs 0.00, GPT 0.95 vs 0.65, Grok 0.15 vs 0.00, Gemini 1.00 vs 0.00 (O4). The explanation parallels the spectrum/deposit pattern: "germination" names the *transformation* from seed to garden, adding directional information; "plant" names something already implied by both endpoints, adding no navigational value. Navigational salience depends on directional information content, not solely on generic associative strength.

Too-central concepts are skipped. "Fire" appeared at near-zero frequency across all configurations in Phase 5B — 0.00 on spark→ash (except GPT 0.15), 0.00 on ambition→motivation, 0.00 on spark→motivation (O6). Both "spark" and "ash" already imply fire; naming it explicitly adds no information. Phase 7C extended this finding: "water" for rain→ocean was 0.00 across all four models (O16). Too-central concepts are navigable *to* but not *through* — a distinct failure mode from off-axis associations.

Forced crossings create near-obligatory bottlenecks. When a polysemous concept is the sole connection between two domains, it becomes a near-obligatory waypoint under unconstrained elicitation. Loan→bank→shore produced bridge frequency of 0.95–1.00 for Claude, GPT, and Grok (O5); Gemini showed 0.05, consistent with its apparent topological disconnection between senses. This is a bank-specific result — the benchmark's strongest forced-crossing case — and generality beyond "bank" remains untested. The forced-crossing mechanism is the inverse of the too-central phenomenon: instead of being redundantly implied, the bridge is the *only* route between otherwise disconnected domains.

A major failed prediction was "metaphor" as the language→thought bridge (graveyard entry G5). Intuitive semantic importance had zero predictive power for navigational bridge behavior. This failure sharpened the operational definition: bridges must name the primary axis of connection between endpoints, not merely be associated with both.

5.5 Cue-Strength Gradient

Phase 5A tested whether bridge frequency decreases monotonically with the bridge concept's cue strength — its associative proximity to both endpoints. Four concept families were constructed, each with three bridge candidates at varying cue levels (very-high, medium, low), and bridge frequency was measured across 4 models (2,040 new runs).

The gradient is real. Twelve of 16 family/model combinations showed monotonic decrease in bridge frequency from highest to lowest cue level. All four failures occurred in a single anomalous family (biological-growth: seed→B→garden), where the "medium" cue bridge ("germination") outperformed the "very-high" bridge ("plant") — the process-naming effect described in Section 5.4. The three well-behaved families — physical-causation (sun→B→desert), compositional-hierarchy (word→B→paragraph), and abstract-affect (emotion→B→melancholy) — showed perfect monotonic gradients across all four models.

Logistic regression revealed similar cue-strength thresholds across models (Claude 2.02, GPT 1.96, Grok 2.16, Gemini 1.79), with the difference CI $[-1.00, 1.07]$ including zero. This falsified the Phase 4 hypothesis that Gemini's bridge failures stemmed from a higher cue-strength threshold. Gemini responds to cue strength at approximately the same threshold as other models; its failures

must be driven by something other than associative sensitivity. R^2 values ranged from 0.311 to 0.640, indicating that cue strength is a real but incomplete predictor — bridge selection is closer to a threshold phenomenon than a smooth gradient.

5.6 Waypoint Distributions Are Non-Uniform

Phase 6A collected 1,200 runs across 8 concept pairs and 4 models to characterize the full waypoint frequency distribution for each (pair, model) condition.

Navigational traffic concentrates in a small number of high-frequency waypoints. Of 32 pair-model conditions (8 pairs $\times$ 4 models), 31 rejected the null hypothesis of uniform waypoint frequency at Bonferroni-corrected significance (chi-squared goodness-of-fit test, threshold $p < 0.0063$). The sole non-rejection was Claude on seed→garden, where only 5 unique waypoints were observed — near-deterministic navigation leaving insufficient diversity to distinguish from uniform.

Entropy varied systematically with gait: Claude showed the lowest mean entropy (2.59), reflecting near-deterministic waypoint selection from a small vocabulary; GPT showed the highest (3.44), reflecting broad exploration across a larger candidate set. This mirrors the gait spectrum from Section 4.1 — high-gait models concentrate traffic on a few landmarks, while low-gait models distribute it across many.

The non-uniformity finding is important because it rules out a class of null models. If waypoints were drawn approximately uniformly from a large vocabulary, the consistency metrics throughout the benchmark could reflect sampling noise rather than navigational structure. The non-uniform, concentrated distributions confirm that models navigate through preferred waypoints, not through arbitrary word generation.

5.7 Bridge Positioning: Early Anchoring, Not Midpoint

Phase 5C tested whether bridge concepts create a convergence peak at the midpoint of waypoint paths (the W-shape hypothesis). The aggregate result was null — bridge-present pairs showed no significant midpoint convergence signal when averaged across models and pairs. Phase 6C revised the approach: instead of assuming the bridge occupies the midpoint, it scanned all positions for bridge frequency peaks.

Bridges anchor early, not at the midpoint. Across 10 pairs and 4 models (40 positional profiles), the cross-model modal bridge position was 1–2 (0-indexed; the 2nd–3rd waypoint out of 7) for 8 of 10 pairs. The peak-detection contrast — bridge frequency at the modal position minus the mean of non-modal positions — was 0.345 (95% CI [0.224, 0.459]), significantly positive. The fixed-midpoint contrast was -0.080 (95% CI [-0.141, -0.024]), confirming that the Phase 5C W-shape null was a methodological artifact of the rigid midpoint assumption, not evidence against bridge positional structure.

The taxonomic exception. The sole departure from early anchoring was the hierarchical pair animal→poodle, where "dog" anchored at positions 4–5 across models. This late positioning reflects the structure of the taxonomic chain: the path from animal to poodle narrows progressively (mammal $\rightarrow$ carnivore $\rightarrow$ canine $\rightarrow$ dog $\rightarrow$ breed), and "dog" is the penultimate step, not an early waypoint. Taxonomic bridges reflect their position in the hierarchy rather than a universal early-anchoring tendency.

Forced-crossing bridges are positionally unstable. The two forced-crossing pairs (loan→shore, deposit→river) showed positional standard deviation of 1.71, compared to 0.52 for the 8 non-forced pairs (O13). The polysemous pivot sits at a frame junction where the timing of sense-crossing is model-dependent: Claude places "bank" at position 1 on loan→shore; Gemini places it at position

4. The bridge is obligatory (it must appear somewhere) but its position is not fixed — a contrast with non-forced bridges like "spectrum" or "sadness," which reliably anchor at positions 1–2.

The seven findings in this section collectively support a characterization of LLM conceptual space as consistent with quasimetric geometry, exhibiting compositional structure and non-uniform navigational traffic. Distances are asymmetric but compose properly (triangle inequality ~91%); hierarchical relationships preserve transitivity; bridge concepts function as structural bottlenecks, not associative co-occurrences; and navigational traffic concentrates at preferred waypoints that anchor early in the path sequence. This geometric characterization provides the foundation for the causal intervention experiments in the next section, which probe whether the observed structure is merely statistical regularity or reflects genuine causal dependencies in the navigation process.

6 Act III — Mechanism: Pre-Filling Reveals Causal Structure

Sections 4 and 5 established that waypoint elicitation measures real navigational structure with consistent geometric properties. But those findings are correlational — they characterize the structure without establishing how it arises. This section presents the benchmark's causal intervention results. By pre-filling the first waypoint position with a specified concept and measuring the downstream effect on bridge concept appearance, we test whether early anchoring causally determines path structure (Phase 7A), whether the semantic relationship between the pre-fill and the bridge modulates the effect (Phase 10B), and whether pre-filling can facilitate rather than displace marginal bridges (Phase 9C).

6.1 Pre-Filling Causally Displaces Bridges

The early-anchoring pattern from Section 5.7 — bridge concepts cluster at positions 1–2 — could reflect either a causal mechanism (the first-generated concept steers the trajectory) or an epiphenomenon (the bridge concept is inherently early in the conceptual path regardless of generation order). Phase 7A distinguished these by pre-filling position 1 with a specified concept, forcing the model to generate waypoints 2–7 with the pre-fill already occupying the first slot.

Pre-filling causally displaces bridge concepts. In the primary incongruent condition, mean displacement across 8 pairs and 4 models was 0.515 (95% CI [0.357, 0.664], excluding zero); congruent and neutral conditions showed comparable magnitudes (0.436 and 0.536 respectively; see Section 6.2 for condition-level analysis). Bridge frequency dropped from 0.807 under unconstrained elicitation to 0.460 under pre-fill conditions — a 43% relative decline. The positional shift exceeded the mechanical +1 that would result from merely inserting a waypoint before the bridge, confirming genuine topological displacement rather than arithmetic position-bumping.

Taxonomic bridges resist displacement. The animal→poodle pair, where "dog" occupies a hierarchically determined mid-path position (Section 5.7), showed displacement of only 0.140 — less than a third of the 0.515 mean for heading-bridge pairs. The taxonomic bridge resists pre-fill perturbation because its position is structurally determined by the hierarchy (mammal → carnivore → canine → dog → breed), not by generation-order primacy. This confirms the distinction from Section 5.7: taxonomic bridges and heading bridges respond to different causal mechanisms.

Forced-crossing bridges do not resist displacement. The forced-crossing bridge "bank" on loan→shore, which appeared at 0.95–1.00 for Claude, GPT, and Grok under unconstrained elicitation in earlier phases (Section 5.4), showed pre-fill survival of only 0.267 in Phase 7A (aggregate

unconstrained baseline 0.717), with a positional shift of 0.791. This falsified the pre-registered prediction that the forced-crossing bottleneck would survive at > 0.90 because it is the only semantic connection between the financial and geographic domains (graveyard entry G19). The forced-crossing bottleneck is maintained by associative primacy — "bank" normally wins position 1 because it is the most strongly cued concept — not by topological necessity. When position 1 is occupied, a bridge that behaved as a near-obligatory bottleneck under unconstrained elicitation can still be displaced.

Bridge survival is bimodal. The pair-level survival rates under pre-fill reveal two distinct modes rather than a continuous distribution:

Pair	Bridge	Unconstrained	Pre-fill Survival	Pattern
emotion→melancholy	sadness	0.950	0.750	Robust
animal→poodle	dog	0.970	0.900	Robust
light→color	spectrum	0.917	0.575	Partially robust
sun→desert	heat	0.917	0.567	Partially robust
bank→ocean	river	0.600	0.300	Fragile
loan→shore	bank	0.717	0.267	Fragile
music→math	harmony	0.550	0.025	Collapsed
seed→garden	germination	0.833	0.000	Collapsed

"Harmony" and "germination" collapse to near-zero under any pre-fill condition; "sadness" and "dog" survive robustly. The pattern is consistent with route exclusivity: "sadness" has no navigational competitors on the emotion→melancholy path, while "harmony" competes with rhythm, pattern, ratio, and frequency on music→mathematics. When the bridge has no competitors, pre-filling merely delays its appearance; when it has competitors, pre-filling creates an opening for alternative routes (H8). This characterization remained descriptive throughout the benchmark — the mechanistic follow-up (route exclusivity as competitor count, G20; dominance ratio, G23) failed in both attempts (Section 7).

6.2 The Mechanism Is Primarily Associative Primacy

Phase 7A tested three pre-fill conditions — incongruent (pointing away from the target), congruent (pointing toward the target), and neutral (semantically unrelated) — to distinguish between two mechanistic theories. The directional-heading theory predicted that incongruent pre-fills would produce the most displacement (they steer away from the bridge), congruent the least (they align with the bridge's heading), and neutral in between. The associative-primacy theory predicted that all pre-fills would produce similar displacement (whatever occupies position 1 anchors the trajectory regardless of its semantic content).

The directional-heading theory fails in its strong form. Displacement magnitudes were statistically indistinguishable across conditions:

Condition	Mean Displacement	95% CI
Incongruent	0.515	[0.357, 0.664]
Congruent	0.436	[0.278, 0.599]
Neutral	0.536	[0.378, 0.692]

The critical comparison — incongruent vs congruent — showed a difference of only 0.079, with overlapping confidence intervals. Neutral pre-fills produced displacement as large as incongruent

ones (0.536 vs 0.515), contradicting the prediction that content-free pre-fills should displace less than actively misdirecting ones.

Bridge survival shows a secondary content effect. While displacement magnitudes did not differentiate conditions, survival rates showed a suggestive pattern: incongruent 0.347 vs congruent 0.631 vs neutral 0.403. The congruent survival advantage (0.631 vs 0.347) suggests that pre-fill content modulates whether the bridge eventually recovers, even though displacement magnitude is similar. The mechanism appears to be primarily associative primacy (occupying position 1 has outsized causal influence on the trajectory) with a possible secondary modulation by content alignment — a weaker claim than the directional-heading theory, but not pure position-driven displacement either.

Per-model displacement. Claude showed the highest displacement (0.567), consistent with the rigid-gait interpretation from Section 4.1: high consistency means the first-generated concept exerts stronger anchoring because subsequent generation follows a more constrained path. GPT showed lower displacement (0.483), consistent with its exploratory gait allowing more recovery after perturbation.

6.3 Relation Class Affects Bridge Survival

Phase 10B refined the content-modulation finding from Section 6.2 by systematically varying the semantic relationship between the pre-fill and the target bridge. Eight pairs were tested under three pre-fill conditions: on-axis substitutes (concepts competing on the same semantic dimension as the bridge), same-domain off-axis (concepts in the same broad domain but not competing on the bridge's dimension), and unrelated (semantically unconnected to both endpoints).

The relation class taxonomy captures real variance. A Friedman test across 8 pairs confirmed that relation class significantly affects bridge survival (chi-squared = 6.750, df = 2, p = 0.034).

Relation Class	Mean Survival	95% CI
On-axis	0.643	[0.330, 0.903]
Same-domain	0.708	[0.482, 0.922]
Unrelated	0.388	[0.191, 0.606]

The predicted ordering was wrong. We expected on-axis substitutes to be most disruptive (competing directly for the bridge's navigational niche), with ordering on-axis $<$ unrelated $<$ same-domain. The observed ordering was unrelated (0.388) $<$ on-axis (0.643) $\approx$ same-domain (0.708). Only 1 of 8 pairs showed the predicted ordering (graveyard entry G27). The data suggest a coarse related-vs-unrelated split as the operationally meaningful distinction (though the pairwise comparisons do not survive multiple-comparison correction; see caveat below): related pre-fills (whether on-axis or same-domain) appear to maintain the navigational context and preserve the bridge, while unrelated pre-fills destroy the context. On-axis and same-domain pre-fills keep the model in the right conceptual neighborhood, making bridge recovery possible; unrelated pre-fills derail navigation to an entirely different region of conceptual space.

Statistical caveat. The Friedman test is significant at $p = 0.034$ with $N = 8$ pairs as blocks. Post-hoc Wilcoxon signed-rank tests show on-axis vs unrelated at $p = 0.025$ and same-domain vs unrelated at $p = 0.036$, but neither survives Bonferroni correction for 3 pairwise comparisons (threshold $p < 0.017$). The on-axis vs same-domain comparison was not significant ($p = 0.889$, Cohen's $d = 0.170$). The qualitative conclusion — unrelated pre-fills are more disruptive than

related ones — is sound; the specific pairwise p-values should be interpreted as suggestive rather than confirmatory.

Replication of Phase 9C results. Two pairs with direct Phase 9C comparisons showed near-perfect replication: warm (hot→cold on-axis) at 0.000 survival in both phases (diff = 0.000), and fermentation (grape→wine same-domain) at 0.991 vs 1.017 (diff = 0.026). This provides cross-phase validation of the pre-fill paradigm's measurement reliability.

6.4 Marginal Bridge Facilitation

The causal intervention paradigm revealed an unexpected bidirectional effect. While pre-filling displaces dominant bridges (Sections 6.1–6.3), it can massively facilitate marginal ones — bridges that appear at low frequency under unconstrained elicitation.

Marginal bridges show large facilitation under aligned pre-fill. Across four marginal pairs (unconstrained frequency 0.125–0.267), mean congruent survival was 3.761×, driven by three strong cases:

Pair	Bridge	Unconstrained Freq	Congruent Survival
science→art	creativity	0.125	8.000×
student→professor	research	0.175	4.000×
problem→solution	analysis (pilot pair)	0.267	2.906×
recipe→meal	cooking	0.183	0.136× (displaced)

The facilitation effect is not universal: recipe→meal "cooking" showed displacement (0.136×) rather than facilitation, indicating that bridge marginality is necessary but not sufficient. The effect is large and model-general for the facilitating cases: 3 of 4 models showed facilitation for the two core marginal pairs (science→art, student→professor). The exception was Gemini, which showed no facilitation — consistent with its overall deficit in bridge generation (Section 4.4) but not with any specific mechanistic characterization of that deficit.

The crossover is qualitative, not a clean regression. Phase 9C tested whether the relationship between unconstrained bridge frequency and pre-fill survival follows a smooth crossover function (H11). The regression slope was in the expected direction (−3.355) but the CI [−6.748, 0.723] included zero, and the crossover point (0.790) was far above the predicted 0.40–0.50 (graveyard entry G25, partial). The data are better described as a qualitative threshold: marginal bridges can be facilitated, dominant bridges are generally displaced, but the transition between regimes is too heterogeneous for a single regression to capture. Beyond the cooking counterexample above, the near-threshold pilot pair sketch→painting ("outline," unconstrained frequency 0.300) also showed displacement rather than facilitation (0.500×), further demonstrating that low unconstrained frequency alone does not guarantee facilitation — the pre-fill must be semantically aligned with the bridge's navigational role.

Theoretical significance. The facilitation finding has implications for the associative-primacy mechanism described in Section 6.2. If position 1 merely anchored the trajectory through recency-weighted attention, all pre-fills should displace all bridges — the mechanism would be purely positional. The fact that congruent pre-fills can *increase* bridge frequency for marginal bridges suggests that position 1 does not just anchor; it primes. A congruent pre-fill activates the same conceptual frame that the marginal bridge occupies, making the bridge more accessible in subsequent generation. This is a content-mediated effect layered on top of the position-driven mechanism — consistent with the secondary content signal observed in Section 6.2 and the related/unrelated distinction in Section 6.3.

The four findings in this section establish that navigational structure is not merely a statistical regularity but reflects genuine causal dependencies in the generation process. Pre-filling position 1 causally determines which bridge concepts appear (Section 6.1), the mechanism is primarily associative primacy with secondary content modulation (Section 6.2), the semantic relationship between pre-fill and bridge systematically affects survival (Section 6.3), and the effect is bidirectional — displacement for dominant bridges, facilitation for marginal ones (Section 6.4). These results also reveal the limits of the causal approach: the bimodal survival pattern (Section 6.1) and the failed crossover regression (Section 6.4) indicate that bridge behavior under perturbation is too heterogeneous for single-variable mechanistic models. The next section pursues this observation to its conclusion.

7 Act IV — Limits: The Mechanism Ceiling

Section 6 established that pre-filling causally displaces bridges and that the mechanism involves associative primacy with secondary content modulation. But Section 6 also exposed limits: bridge survival under perturbation is bimodal and heterogeneous (Section 6.1), the directional-heading theory failed (Section 6.2), and the facilitation crossover regression did not reach significance (Section 6.4). This section follows those limits to their conclusion. Phases 8 and 9 tested six single-variable mechanistic hypotheses, and all six failed at the primary-test level. Phase 10 added one additional falsification (G27, effect direction) and one initially falsified result later resurrected with more data (G26). The mechanism ceiling — the finding that conceptual navigation has qualitative geometric structure that current tools can characterize but cannot yet explain through simple mechanistic models — is itself the central result of these two phases.

7.1 The Mechanistic Prediction Collapse

Prediction accuracy across the benchmark follows a distinctive trajectory. The early characterization phases achieved high accuracy: Phase 4 at 81.3% (26/32), Phase 5A at 64.6% (31/48). These predictions asked whether known structural patterns would replicate in new configurations — "will spectrum appear as a bridge for light→color?" — and the answer was usually yes. As the benchmark shifted from characterization to mechanism, accuracy collapsed.

Phase 8 achieved 24% (6/25 evaluable predictions), the lowest in the benchmark's history. Phase 9 achieved 20% (5/25). Combined across 50 evaluable predictions in the two most mechanism-focused phases, accuracy was 22%, with zero novel mechanistic predictions confirmed. The confirmed predictions clustered as replications of established structural facts (known bridges appeared at known frequencies, cross-model distance correlation remained low) and qualitative directional calls (Claude retained the most compact navigational profile, marginal bridges showed facilitation under congruent pre-fill). Every prediction that required a specific variable to explain a specific pattern — competitor count predicts fragility, gradient type predicts Gemini's deficit, gait normalization rescues cross-model distance — failed.

The decline is not random degradation. It tracks a systematic shift in prediction type. Replication predictions — "will X hold in new data?" — succeed at approximately 80% across all phases. Structural predictions — "will X pattern generalize to new pairs/models?" — succeed at approximately 50%. Mechanistic predictions — "will variable Y explain pattern X?" — succeed at approximately 0–15%. The benchmark maps coarse geometry reliably but cannot predict fine-grained mechanism with single-variable models.

7.2 The Falsified Hypotheses

Across Phases 8–10, seven major follow-up predictions were falsified and one early false negative was later resurrected with additional data. These span different types — genuinely mechanistic (predicting which variable explains a pattern), model-deficit characterizations (predicting the mechanism behind Gemini's divergence), methodological (attempting to rescue a measurement approach), structural (testing whether a finding generalizes), and effect-direction (predicting the ordering of a known effect). Presenting them as a uniform class of "failed mechanism predictions" would be misleading; they failed for different reasons and teach different lessons.

#	Hypothesis	Phase	Predicted	Observed	Type
1	Route exclusivity predicts fragility (G20)	8A	rho < -0.70	rho $= 0.116$	Mechanistic
2	Gemini gradient blindness (G21)	8B	interaction ≥ 0.20	0.046	Model deficit
3	Gait normalization rescues distance (G22)	8C	normalized r > 0.50	r $= 0.212$ (0.000 improvement)	Methodological
4	Dominance ratio predicts fragility (G23)	9A	rho > 0.50	rho $= 0.157$	Mechanistic
5	Gemini transformation blindness (G24)	9B	transformation deficit	transformation *advantage*	Model deficit
6	Pre-fill crossover regression (G25)	9C	significant slope	CI includes zero	Mechanistic (partial)
7	Bridge generalization fails (G26)	10A	CI excludes zero	**Resurrected** — CI included zero with more data	Structural
8	Predicted relation class ordering (G27)	10B	on-axis $<$ unrelated $<$ same-domain	unrelated $<$ on-axis $\approx$ same-domain	Effect direction

Honest accounting. G26 (bridge bottleneck generalization) was initially falsified based on data from a single model (Llama 3.1 8B) under aggressive 60-second timeouts that excluded 4 of 5 new models on infrastructure latency, not capability. When timeouts were relaxed to 300 seconds and 3 additional models were tested, the bridge frequency CI shifted from excluding zero to including it — the structural finding generalized. This is the benchmark's strongest example of infrastructure limitations masquerading as substantive findings. G26 is retained in the graveyard as a historical record but is functionally resurrected, reducing the clean falsification count to 7.

The two mechanistic predictors. G20 (competitor count) and G23 (dominance ratio) represent two successive attempts to explain the bimodal bridge survival pattern from Section 6.1. Both failed: competitor count showed rho $= 0.116$ (the bridge with the most competitors — "sadness" at 8 — showed the highest survival), and dominance ratio showed rho $= 0.157$ (warm at ratio 1.00 was destroyed while fermentation at ratio 1.07 was bulletproof). After two attempts, both single-variable structural predictors of bridge fragility are falsified. The strongest surviving lesson is that survival depends on an interaction between pre-fill content and bridge structural role; content appears more

predictive than raw competitor count or dominance ratio, but no single-variable explanation has succeeded (Sections 6.3, 6.4).

The two Gemini characterizations. G21 (gradient blindness) and G24 (transformation-chain blindness) represent two successive attempts to characterize Gemini's overall deficit (mean bridge frequency 0.480 vs ~0.67 for non-Gemini models). Both produced results opposite to the predicted direction: G21 found Gemini's zeros concentrated on causal-chain pairs (6/10) rather than gradient pairs (1/10); G24 found Gemini's transformation mean (0.667) exceeded its gradient mean (0.293). The pooled meta-analytic interaction combining Phases 8B and 9B was −0.113 (CI includes zero). See Section 7.5 for the broader implications.

7.3 The Pattern: Qualitative Structure, Quantitative Opacity

The prediction-type analysis reveals a characteristic pattern in the data. Conceptual navigation has robust qualitative structure that replicates across models, phases, and experimental designs. Bridges exist, asymmetry is high, hierarchical triples compose, triangle inequality holds at ~91%. These properties are discoverable, confirmable, and consistent with a quasimetric interpretation (though based on two complementary distance operationalizations rather than a single unified metric; see Section 5). But the quantitative mechanisms behind that structure — which specific variable explains bridge fragility, what interaction accounts for Gemini's deficit, how gait differences relate to distance metrics — resist single-variable models with mathematical consistency.

This is not a failure of the benchmark. It is a finding about the phenomenon. Conceptual navigation is governed by interaction effects between bridge type, pair semantics, model architecture, and pre-fill content that no single variable can capture. The mechanism ceiling does not mean the phenomenon is random or unexplainable; it means the explanation is multi-variable, and the benchmark's experimental framework (testing one variable at a time) has reached its resolving power. Future work would need to test joint models — bridge role × pre-fill content × dominance, for instance — but the combinatorial explosion of such designs exceeds the benchmark's current scope.

The mechanism ceiling also provides a calibration for the benchmark's claims. The repeated gap between strong replication performance (~80%) and weak mechanistic prediction performance (~15%) suggests that the benchmark captures real coarse structure while leaving the underlying mechanism unresolved. The core structural findings (R1–R4, R6, R7, with R5 explicitly qualified) are genuine properties of navigational geometry; the mechanism that produces them is multi-variable and beyond the resolving power of single-factor designs.

7.4 Cross-Model Distance Is Fundamentally Model-Dependent

A separate line of failure concerns the navigational distance metric itself. Phase 7B computed pairwise distances (1 minus mean within-direction Jaccard similarity) across 8 concept triangles and 4 models, then tested whether models agree on which concepts are "near" and which are "far."

Models do not agree. The cross-model distance correlation was r = 0.170, far below the 0.50 validity threshold. Claude and GPT/Grok occupy very different quantitative regimes — Phase 7B showed Claude's overall mean triangle excess at 0.225 vs GPT's at 0.680, reflecting the gait difference from Section 4.1. The gait difference is not merely a consistency metric — it contaminates all distance-based measurements, placing models on incomparable scales.

Gait normalization produces zero improvement. Phase 8C attempted to rescue cross-model geometry by dividing raw Jaccard distances by model-specific baselines. The result was mathematically definitive: normalized r = 0.212, identical to the raw r = 0.212, with 0.000 im-

provement on every model pair. The gait difference is not a scalar factor that can be divided out; it is a structural reorganization of conceptual space. Some model pairs actively anti-correlate: Grok-Gemini at r = −0.580, GPT-Gemini at r = −0.210. When Grok says winter and summer are far apart (d = 0.925), Gemini says they are close (d = 0.273).

Implication. Model-independent navigational geometry cannot be recovered from path-based measurements. The distance metric is model-specific, not a property of the concept pair. This finding blocks the cross-model curvature program proposed in Phase 5 and first attempted in Phase 7B. Within-model distance analysis remains viable in principle — across the three tested samples, the triangle inequality (under within-direction Jaccard distance) holds in about 91% of triangle-model combinations, though compliance is heterogeneous across models (Gemini accounts for most violations; see Section 5.2). Individual gait profiles are interpretable. But cross-model distance comparisons are fundamentally contaminated by gait.

7.5 The Gemini Mystery

Gemini's overall deficit in bridge generation is one of the benchmark's most consistent observations. Across the Phases 8–9 pair sets, Gemini's mean bridge frequency was 0.480 compared to approximately 0.67 for non-Gemini models — a roughly 30% reduction. The deficit is real and stable across phases. In Phase 8B specifically, Gemini also produced more zero-frequency pairs (7/20 vs 6/60 for non-Gemini models combined), with zeros concentrated on causal-chain pairs rather than distributed across types. But despite three successive mechanistic characterizations, the deficit remains unexplained.

Three failed characterizations:

1. **Frame-crossing threshold (Phase 5A):** Hypothesized that Gemini has a higher cue-strength threshold for activating bridges that cross between conceptual frames. Falsified: Gemini's logistic threshold (1.79) was indistinguishable from other models (CI includes zero).

2. **Gradient blindness (Phase 8B, G21):** Hypothesized that Gemini selectively fails on gradient-midpoint bridges. Falsified with a reversal: Gemini's zeros concentrated on causal-chain pairs (6/10 causal zeros vs 1/10 gradient zeros), the opposite of the prediction.

3. **Transformation-chain blindness (Phase 9B, G24):** Hypothesized that Gemini selectively fails on material/biological transformation intermediaries. Falsified with a reversal: Gemini's transformation mean (0.667) exceeded its gradient mean (0.293), the opposite of the prediction. The pooled interaction across Phases 8B and 9B was −0.113, null.

The pattern across these three attempts is telling. Each hypothesized a type-based deficit: Gemini fails specifically on X-type bridges. Each found that Gemini's performance on the targeted type was either comparable to or better than its performance on the control type. Whatever pair type is designated as "hard" for Gemini, Gemini does relatively better on it. The deficit is model-level — Gemini shows lower bridge frequency across the board — not type-level. A type-based approach to characterizing the deficit may be fundamentally misguided.

The distance-metric anti-correlations from Section 7.4 provide additional context. Gemini's distance structure is not merely different from the other three models; it is inversely related to Grok's (r = −0.580) and weakly inverse to GPT's (r = −0.210). This suggests that Gemini may organize conceptual space with different salience weightings, producing systematically different navigational priorities rather than a deficit on any particular bridge type.

The Gemini mystery remains open. The divergence is documented empirically in Section 4.4 and confirmed across Phases 8–9, but remains mechanistically unexplained. Future work might approach it through representational analysis (comparing Gemini's internal embeddings to other models) rather than the behavioral type-classification approach that has failed three times.

The five findings in this section collectively establish a mechanism ceiling for the waypoint elicitation paradigm. Single-variable mechanistic models fail with mathematical consistency (6 of 6 Phases 8–9 hypotheses falsified, plus 1 effect-direction failure in Phase 10; Section 7.2); the failure follows a systematic pattern where qualitative structure replicates but quantitative mechanism does not (Section 7.3); cross-model distance is fundamentally model-dependent (Section 7.4); and model-specific deficits resist type-based characterization (Section 7.5). These limits are not failures of the benchmark — they are findings about the nature of conceptual navigation. The qualitative geometric structure documented in Sections 4–6 is real and replicable; the mechanism that produces it is multi-variable and beyond the resolving power of single-factor experimental designs. The next two sections show that this structure generalizes across architectures (Section 8) and survives protocol variation (Section 9).

8 Act V — Generality: Structure Scales Across Architectures

The structural findings in Sections 4–6 were established on four models from four providers. A natural concern is that the properties might reflect idiosyncrasies of those specific models rather than general features of language model navigation. Phases 10A and 11A addressed this by testing 8 additional models from 7 independent training pipelines, bringing the total to 12 models from 11 independent training pipelines. The central result is that both navigational structure and navigational content generalize across architectures — with one scale-dependent exception in content generalization, alongside separate measurement caveats discussed in Sections 5 and 9.

8.1 Replication Across 12 Models

Phase 10A tested 5 models (Qwen 3.5 397B-A17B, MiniMax M2.5, Kimi K2.5, GLM 5, Llama 3.1 8B Instruct) across the same 12 concept pairs used in earlier phases. Four passed the reliability gate; GLM 5 was blocked by upstream rate limiting and excluded. Phase 11A tested 4 additional models (DeepSeek V3.2, Mistral Large 3, Cohere Command A, Llama 4 Maverick), all of which passed reliability.

Methodological note. The initial Phase 10A run used 60-second timeouts and blocked 4 of 5 models on OpenRouter latency, not model capability — all four parsed correctly with 100% connectivity. After relaxing timeouts to 300 seconds via a `--patient` CLI flag and re-running overnight, 4 of 5 models passed. This infrastructure episode is instructive: aggressive timeout thresholds can masquerade as model failures, and a consequential methodological change for the benchmark's generality results was a timeout relaxation, not a prompt revision.

R1 (gait) replicates universally. All 12 models show characteristic conceptual gaits — stable, model-specific consistency levels measured by intra-model Jaccard similarity. The full spectrum:

Model	Gait (Mean Jaccard)	Cohort
Mistral Large 3	0.747	Phase 11A

Model	Gait (Mean Jaccard)	Cohort
Claude Sonnet 4.6	0.578	Original
DeepSeek V3.2	0.540	Phase 11A
Llama 4 Maverick	0.539	Phase 11A
Qwen 3.5 397B-A17B	0.508	Phase 10A
Cohere Command A	0.502	Phase 11A
MiniMax M2.5	0.419	Phase 10A
Kimi K2.5	0.414	Phase 10A
Gemini 3 Flash	0.372	Original
Llama 3.1 8B Instruct	0.298	Phase 10A
Grok 4.1 Fast	0.293	Original
GPT-5.2	0.258	Original

The range spans 2.9× from GPT (0.258) to Mistral (0.747). Gait is not bounded at Claude's level (~0.58); architectural and training differences can produce substantially more rigid navigation. The property is universal — every model tested has a measurable, characteristic gait — but the specific value is model-specific.

R2 (asymmetry) replicates universally. All 12 models exceed the 0.60 directional asymmetry threshold: Phase 10A models ranged from 0.638 (MiniMax) to 0.785 (Llama 8B); Phase 11A models from 0.673 (Llama 4 Maverick) to 0.729 (Mistral). The quasimetric property — that forward and reverse paths share few waypoints — is universal across models from 11 independent training pipelines. The resolution-dependence caveat from Section 5.1 (asymmetry falls below threshold at 5 waypoints in the Phase 11C robustness subset) applies to the measurement, not the phenomenon.

8.2 The Structure/Content/Scale Hierarchy

The most informative result from the generality phases is not that structure replicates — that was expected — but that navigational *content* (which specific bridge concepts models use) also mostly generalizes. The hierarchy:

Structure generalizes across tested models under the primary elicitation protocol. Gait and asymmetry are universal properties across all 12 models from 11 independent training pipelines. No model fails either property, though asymmetry detectability requires sufficient waypoint count (Section 5.1).

Content mostly generalizes among large models. Aggregate bridge frequency for the benchmark's pre-specified bridges is statistically indistinguishable between cohorts, with some bridges near-universal and others remaining model-dependent. Phase 10A: new cohort mean 0.721 vs original 0.817, diff −0.096, CI including zero. Phase 11A combined: new cohort 0.717 vs original 0.817, diff −0.100, CI including zero. The strongest bridges — "spectrum" for light→color, "dog" for animal→poodle, "warm" for hot→cold — appear consistently across providers, but others show substantial heterogeneity: "sadness" ranges from 0.000 to 1.000 across models, "harmony" from 0.133 to 0.600, and "leather" from 0.267 to 1.000. The aggregate pattern is convergence; the bridge-level pattern is a mixture of universal and model-dependent landmarks.

Scale differentiates. Llama 3.1 8B Instruct is the sole outlier in bridge frequency: mean 0.200, compared to 0.717–0.817 for all frontier-scale models. Phase 11A provides a within-family comparison: Llama 4 Maverick (frontier-scale) shows bridge frequency of 0.724, while Llama 3.1 8B shows 0.200 — a 3.6× difference within the same model family. Both models have navigational structure (gait and asymmetry are present), but the small model navigates through different land-

marks. Scale is the clearest observed differentiator in current data, though the threshold and the role of architecture remain open.

Connection to Karkada and others (2025). Karkada et al. proved that geometric structure in LLM representations arises inevitably from translation symmetry in co-occurrence statistics — structure that all models share because they train on similar data distributions. Our empirical finding of structure/content generality is consistent with this prediction: if geometric structure is determined by data statistics, then navigational structure derived from that geometry should also generalize. The scale effect suggests a minimum capacity threshold for extracting the full navigational structure from shared data statistics.

Connection to the Platonic Representation Hypothesis. The Platonic Representation Hypothesis (Huh et al., 2024) predicts that larger models converge toward shared representations, supported by measures like CKA alignment. Our behavioral test partially confirms this: navigational *structure* converges (all models have gait and asymmetry), navigational *landmarks* converge for large models (same bridge concepts), but navigational *gaits* remain model-specific (the 2.9× gait range does not collapse with scale). The models share a map but prefer different routes — partial confirmation of representational convergence, with a clear boundary where convergence stops.

8.3 Gait Profiles of New Models

Phase 11A extended the gait spectrum beyond the original four models, revealing that the range is wider than initially characterized.

Mistral sets the record at 0.747. Mistral Large 3 showed near-deterministic navigation on several pairs: 0.936 on music→mathematics (producing nearly identical waypoint sets across all repetitions) and 0.934 on hot→cold. This exceeds Claude's 0.578 by a substantial margin and demonstrates that rigid, near-deterministic navigation is an achievable endpoint of the gait spectrum.

DeepSeek, Cohere, and Llama 4 Maverick cluster at moderate consistency. DeepSeek V3.2 at 0.540, Llama 4 Maverick at 0.539, and Cohere Command A at 0.502 occupy the middle of the gait spectrum — more consistent than GPT/Grok but less rigid than Claude/Mistral. The Llama within-family comparison is particularly informative: Maverick (0.539) vs 8B (0.298) shows that gait increases with scale within a model family, suggesting that scale may contribute to navigational consistency within the Llama family, though broader causal attribution remains open.

The 12-model gait spectrum reveals that architectural and training differences produce qualitatively different navigation strategies. This is not merely quantitative variation — a model with gait 0.747 (Mistral) generates near-identical paths across runs, while a model with gait 0.258 (GPT) explores broadly across a large waypoint vocabulary. The gait metric captures a genuine behavioral dimension that spans the full range from near-deterministic to broadly exploratory.

The generality results establish that the benchmark's structural claims are not artifacts of the original four-model cohort. Gait and asymmetry are universal across 12 models from 11 independent training pipelines. Frontier-scale models share the benchmark's strongest bridges and similar aggregate bridge frequencies, but several bridges remain model-dependent. The scale effect provides a meaningful boundary condition: frontier-scale models converge on the same geometric structure and share the strongest navigational landmarks, while small models retain the structure but navigate through different waypoints. The next section tests whether these findings are also robust to protocol variation.

9 Act VI — Robustness: The Benchmark Survives Its Own Tests

The structural claims in Sections 4–8 rest on a specific elicitation protocol: 7 waypoints, temperature 0.7, single-direction prompting. A natural concern is that the findings reflect idiosyncratic interactions between specific models and these specific protocol choices. Phase 11C addressed this directly with a multiverse robustness analysis: 3 models tested across a 2×2 grid of waypoint count (5, 9) and sampling temperature (0.5, 0.9), plus the 7-waypoint/0.7-temperature baseline. Phase 11B addressed a separate vulnerability — the control pair problem — by attempting to find replacement controls. Together, these experiments establish where the benchmark's claims are robust and where they require qualification.

9.1 Protocol Independence: Model Identity Drives Structure

The factorial analysis testing gait (intra-model Jaccard) against model identity, waypoint count, temperature, and their interaction produces the cleanest result in the benchmark:

Factor	η^2 (SS Proportion)	p-value (approx.)
Model identity	0.242	≈ 0.001
Waypoint count	0.008	≈ 0.520
Temperature	0.002	≈ 0.743
Waypoint × Temperature	≈ 0.000	≈ 0.886

Model identity accounts for 30× more variance than waypoint count and 120× more than temperature. The interaction term is as null as this analysis can produce. For gait, the structural claims are not artifacts of the 7-waypoint, 0.7-temperature protocol. Bridge frequency is similarly stable (Section 9.2). Asymmetry, however, is waypoint-count sensitive (Section 9.4) — the ANOVA result applies to gait specifically, not to all structural metrics uniformly.

Statistical caveat. The p-values use a chi-squared approximation to the F-distribution (ignoring denominator degrees of freedom) and treat per-pair cells as independent without repeated-measures modeling. Only 3 models were tested (Claude, GPT, DeepSeek). The effect sizes (η^2) are more reliable than the exact p-values. The qualitative conclusion — that model identity dominates while protocol parameters are null — is robust to the approximation, but the analysis should be understood as descriptive/exploratory rather than confirmatory.

Baseline data limitation. The 7wp-t0.7 baseline is assembled from prior-phase data (Phases 1, 2, 6A). DeepSeek, added in Phase 11A, has no prior-phase data — its baseline cells are empty. Additionally, reverse-direction data for some pairs is missing from prior phases. Baseline asymmetry and gait values should be interpreted with this limitation in mind; the four new conditions (which have complete data) provide more reliable comparisons.

9.2 Bridge Frequency Is the Most Robust Property

Bridge frequency across all conditions:

Condition	Mean Bridge Frequency	95% CI
7wp-t0.7 (baseline)	0.992	[0.983, 1.000]
5wp-t0.5	0.993	[0.978, 1.000]
5wp-t0.9	0.978	[0.933, 1.000]

Condition	Mean Bridge Frequency	95% CI
9wp-t0.5	0.993	[0.978, 1.000]
9wp-t0.9	0.985	[0.956, 1.000]

All conditions exceed 0.97. The lowest (5wp-t0.9 at 0.978) — the condition most likely to produce variability, with fewer waypoint slots and higher sampling randomness — is still essentially at ceiling. Bridge bottleneck structure is insensitive to both waypoint count and temperature. "Spectrum" for light→color and "warm" for hot→cold appear as obligatory landmarks regardless of how many total waypoints are available or how much randomness is introduced. "Sadness" for emotion→melancholy is near-obligatory, dipping to 0.933 in one condition (Claude at 9wp-t0.5) and 0.867 in another (GPT at 9wp-t0.9) but never absent.

This result strengthens R6 (bridge bottlenecks) as the benchmark's most protocol-robust claim. The bottleneck phenomenon is not an artifact of long paths or low temperature — it persists even with short paths at high temperature.

9.3 Gait Rankings Are Largely Stable

Kendall's W = 0.840 across conditions, confirming that the rank ordering of models by gait consistency is stable under protocol variation. Claude is the most consistent model in every condition. The second and third positions show one swap:

Condition	Rank Order
7wp-t0.7 (baseline)	Claude > GPT > DeepSeek
5wp-t0.5	Claude > DeepSeek > GPT
5wp-t0.9	Claude > DeepSeek > GPT
9wp-t0.5	Claude > DeepSeek > GPT
9wp-t0.9	Claude > DeepSeek > GPT

The baseline is the outlier, placing GPT above DeepSeek. This likely reflects DeepSeek's sparse baseline data (no prior-phase runs) rather than genuinely lower consistency; under actual elicitation in the four new conditions, DeepSeek shows Jaccard of 0.492–0.597, comfortably above GPT's 0.365–0.477 range. Baseline sparsity may depress the observed concordance, so W = 0.840 is best interpreted conservatively.

Claude's absolute gait values are condition-dependent — 0.794 at 5wp-t0.5 down to 0.624 at 9wp-t0.9. For Claude in this subset, consistency is lower at 9wp-t0.9 than at 5wp-t0.5, but Phase 11C does not show a reliable general main effect of waypoint count or temperature across all models. The rank ordering, however, is preserved: Claude remains the most consistent under all tested conditions, and the relative gaps between models are stable. Gait is a model property that varies in magnitude with protocol but maintains its relative structure.

9.4 Asymmetry Is Resolution-Dependent

Mean asymmetry per condition:

Condition	Mean Asymmetry	> 0.60 Threshold?
7wp-t0.7 (baseline)*	0.599	no

Condition	Mean Asymmetry	> 0.60 Threshold?
5wp-t0.5	0.594	no
5wp-t0.9	0.593	no
9wp-t0.5	0.669	**yes**
9wp-t0.9	0.684	**yes**

*Baseline mean from 2 valid cells only (Claude and GPT on hot→cold); other cells lack reverse or prior-phase data.

The pattern is consistent: 9-waypoint conditions pass the 0.60 threshold; 5-waypoint conditions fall just short. This is theoretically interpretable — more waypoints provide more positions where directional differences can manifest. At 5 waypoints, the path is too short for the quasimetric property to be reliably detected above the threshold.

Temperature effects on asymmetry are negligible: 5wp-t0.5 (0.594) vs 5wp-t0.9 (0.593); 9wp-t0.5 (0.669) vs 9wp-t0.9 (0.684). This is consistent with the ANOVA finding that temperature has no significant effect on navigational structure.

This finding qualifies R2: the quasimetric property is real — individual model × pair asymmetries are consistently positive at all waypoint counts — but the aggregate asymmetry index is resolution-dependent. The 0.60 threshold is a measurement sensitivity boundary, not a physical boundary. The asymmetry finding from Section 5.1 (mean 0.811 in the Phase 2 cohort at 5 waypoints) was measured with a larger pair set and model cohort; the robustness subset uses only 2 pairs and 3 models, so the lower absolute values reflect the reduced sample as much as any resolution effect.

9.5 The Control Pair Problem

The most uncomfortable finding in the robustness analysis concerns the benchmark's control validation. Phase 11B tested 4 replacement control candidates — accordion–stalactite, turmeric–trigonometry, barnacle–sonnet, and magnesium–ballet — across 6 models. The result was decisive: 0 of 24 model–candidate cells passed either the frequency gate (< 0.15) or the entropy gate (> 5.0).

Candidate	Max Top Freq	Min Entropy	Passes?
accordion–stalactite	1.000	3.29	fail
turmeric–trigonometry	1.000	3.44	fail
barnacle–sonnet	1.000	3.44	fail
magnesium–ballet	1.000	3.65	fail

Each pair reveals a different failure mode. Accordion–stalactite converges on "bellow" (an accordion feature) at 0.60–1.00 frequency across 5 of 6 models. Turmeric–trigonometry shows the most creative routing — Claude goes through "ancient India," GPT through "spice," Gemini through "curcumin," DeepSeek through "angle," Mistral through "golden ratio" — yet each individual model still converges on a single dominant waypoint. Barnacle–sonnet finds ocean/poetry bridges ("shell," "sea shanty," "verse," "meter"). Magnesium–ballet finds movement/grace bridges ("choreography," "swan lake," "grace").

Retrospective analysis. Stapler–monsoon itself — the benchmark's original control pair — fails the strict R5 criteria for all 12 models when evaluated under the same gates. Top-waypoint frequencies range from 0.650 (GPT) to 1.000 (Mistral, Cohere, Llama 4 Maverick). The original 4

models show frequencies of 0.650–0.775, which are lower than the newer models but still dramatically above the 0.15 threshold. Under strict criteria, stapler–monsoon has never been a valid control.

What this reveals. Across the tested control candidates and stapler–monsoon, models consistently found structured routes, suggesting that truly unstructured single-pair controls are hard to obtain at 7 waypoints. This is not noise — it is a finding about the richness of the associative landscape in trained language models. The models are creative associators: given two concepts and 7 waypoint slots, they construct plausible, structured semantic routes.

Reframing. The benchmark's validity does not rest on controls being truly unstructured. It rests on the predictability gap: experimental pairs show bridge frequencies of 0.60–1.00 for specific bridges *predicted from the endpoint relationship* (spectrum for light→color, dog for animal→poodle), while control pairs show comparable frequencies for *unpredicted* concepts. The experimental pairs' structure is theory-derived; the control pairs' structure is ad hoc. The difference is predictability, not presence versus absence of structure.

R5 is retained but qualified as the benchmark's weakest robust claim. The practical implications are threefold: (1) the control limitation should be acknowledged prominently; (2) the evidence base should be framed as relative performance gaps, not absolute control baselines; and (3) future work should explore fundamentally different control designs — multi-pair batteries, entropy-based criteria, or relative consistency ratios rather than single-pair strict-threshold validation.

The robustness results establish that the benchmark's core structural claims survive protocol variation, with metric-specific qualifications. For gait, model identity dominates protocol parameters (ANOVA $\eta^2 = 0.242$ for model vs ≈ 0 for protocol factors). Bridge frequency is the most robust property (> 0.97 across all conditions). Gait rank ordering is largely stable ($W = 0.840$). Asymmetry is real but its detectability is waypoint-count sensitive — a measurement sensitivity finding that qualifies R2 without undermining it. The control pair problem is real and acknowledged: across tested candidates, models consistently find structured routes, and the benchmark's validity rests on the predictability of those routes, not on their absence in controls. Together with the generality results in Section 8, these findings confirm that the benchmark's central claims — characteristic gaits, quasimetric asymmetry, and bottleneck bridge structure — are properties of language models, not properties of a particular experimental protocol.

10 Discussion

The preceding six acts established that LLM conceptual navigation has measurable geometric structure (Section 4), consistent with quasimetric topology (Section 5), causally sensitive to early anchoring (Section 6), resistant to single-variable mechanistic explanation (Section 7), generalizable across 12 models from 11 independent training pipelines (Section 8), and robust to protocol variation (Section 9). This section synthesizes those findings, states their limitations directly, and identifies the directions they open. Across approximately 21,540 runs, 12 models, 11 phases, and 29 documented dead ends, the benchmark reveals a consistent empirical picture: conceptual navigation is structured, asymmetric, model-specific in dynamics, and opaque in mechanism.

10.1 Summary of the Six-Act Narrative

Structure (Section 4). Models have distinct, stable conceptual gaits spanning a 2.9x range from GPT (0.258) to Mistral (0.747) [robust] R1. Navigation is self-consistent across time and resolution

— cross-batch Jaccard falls within 0.05 of within-batch (O9), and 5-waypoint paths appear as proper subsequences of 10-waypoint paths 67.9% of the time (O10). The dual-anchor effect confirms that both endpoints shape the trajectory, ruling out pure forward-chaining (O2).

Topology (Section 5). Conceptual space is consistent with quasimetric structure. The symmetry axiom fails comprehensively — mean directional asymmetry is 0.811 in the Phase 2 cohort, exceeding 0.60 for all 12 models tested [robust] R2. The triangle inequality holds at approximately 91% across three independent samples, tested on within-direction Jaccard distance. These two properties are measured on different distance operationalizations (cross-direction Jaccard for asymmetry, within-direction Jaccard for triangle inequality), so the quasimetric characterization is convergent rather than formally established on a single unified metric. Bridge concepts function as structural bottlenecks — obligatory intermediaries rather than mere associations [robust] R6. Hierarchical paths are compositional, with 4.9x higher transitivity than random controls [robust] R4.

Mechanism (Section 6). Pre-filling the first waypoint causally displaces bridges (O15, mean displacement 0.515, CI [0.357, 0.664] excluding zero). The primary mechanism is associative primacy: whatever occupies position 1 anchors the trajectory regardless of its semantic content, with displacement magnitudes statistically indistinguishable across incongruent, congruent, and neutral conditions. Relation class matters as a secondary modulator (O26): related pre-fills (on-axis or same-domain) preserve the navigational context and allow bridge recovery, while unrelated pre-fills destroy it (Friedman $p = 0.034$). Bridge survival under perturbation is bimodal — "sadness" and "dog" survive robustly; "harmony" and "germination" collapse to near-zero.

Limits (Section 7). Seven follow-up hypotheses were falsified — spanning mechanistic, model-deficit, methodological, and effect-direction types; one initially falsified result was resurrected with additional data. Prediction accuracy descended from 81.3% in the characterization phases to 20-24% in the mechanism phases (O24). The pattern is systematic: qualitative structure is characterizable and replicable at approximately 80%; quantitative mechanism is opaque, with single-variable models succeeding at approximately 0-15%.

Generality (Section 8). Gait and asymmetry replicate universally across 12 models from 11 independent training pipelines [robust] R1, R2. Bridge frequency generalizes among frontier-scale models (aggregate difference CI includes zero) but not to Llama 3.1 8B, which navigates through different landmarks (bridge frequency 0.200 vs 0.717-0.817 for frontier models). The within-family Llama comparison is particularly informative: Maverick (frontier-scale) shows bridge frequency 0.724 while 8B shows 0.200, a 3.6x difference. The hierarchy is structure > content > scale: structural properties are universal, navigational content is shared among large models, and scale is the clearest differentiator [robust] R6 extended.

Robustness (Section 9). Model identity dominates protocol variation in an ANOVA decomposition (eta-squared = 0.242 for model vs approximately 0 for waypoint count and temperature) (O30). Bridge frequency is the most robust property, exceeding 0.97 across all conditions including the most challenging (5 waypoints, temperature 0.9) (O31). Gait rank ordering is largely stable (Kendall's $W = 0.840$), with Claude maintaining the top position in every condition. Asymmetry is resolution-dependent: 9-waypoint conditions clear the 0.60 threshold (0.669-0.684) while 5-waypoint conditions fall just short (0.593-0.594) (O32). Temperature effects on all metrics are negligible.

10.2 What the Benchmark Reveals About Conceptual Space

Four interpretive points emerge from the empirical results.

Consistent with quasimetric structure. Symmetry fails comprehensively (mean 0.811). The triangle inequality holds at approximately 91% across three independent samples. These two findings use different operationalizations of distance — cross-direction Jaccard for asymme-

try, within-direction Jaccard for triangle inequality — so the formal quasimetric claim is "consistent with" rather than deductively established on a single metric. The asymmetry finding parallels two independent results from different literatures: Tversky's (1977) demonstration that human similarity judgments are inherently asymmetric (North Korea is judged more similar to China than China is to North Korea), and Berglund et al.'s (2024) reversal curse showing that autoregressive models cannot reverse learned associations. Our finding is consistent with both: navigational asymmetry parallels the reversal curse and Tversky's human similarity asymmetry. The connection is thematic — all three concern directional asymmetry in conceptual structure — rather than mechanistic; we do not demonstrate that the reversal curse causes the specific asymmetry we observe.

Bottleneck topology. The most reliable navigational landmarks are not the most strongly associated concepts but obligatory intermediaries: "spectrum" for light-to-color (frequency 1.00 across all four original models), "dog" for animal-to-poodle, "warm" for hot-to-cold. "Metaphor" — intuitively central to the language-to-thought relationship — appeared at 0.00 frequency (graveyard entry G5). "Energy" — associated with both hot and cold — appeared at 0.00 on hot-to-cold. Process-naming concepts outperformed object-naming concepts: "germination" outperformed "plant" for seed-to-garden in all four models. This echoes a finding from knowledge graph research: shortest-path intermediaries are more informative than high-degree hubs. Navigational salience depends on directional information content, not on generic associative strength.

Model-specific gaits on shared geometry. All frontier-scale models navigate the same basic geometry through the same strongest landmarks, but with different route preferences. Mistral (gait 0.747) produces near-deterministic paths — 0.936 Jaccard on music-to-mathematics, producing nearly identical waypoint sets across all repetitions. GPT (0.258) explores broadly, sampling different landmarks on each traversal. The aggregate bridge frequency difference between original and expanded cohorts has a CI including zero, yet gait varies 2.9x. This is the behavioral analog of a refinement to the Platonic Representation Hypothesis: representations may converge (Huh et al., 2024), but navigation diversifies. Same map, different routes. Structure converges; gaits remain model-specific. Neither extreme is intrinsically superior — they represent genuinely different navigational strategies that a model-agnostic benchmark must accommodate.

Navigational structure is not compositional from simple variables. The mechanism ceiling — 7 of 8 hypotheses falsified at primary test — suggests that navigation emerges from the interaction of multiple factors rather than any single structural variable. Competitor count (G20), dominance ratio (G23), gradient type (G21, G24), and gait normalization (G22) all failed individually. The prediction accuracy descent from 81.3% to 20% tracks a shift in prediction type, not a loss of signal: replication predictions succeed at approximately 80% throughout, while mechanistic predictions hover near 0-15%. This parallels findings in computational neuroscience, where spatial navigation in rodents and humans depends on multi-factorial interactions between grid cells, place cells, head-direction cells, and boundary cells. The analogy is structural, not mechanistic: in both cases, the emergent navigational behavior resists decomposition into single-variable explanations.

Statistical caveat on the quasimetric characterization. The two tests supporting the quasimetric claim — asymmetry and triangle inequality — use different distance operationalizations. Asymmetry uses cross-direction Jaccard (comparing A-to-B paths against B-to-A paths); triangle inequality uses within-direction Jaccard (consistency of repeated same-direction runs). Additionally, all metrics are lexical — synonymous waypoints are treated as distinct. The magnitude of the resulting bias is bounded by the canonicalization pipeline but is otherwise unknown. These caveats do not undermine the qualitative finding (systematic asymmetry, triangle inequality holding at approximately 91%), but the formal quasimetric claim should be understood as "consistent with" rather than "formally established."

10.3 Limitations

The benchmark's limitations divide into two distinct categories: design issues with the benchmark itself, and findings about the phenomenon that constrain interpretation. Conflating these would obscure the distinction between things we should have done better and things that turned out to be harder than expected.

10.3.1 Benchmark Limitations

1. Control pair failure [robust] R5 (qualified). This is the benchmark's most serious design limitation. The original control pair stapler-monsoon fails strict criteria for all 12 models (top waypoint frequency 0.650-1.000). Phase 11B tested 4 replacement candidates (accordion-stalactite, turmeric-trigonometry, barnacle-sonnet, magnesium-ballet) across 6 models: 0 of 24 cells passed either the frequency gate (< 0.15) or the entropy gate (> 5.0). Each candidate revealed creative structured routing — turmeric-trigonometry produced "ancient India" (Claude), "spice" (GPT), "golden ratio" (Mistral) — yet each model still converged on a single dominant waypoint. LLMs find navigable routes between any concept pair at 7 waypoints, defeating single-pair strict-threshold control validation. The benchmark's validity rests on the predictability gap — experimental bridges are predicted from the endpoint relationship; control bridges are ad hoc — not on presence versus absence of structure. This limitation should be weighted prominently in any evaluation of the benchmark's claims.

2. Self-grounding circularity. Model outputs generate their own ground truth. The concern is that structural findings could be artifacts of the generation process rather than properties of conceptual organization. Three mitigations partially address this:

(a) Testing against external mathematical properties that have model-independent definitions (metric axioms — the triangle inequality either holds or it does not, independent of the generation process).
(b) Cross-model comparison across 12 independent models (if all 12 show the same property, it cannot be attributed to idiosyncrasies of any single model's decoding).
(c) Causal intervention via the pre-fill experiments (bridge displacement demonstrates that structure is causally sensitive, not merely a statistical regularity).

These mitigations reduce the concern but never fully eliminate it. The benchmark cannot distinguish between "models have navigational geometry" and "the generation process has statistical regularities that mimic navigational geometry" — though the causal sensitivity of bridge displacement (Section 6.1) makes the second interpretation less parsimonious.

3. Lexical metrics only. All metrics use Jaccard similarity after lemmatization. "Melody" and "tune" are treated as entirely distinct. The asymmetry mean of 0.811 is lexical asymmetry; true conceptual asymmetry is likely lower because some apparent non-overlap reflects synonymous waypoints scored as different. The magnitude of this bias is unknown but bounded by the canonicalization pipeline, which handles morphological variants (running/run, cities/city).

Embedding-based semantic similarity was implemented but deferred due to API cost. This is the benchmark's most pervasive methodological limitation, present in every metric and every phase. It affects all structural claims: gait underestimates true conceptual consistency, asymmetry overestimates true conceptual asymmetry, and bridge frequency may miss bridges expressed through synonyms rather than exact tokens.

4. API-mediated access. Internal model state cannot be controlled, temperature settings may be approximate for some providers, and silent model updates may occur during multi-day data

collection. Partially addressed by the no-temporal-drift finding (O9, cross-batch Jaccard within 0.05 of within-batch) and the robustness to temperature variation (Phase 11C, eta-squared = 0.002 for temperature). These mitigations apply to the tested conditions; they do not rule out sensitivity to untested API-level changes.

5. Concept pair selection bias. Pairs were selected by the experimenters based on semantic interest and hypothesis-driven design, not sampled from a principled distribution over conceptual space. The benchmark tests navigational structure on the pairs it tests, not on a representative sample of all possible concept relationships. Partially addressed by the breadth of the pair inventory (100+ unique pairs across 11 phases) and the generality analysis (Section 8), but the selection is not random and the results should not be extrapolated to arbitrary concept pairs without further testing.

6. Resolution dependence. Asymmetry requires 9 or more waypoints to reliably exceed the 0.60 threshold in the robustness subset (O32): the 5-waypoint conditions fall just short (0.593-0.594), the 7-waypoint baseline also fell short (0.599) in the sparse 3-model subset, and only the 9-waypoint conditions clear it (0.669-0.684). The full Phase 2 cohort at 5 waypoints produced 0.811, but that used a larger pair set and model cohort. Other structural properties may have similar resolution thresholds that have not been identified — the benchmark tested waypoint counts of 5, 7, and 9 but not the full range. This is a measurement sensitivity finding, not evidence against asymmetry, but it means that claims about asymmetry magnitude are conditioned on sufficient path length.

10.3.2 Phenomenon Limitations

7. Mechanism ceiling. Seven of eight hypotheses were falsified at the primary-test level (Section 7.2). This is not a design flaw — it is the benchmark's most informative negative result. The mechanism ceiling establishes that conceptual navigation resists single-variable explanation and that the resolving power of one-factor experimental designs has been exhausted for this phenomenon. The prediction accuracy trajectory — 81.3% (Phase 4) to 20% (Phase 9) — tracks a systematic shift in prediction type, not random degradation.

8. Cross-model distance failure (O18, O19). Models rank-order navigational distances differently. The cross-model distance correlation was r = 0.170, far below the 0.50 validity threshold. Gait normalization produced zero improvement (r = 0.212 before and after, with 0.000 change on every model pair, G22). Some model pairs actively anti-correlate: Grok-Gemini at r = -0.580. No universal distance function is recoverable from path-based measurements. Within-model distance analysis remains viable; cross-model distance comparison is blocked.

9. Gemini characterization failure. Gemini's approximately 30% bridge frequency deficit relative to the non-Gemini mean (0.480 vs approximately 0.67) is real and stable across the tested phases. Three successive mechanistic characterizations — frame-crossing threshold (G7), gradient blindness (G21), transformation-chain blindness (G24) — all failed, with two producing results in the opposite direction from predicted. Whatever pair type is designated as "hard" for Gemini, Gemini does relatively better on it. The type-based approach to characterizing the deficit appears fundamentally misguided; the deficit is model-level, not type-level. Future approaches might require representational analysis rather than behavioral type-classification.

10.4 The Graveyard as Methodology

The benchmark documents 29 dead ends across 11 phases — a transparent record of the benchmark's learning curve.

Four lessons emerge from the graveyard.

First, intuitive semantic importance has zero predictive power for navigational behavior. "Metaphor" — the concept most intuitively connected to both language and thought — appeared at 0.00 frequency as a bridge for language-to-thought (G5). "Energy" — associated with both hot and cold — appeared at 0.00 on hot-to-cold. Navigational salience and human intuition about conceptual centrality are orthogonal; bridges must name the primary axis of connection, not merely inhabit the shared associative neighborhood.

Second, qualitative models can be correct while their quantitative predictions fail. Gemini's deficit is real (approximately 30% reduction in bridge frequency), but three quantitative characterizations of its mechanism failed (G7, G21, G24). The qualitative observation replicates across every phase in which Gemini was tested; the mechanism remains opaque. This pattern — robust qualitative finding, intractable quantitative mechanism — characterizes the benchmark as a whole.

Third, retrospective signals on small samples do not generalize. The dominance ratio showed a promising correlation (rho = 0.60) on the original 8-pair sample but collapsed to rho = 0.157 on an independent sample (G23). The competitor count predictor showed a similar failure (G20, rho = 0.116). Both variables were plausible on the original data; neither survived replication.

Fourth, infrastructure limitations can masquerade as substantive findings. The bridge generalization failure (G26) was initially attributed to the phenomenon based on data from a single small model (Llama 8B) under aggressive 60-second timeouts that excluded most new models on latency, not capability. After timeout relaxation to 300 seconds and testing of 3 additional frontier-scale models, the aggregate finding reversed — bridge frequency CI shifted from excluding zero to including it. Llama 8B's low bridge frequency (0.200) remains real as a scale effect, but the original conclusion that bridge structure does not generalize was wrong. G26 is the sole clean resurrection in the graveyard and a caution against both premature interpretation of negative results and confounding infrastructure limitations with phenomenon limitations.

Transparent dead-end documentation should be standard practice in benchmarking work. The incentive structure in machine learning research favors reporting only successful experiments, but the graveyard entries in this benchmark — particularly G5 (metaphor), G22 (gait normalization), and G26 (resurrected) — are as methodologically informative as the successful findings. We recommend that future benchmark papers adopt a similar practice of documenting falsified hypotheses alongside confirmed ones.

10.5 Connections to Theory

Karkada and others (2025). They proved that static geometric structure in LLM representations arises inevitably from translation symmetry in co-occurrence statistics — the manifold shape is determined by the data distribution, not by architectural choices. Our benchmark tests the dynamic complement: whether that geometry supports structured, consistent behavioral navigation. Together, the contributions are complementary.

Their result explains why static geometry exists; ours shows that it supports navigation that is structured and asymmetric at the universal level, but model-specific in its dynamics. Geometry comes from data statistics (their contribution); navigational behavior on that geometry is universal in structure but model-specific in gait (ours). The scale effect from Section 8 is consistent with their framework: if geometric structure requires sufficient capacity to extract translation symmetries from training data, smaller models (Llama 8B) may have the structure but lack the resolution to navigate through the same landmarks as frontier models.

Gardenfors operationalized. Gärdenförs's (2000) conceptual spaces theory posits that concepts live in navigable geometric spaces with measurable distance. The benchmark provides a large-scale computational probe of this theory across 12 models and 100+ concept pairs. The par-

tial confirmation is substantive: concepts do live in navigable spaces with measurable, compositional distance (triangle inequality approximately 91%, hierarchical transitivity 4.9x over random). Polysemy sense differentiation (cross-pair Jaccard near zero for different senses of "bank," "bat," and "crane") confirms that paths reflect conceptual structure rather than surface word statistics. But the pervasive asymmetry (mean 0.811) challenges the standard metric assumption in conceptual spaces theory, which typically presumes symmetric distance functions. The data suggest that quasimetric structure — metric-like but inherently directional — may be a more appropriate formal framework for computational conceptual spaces.

Platonic Representation Hypothesis refined. Huh et al. (2024) predicted that larger models converge toward shared representations. Our behavioral test partially confirms this: navigational structure converges (all 12 models have gait and asymmetry), navigational landmarks converge for frontier-scale models (aggregate bridge frequency difference CI includes zero), but navigational gaits remain model-specific (the 2.9x gait range does not collapse with scale). The result establishes a boundary for the hypothesis: representational convergence does not imply behavioral convergence. Structure converges; gaits diversify. Partial confirmation with a clear boundary condition.

Wyss and others (2025). Spectral analysis of semantic attractors predicts discrete basins in LLM conceptual space. Our bridge bottlenecks and the "deep basins" observed in the word convergence game that inspired this benchmark (Hodge, 2025) are plausibly analogous to those spectral attractors — the behavioral manifestation of the basin structure their theory predicts. Bridge concepts like "spectrum" and "dog" behave as attractor basins — navigational points that paths converge toward regardless of starting trajectory. The pre-fill displacement results (Section 6.1) demonstrate that these basins are causally active: they anchor paths when present and paths reorganize when they are displaced. The bridge frequency robustness finding (> 0.97 across all protocol conditions, O31) further supports the attractor interpretation — these are deep structural features of the navigational landscape, not artifacts of a particular elicitation protocol.

10.6 Future Work

The benchmark's findings and limitations suggest ten specific directions, ordered from the most directly motivated by current data to the most speculative.

1. **Multi-variable bridge fragility model.** The mechanism ceiling (Section 7) was reached by testing single variables in isolation. A joint model — bridge role x pre-fill content x dominance ratio x competitor count — could capture the interaction effects that single-variable designs miss. The bimodal survival pattern (Section 6.1) suggests that at least two factors interact: "sadness" survives because it has no competitors and sits on an obligatory navigational axis, while "harmony" collapses because it competes with rhythm, pattern, and frequency in a crowded navigational space. The combinatorial cost is high (likely 5,000+ runs for a 4-factor design), but this is the only known path past the mechanism ceiling.

2. **Embedding-based distance validation.** Replace Jaccard with embedding cosine similarity to rescue the lexical limitation (Section 10.3, limitation 3). This would also test whether the cross-model distance failure (O18, O19) is a property of lexical metrics or of the navigational distances themselves.

3. **Multilingual conceptual topology.** Test whether the navigational geometry holds for non-English prompts and multilingual models. If the structure arises from data statistics (Karkada and others, 2025), language-specific training distributions may produce different geometries.

4. **Scale threshold investigation.** The Llama family provides a natural within-family comparison: 8B (bridge frequency 0.200) vs Maverick at frontier scale (0.724). Intermediate checkpoints (Llama 70B, Llama 405B) would locate the scale threshold at which bridge content converges with frontier models and determine whether the transition is gradual or abrupt.

5. **Temporal stability under model updates.** The no-drift finding (O9) covers the data collection window. Longitudinal monitoring — re-running the core pair set on the same model IDs at 3-month intervals — would test whether navigational structure is preserved across training updates and provider-side model swaps.

6. **Human waypoint comparison.** A calibration subset of 10-15 concept pairs administered to human participants under the same prompt format would establish whether the structural properties (asymmetry, bridge bottlenecks, gait variation) are shared between human and LLM navigation or are specific to autoregressive generation. Given Tversky's (1977) finding that human similarity is asymmetric and Constantinescu et al.'s (2016) evidence for grid-cell-based conceptual navigation, partial overlap is plausible but untested.

7. **External anchoring.** Validate bridge predictions against ConceptNet relation strength, Small World of Words association norms, or Wikipedia link structure. This would partially address the self-grounding circularity (limitation 2) by grounding navigational landmarks in independently measured conceptual structure.

8. **Control pair redesign.** The single-pair strict-threshold approach has failed (limitation 1). Three alternatives: (a) relative consistency metrics comparing experimental vs control pairs on predictability rather than presence of structure; (b) pair batteries testing 20+ control pairs to characterize the distribution of spurious bridge frequencies; (c) difficulty-graded controls sampling from the full spectrum of conceptual distance.

9. **Chain-of-thought as intervention.** Eliciting waypoints with and without chain-of-thought reasoning would test whether explicit reasoning changes navigational structure or merely rationalizes the same implicit routes. This connects to the compositionality gap literature: if chain-of-thought improves multi-hop reasoning, it may also alter bridge selection, path composition, and the gait spectrum.

10. **Curvature estimation.** Within-model curvature is estimable from the triangle inequality data — the distribution of triangle excess (d(A,B) + d(B,C) - d(A,C)) characterizes local geometry, with positive excess suggesting positive curvature and near-zero excess suggesting flatness. Cross-model curvature comparison is blocked by the distance failure (O18, O19) and should not be attempted until embedding-based distances are validated (item 2 above).

The discussion synthesizes a benchmark that achieves high replication accuracy (approximately 80%) for structural characterization while reaching a mechanism ceiling (approximately 15-20%) for quantitative explanation. The core empirical contribution is a set of structural properties — characteristic gaits spanning 2.9x across 12 models, quasimetric asymmetry at 0.811, bottleneck bridge topology with frequencies exceeding 0.97 under robustness testing, and compositional hierarchy with 4.9x transitivity over random controls — that generalize across architectures and survive protocol variation. The core negative contribution is equally important: 7 falsified mechanistic hypotheses and 29 documented dead ends establish that the qualitative geometry of conceptual navigation is

accessible to current methods while the quantitative mechanism is not. The benchmark's most serious limitation — the control pair failure, with 0 of 24 replacement cells passing — is acknowledged directly, and its validity is reframed as resting on predictability gaps rather than absolute baselines. The six theoretical connections — to quasimetric topology, Tversky's asymmetry, Karkada et al.'s geometric inevitability, Gardenfors's conceptual spaces, the Platonic Representation Hypothesis, and spectral semantic attractors — position the findings within a broader landscape where static geometry is well-characterized but dynamic navigation is only beginning to be understood. The most productive future directions involve multi-variable mechanism models, embedding-based distance validation, and human comparison — each addressing a specific limitation identified across the six acts.

11 Conclusion

Across approximately 21,540 runs spanning 12 models from 11 independent training pipelines, the Conceptual Topology Mapping Benchmark establishes that LLMs have consistent, measurable geometric structure in how they navigate between concepts. This structure is consistent with quasimetric geometry — systematically asymmetric (mean 0.811), with the triangle inequality holding at approximately 91%. Navigation is organized around bottleneck bridges that function as structural landmarks, not mere associations, and expressed through model-specific gaits spanning a 2.9x range from GPT (0.258) to Mistral (0.747). These findings generalize across all 12 models and resist protocol variation.

The mechanism ceiling is itself the most important finding. Seven follow-up hypotheses — spanning mechanistic, model-deficit, methodological, and effect-direction types — were falsified across Phases 8–10; one was resurrected with additional data. Prediction accuracy descended from 81.3% for characterization questions to 20–24% for mechanistic ones — not because signal disappeared, but because qualitative structure (characterizable at ~80%) resists quantitative mechanistic decomposition (opaque at 0–15%). The dynamics generating navigational regularity do not reduce to any individual variable tested. This parallels the broader challenge in interpretability: structure is detectable before it is explainable.

Static embedding geometry is well-studied; navigation on that geometry is not. Karkada and others (2025) proved why the geometric structure exists — translation symmetry in co-occurrence statistics determines manifold shape. We showed what that geometry supports: structured, asymmetric, model-specific behavioral navigation. The Conceptual Topology Mapping Benchmark provides a different evaluation paradigm — not testing what models know, but how they navigate what they know. The 29 documented dead ends and the mechanism ceiling are not failures of the research program but honest accounting of where the field's explanatory tools fall short, and where the next generation of multi-factorial mechanistic models will need to reach.

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A Concept Pair and Triple Inventory

This appendix provides the complete inventory of concept pairs, triples, and phase-specific stimuli used across the benchmark's 11 experimental phases.

A.1 A.1 Core Concept Pairs (Phase 1)

36 pairs across 9 categories, split into holdout (15, used only for prompt selection) and reporting (21, used for all published analyses).

A.1.1 Anchor Pairs

Known basin structure from the word convergence game experiments.

ID	From	To	Concreteness	Basin	Depth	Set
anchor-beyonce-erosion	Beyonce	erosion	concrete–abstract	formation	deep	reporting
anchor-tesla-mycelium	Tesla	mycelium	concrete–concrete	network	deep	reporting

ID	From	To	Concreteness	Basin	Depth	Set
anchor-shadow-melody	shadow	melody	concrete–abstract	echo	moderate	holdout
anchor-skull-garden	skull	garden	concrete–concrete	—	moderate	reporting
anchor-kanye-lattice	Kanye West	lattice	concrete–abstract	—	flat	holdout

A.1.2 Hierarchy Pairs

Hypernym–hyponym relationships at varying abstraction levels.

ID	From	To	Concreteness	Set
hierarchy-animal-poodle	animal	poodle	concrete–concrete	reporting
hierarchy-vehicle-skateboard	vehicle	skateboard	concrete–concrete	holdout
hierarchy-emotion-nostalgia	emotion	nostalgia	abstract–abstract	reporting
hierarchy-structure-lattice	structure	lattice	abstract–abstract	holdout

A.1.3 Cross-Domain Pairs

Concepts from different semantic domains.

ID	From	To	Concreteness	Set
cross-music-mathematics	music	mathematics	abstract–abstract	reporting
cross-cooking-architecture	cooking	architecture	concrete–concrete	holdout
cross-justice-erosion	justice	erosion	abstract–concrete	reporting
cross-time-crystal	time	crystal	abstract–concrete	holdout

A.1.4 Polysemy Pairs

Same ambiguous word paired with different targets to steer sense activation.

ID	From	To	Polysemy Group	Target Sense	Set
polysemy-bank-river	bank	river	bank	geographic	reporting
polysemy-bank-mortgage	bank	mortgage	bank	financial	holdout
polysemy-bat-cave	bat	cave	bat	animal	holdout
polysemy-bat-baseball	bat	baseball	bat	sports	reporting
polysemy-crane-construction	crane	construction	crane	machine	reporting
polysemy-crane-wetland	crane	wetland	crane	bird	holdout

A.1.5 Near-Synonym Pairs

Dense semantic neighborhoods testing navigational friction.

ID	From	To	Concreteness	Set
synonym-cemetery-graveyard	cemetery	graveyard	concrete–concrete	reporting
synonym-forest-woods	forest	woods	concrete–concrete	holdout
synonym-happy-joyful	happy	joyful	abstract–abstract	reporting
synonym-anger-rage	anger	rage	abstract–abstract	holdout

A.1.6 Antonym Pairs

Oppositional continua testing axis traversal.

ID	From	To	Concreteness	Set
antonym-hot-cold	hot	cold	concrete–concrete	reporting
antonym-order-chaos	order	chaos	abstract–abstract	holdout

A.1.7 Control Pairs

Type	ID	From	To	Set
Identity	control-identity-apple	apple	apple	reporting
Identity	control-identity-justice	justice	justice	holdout
Random	control-random-stapler-monsoon	stapler	monsoon	reporting
Random	control-random-parliament-cinnamon	parliament	cinnamon	holdout
Random	control-random-telescope-jealousy	telescope	jealousy	reporting
Random	control-random-shoelace-democracy	shoelace	democracy	reporting
Random	control-random-flamingo-calculus	flamingo	calculus	reporting
Random	control-random-umbrella-photosynthesis	umbrella	photosynthesis	reporting
Random	control-random-origami-gravity	origami	gravity	reporting
Nonsense	control-nonsense-xkplm-qrvzt	xkplm	qrvzt	reporting
Nonsense	control-nonsense-bwfnj-tlmgk	bwfnj	tlmgk	holdout

A.2 A.2 Phase 3B Triples (Transitive Path Structure)

8 triples testing transitivity, triangle inequality, and bridge frequency. Each triple (A, B, C) requires 6 directional legs; some legs were reused from Phase 1/2 data.

ID	A	B	C	Type	Target Bridge
triple-animal-dog-poodle	animal	dog	poodle	hierarchical	dog
triple-emotion-nostalgia-melancholy	emotion	nostalgia	melancholy	hierarchical	nostalgia
triple-music-harmony-mathematics	music	harmony	mathematics	semantic-chain	harmony
triple-hot-energy-cold	hot	energy	cold	semantic-chain	energy
triple-beyonce-justice-erosion	Beyonce	justice	erosion	existing-pair	justice
triple-bank-river-ocean	bank	river	ocean	polysemy-extend	river
triple-music-stapler-mathematics	music	stapler	mathematics	random-control	stapler
triple-hot-flamingo-cold	hot	flamingo	cold	random-control	flamingo

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A.3 A.3 Phase 4 Triples (Cross-Model Bridge Topology)

8 targeted triples testing specific bridge predictions, with focus on Gemini fragmentation.

ID	A	B	C	Diagnostic Type	Bridge
p4-bank-river-ocean	bank	river	ocean	polysemy-retest	river
p4-bank-deposit-savings	bank	deposit	savings	polysemy-financial	deposit

ID	A	B	C	Diagnostic Type	Bridge
p4-light-spectrum-color	light	spectrum	color	cross-domain-concrete	spectrum
p4-emotion-nostalgia-melancholy	emotion	nostalgia	melancholy	abstract-retest	nostalgia
p4-language-metaphor-thought	language	metaphor	thought	abstract-bridge	metaphor
p4-tree-forest-ecosystem	tree	forest	ecosystem	concrete-hierarchical	forest
p4-light-chandelier-color	light	chandelier	color	random-control	chandelier
p4-emotion-calendar-melancholy	emotion	calendar	melancholy	random-control	calendar

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A.4 A.4 Phase 5 Stimuli (Cue-Strength, Dimensionality, Convergence)

A.4.1 Part A: Cue-Strength Triples

14 triples across 4 concept families at 3–4 cue levels, plus 2 random controls.

Family	A	C	Cue Level	B (Bridge)
Physical causation	sun	desert	very-high	heat
Physical causation	sun	desert	medium	radiation
Physical causation	sun	desert	low	solstice
Biological growth	seed	garden	very-high	plant
Biological growth	seed	garden	medium	germination
Biological growth	seed	garden	low	husk
Compositional hierarchy	word	paragraph	very-high	sentence
Compositional hierarchy	word	paragraph	medium	clause
Compositional hierarchy	word	paragraph	low	syllable
Abstract affect	emotion	melancholy	high	sadness
Abstract affect	emotion	melancholy	medium	nostalgia
Abstract affect	emotion	melancholy	low	apathy
Control	sun	desert	—	umbrella
Control	seed	garden	—	telescope

A.4.2 Part B: Dimensionality Triples

14 triples probing polysemous focal concepts across same-axis vs. cross-axis configurations.

Focal Concept	A	C	Axis Pattern	Bridge
light	photon	color	same-axis (physics)	light
light	feather	heavy	same-axis (weight)	light
light	candle	darkness	same-axis (illumination)	light
light	photon	heavy	cross-axis	light
light	candle	color	partial-overlap	light
light	prayer	darkness	cross-axis	light
bank	loan	savings	same-axis (financial)	bank
bank	river	shore	same-axis (geographic)	bank
bank	loan	shore	cross-axis	bank
fire	spark	ash	same-axis (combustion)	fire
fire	ambition	motivation	same-axis (passion)	fire
fire	spark	motivation	cross-axis	fire
control	photon	color	—	bicycle
control	loan	savings	—	penguin

A.4.3 Part C: Convergence Pairs

8 pairs for W-shape positional convergence testing.

From	To	Type	Expected Bridge
light	color	bridge-present	spectrum
bank	savings	bridge-present	deposit
animal	poodle	bridge-present	dog
tree	ecosystem	bridge-present	forest
language	thought	bridge-absent	—
hot	cold	bridge-absent	—
music	mathematics	bridge-variable	harmony
telescope	jealousy	no-bridge-control	—

A.5 A.5 Phase 6 Pairs (Salience, Forced-Crossing, Position)

A.5.1 Part A: Salience Mapping (8 pairs, 30–40 reps each)

From	To	Known Bridge	Category
music	mathematics	harmony	bridge-present
sun	desert	heat	bridge-present
seed	garden	germination	bridge-present
light	color	spectrum	bridge-present
bank	ocean	river	bridge-present

From	To	Known Bridge	Category
emotion	melancholy	sadness	bridge-present
language	thought	—	bridge-absent
hot	cold	—	bridge-absent

A.5.2 Part B: Forced-Crossing Asymmetry (8 pairs)

From	To	Bridge	Type
loan	shore	bank	forced-crossing
deposit	river	bank	forced-crossing
savings	cliff	bank	forced-crossing
photon	heavy	light	forced-crossing
loan	savings	—	same-axis comparison
river	shore	—	same-axis comparison
sun	desert	—	same-axis comparison
seed	garden	—	same-axis comparison

A.5.3 Part C: Positional Bridge Scanning (10 pairs)

From	To	Bridge	Expected Position
light	color	spectrum	middle (3–5)
bank	savings	deposit	middle (3–5)
animal	poodle	dog	middle (3–5)
music	mathematics	harmony	middle (3–5)
tree	ecosystem	forest	middle (3–5)
loan	shore	bank	middle (3–5)
deposit	river	bank	middle (3–5)
sun	desert	heat	early (2–3)
seed	garden	germination	middle (3–5)
emotion	melancholy	sadness	late (4–6)

A.6 A.6 Phase 7 Pairs and Triples (Anchoring, Curvature, Too-Central)

A.6.1 Part A: Anchoring-Effect Pairs (8 pairs, 3 pre-fill conditions each)

From	To	Bridge	Incongruent	Congruent	Neutral	Role
music	mathematics	harmony	symmetry	rhythm	element	heading-bridge
sun	desert	heat	drought	warmth	element	heading-bridge

From	To	Bridge	Incongruent	Congruent	Neutral	Role
seed	garden	germination	sprout	growth	element	heading-bridge
light	color	spectrum	wavelength	prism	element	heading-bridge
bank	ocean	river	estuary	stream	element	heading-bridge
emotion	melancholy	sadness	mood	grief	element	heading-bridge
loan	shore	bank	interest	credit	element	forced-crossing
animal	poodle	dog	mammal	canine	element	taxonomic

A.6.2 Part B: Curvature Triangles (8 triangles)

A	B	C	Vertex Type	Relationship
loan	bank	river	polysemous (homonym)	financial/geographic
deposit	bank	shore	polysemous (homonym)	financial/geographic
photon	light	heavy	polysemous (homonym)	electromagnetic/weight
candle	light	feather	polysemous (homonym)	illumination/weight
sun	heat	desert	non-polysemous	causal-chain
seed	germination	garden	non-polysemous	process-chain
music	harmony	mathematics	non-polysemous	cross-domain
word	sentence	paragraph	non-polysemous	compositional

A.6.3 Part C: Too-Central Pairs (10 pairs)

From	To	Candidate Bridge	Category
spark	ash	fire	too-central
acorn	timber	tree	too-central
flour	bread	dough	too-central
hot	cold	warm	obvious-useful
infant	elderly	adolescent	obvious-useful
whisper	shout	speak	obvious-useful
rain	ocean	water	boundary
egg	chicken	embryo	boundary
ice	steam	water	boundary
dawn	dusk	noon	boundary

A.7 A.7 Phase 8 Pairs (Fragility, Gradient, Distance)

A.7.1 Part A: Bridge Fragility (8 new pairs)

From	To	Predicted Bridge	Competitor Level	Pre-fill
question	answer	reasoning	low (0–1)	inquiry
parent	child	birth	low (0–1)	family
cause	effect	mechanism	low (1–2)	reason
winter	summer	spring	medium (3–4)	snow
student	professor	research	medium (3–5)	university
science	art	creativity	high (6–10)	experiment
ocean	mountain	landscape	high (6–10)	wave
brain	computer	algorithm	high (8–12)	neuron

A.7.2 Part B: Gradient vs. Causal-Chain Pairs (20 pairs)

Gradient-midpoint (10):

From	To	Bridge	Dimension
whisper	shout	speak	vocal intensity
hot	cold	warm	temperature
dawn	dusk	noon	time of day
infant	elderly	adolescent	life stage
crawl	sprint	walk	locomotion speed
pond	ocean	lake	water body scale
pebble	boulder	rock	stone size
drizzle	downpour	rain	precipitation
village	metropolis	city	settlement scale
murmur	scream	speech	vocal intensity

Causal-chain (10):

From	To	Bridge	Process
grape	wine	fermentation	winemaking
ore	jewelry	metal	smelting-crafting
caterpillar	butterfly	cocoon	metamorphosis
clay	vase	pottery	crafting
wool	sweater	yarn	textile
milk	cheese	curd	dairy
seed	fruit	flower	reproduction
sand	glass	heat	manufacturing
acorn	timber	tree	growth-harvest
flour	bread	dough	baking

A.7.3 Part C: Gait-Normalized Distance (16 pairs)

8 reference pairs (short to very-long expected distance) and 8 test pairs spanning the distance range. See Section 7.4 for the finding that cross-model distance normalization produces zero improvement.

A.8 A.8 Phase 9–11 Pairs

Phase 9 tested 6 dominance pairs (selected from Phase 8B data), 10 transformation/gradient pairs (retesting Gemini deficit), and 8 facilitation pairs (testing the pre-fill facilitation crossover).

Phase 10A used a 12-pair battery (8 forward + 4 reverse from the core pairs above) across 5 new models. Phase 10B tested 8 pairs under 3 pre-fill relation classes (on-axis, same-domain, unrelated).

Phase 11A reused the same 12-pair battery across 4 additional new models. Phase 11B screened 4 control candidates (accordion–stalactite, turmeric–trigonometry, barnacle–sonnet, magnesium–ballet) across 6 models. Phase 11C tested 6 pairs across a 2×2 waypoint/temperature grid (3 forward pairs: light–color, hot–cold, emotion–melancholy; plus their reverses) on 3 models (Claude, GPT, DeepSeek).

A.9 A.9 Model Inventory

ID	Display Name	Provider	OpenRouter ID	Cohort
claude	Claude Sonnet 4.6	Anthropic	anthropic/claude-sonnet-4.6	Original
gpt	GPT-5.2	OpenAI	openai/gpt-5.2	Original
grok	Grok 4.1 Fast	xAI	x-ai/grok-4.1-fast	Original
gemini	Gemini 3 Flash	Google	google/gemini-3-flash-preview	Original
minimax	MiniMax M2.5	MiniMax	minimax/minimax-m2.5	Phase 10A
kimi	Kimi K2.5	Moonshot AI	moonshotai/kimi-k2.5	Phase 10A
glm	GLM 5	Zhipu AI	z-ai/glm-5	Phase 10A (rate-limited)
qwen	Qwen 3.5 397B-A17B	Alibaba Cloud	qwen/qwen3.5-397b-a17b	Phase 10A
llama	Llama 3.1 8B Instruct	Meta	meta-llama/llama-3.1-8b-instruct	Phase 10A
deepseek	DeepSeek V3.2	DeepSeek	deepseek/deepseek-v3.2	Phase 11A
mistral	Mistral Large 3	Mistral AI	mistralai/mistral-large-2512	Phase 11A
cohere	Cohere Command A	Cohere	cohere/command-a	Phase 11A
llama4	Llama 4 Maverick	Meta	meta-llama/llama-4-maverick	Phase 11A

All models accessed via OpenRouter API. GLM 5 was blocked by upstream rate limiting and excluded from analysis. Llama 3.1 8B Instruct is the sole small-scale model, included to probe scale effects.

B Phase-by-Phase Results Summary

This appendix provides a condensed summary of each phase's design, findings, predictions, and prediction accuracy. It is intended for readers who want to trace the exploration-first methodology in detail.

B.1 Phase 1: Pilot and Prompt Selection

Runs: ~2,480 (4 models, 36 pairs, 20 runs each)

Key question: Do models produce consistent waypoint paths?

Design: Evaluated two prompt formats (semantic, direct) on 15 holdout pairs. Collected 20-run batches on 21 reporting pairs across 4 models (Claude Sonnet 4.6, GPT-5.2, Grok 4.1 Fast, Gemini 3 Flash). Tested both 5-waypoint and 10-waypoint conditions.

Primary findings: - Models have distinct conceptual gaits: Claude 0.578 vs GPT 0.258 (2.2x gap) [R1]. - Control validation: nonsense pairs produce near-zero Jaccard (0.062). - Polysemy sense differentiation: cross-pair Jaccard 0.011–0.062 [R3]. - 5-waypoint paths are genuine coarse-grainings of 10-waypoint paths (70.5% shared, 67.9% subsequence rate) [O10]. - Semantic prompt format selected over direct format.

Prediction accuracy: N/A (exploratory phase).

B.2 Phase 2: Reversals and Path Consistency

Runs: ~960 (840 reverse + 120 polysemy supplementary)

Key question: Is A-to-B navigation symmetric with B-to-A?

Design: Reversed all 21 reporting pairs across 4 models (20 runs each). Added 120 supplementary runs to correct the Phase 1 polysemy artifact (G1).

Primary findings: - Navigation is fundamentally asymmetric: mean directional asymmetry 0.811 [R2]. - 87% of 84 pair/model combinations show significant asymmetry (permutation test, $p < 0.05$). - Dual-anchor effect: U-shaped convergence profile [O2]. - Category symmetry predictions failed for 4 of 8 categories [G3].

Prediction accuracy: N/A (exploratory phase).

B.3 Phase 3: Positional Convergence and Transitive Path Structure

Runs: ~1,260 (~660 Phase 3B new runs; Phase 3A was pure analysis)

Key question: Do paths compose? Does the triangle inequality hold?

Design: Part A: pure analysis of Phase 2 reversal data for positional convergence. Part B: 8 concept triples (hierarchical, semantic-chain, polysemy-extend, random-control) with 10 runs per leg, testing transitivity and triangle inequality.

Primary findings: - Hierarchical transitivity 0.175 vs random 0.036 (4.9x, non-overlapping CIs) [R4]. - Triangle inequality holds in 90.6% of cases (29/32 triple/model combinations). - Bridge frequency model-dependent: "dog" 15–100%, "harmony" 0–100%. - "Energy" fails as hot–cold bridge [G4].

Prediction accuracy: N/A (exploratory phase).

B.4 Phase 4: Cross-Model Bridge Topology

Runs: ~1,600 (Part A: 0 new runs, pure analysis; Part B: ~1,520 new runs)

Key question: Do models agree on bridge concepts?

Design: Part A: analyzed inter-model bridge agreement across 6 non-control triples from Phase 3B. Part B: 8 targeted triples with 20 runs per leg across 4 models, testing specific bridge predictions.

Primary findings: - Three models agree, Gemini diverges (isolation index 0.136) [O1]. - Bridges are structural bottlenecks: "spectrum" 1.00 all models [R6]. - "Metaphor" fails at 0.00 all models [G5]. - Universal concrete bridging (spectrum 100%), universal abstract failure (metaphor 0%). - Triangle inequality replicates at 93.8%.

Prediction accuracy: 81.3% (26/32). The highest in the benchmark — characterization questions.

B.5 Phase 5: Cue-Strength and Dimensionality

Runs: ~3,720 new + 200 reused

Key question: Is bridge frequency graded by semantic cue strength?

Design: Part A: 4 concept families x 3 cue levels x 4 models. Part B: polysemy dimensionality with same-axis vs cross-axis triples. Part C: triple-anchor convergence (W-shape test).

Primary findings: - Cue-strength gradient real: 12/16 monotonic [R7]. All 4 failures in biological-growth family. - "Germination" > "plant": process > object [O4, G8]. - Forced crossing discovery (loan–bank–shore) [O5]. - "Fire" dead as bridge (too-central) [O6, G10]. - W-shape aggregate null [G11]. Gemini threshold hypothesis fails [G7].

Prediction accuracy: 42.9% (Phase 5A: 64.6%; Phase 5B: 31/48 but overall combined lower).

B.6 Phase 6: Navigational Salience and Bridge Position

Runs: ~2,080 new + 280 reused

Key question: Where do bridges appear in paths, and how does salience vary?

Design: Part A: salience mapping (40 runs/condition, 8 pairs, 4 models). Part B: forced-crossing asymmetry test (8 pairs). Part C: positional bridge scanning (10 pairs, 4 models).

Primary findings: - Salience distributions non-uniform: 31/32 conditions reject uniformity [O11]. - Forced-crossing asymmetry matches baseline (0.817 vs 0.811) [G12]. - Bridges anchor early (positions 1–2), not midpoint [O12]. - Peak-detection contrast 0.345 (CI excludes zero) [O13]. - Forced-crossing bridges positionally unstable (SD 1.71 vs 0.52) [G14]. - GPT highest entropy (3.44); Claude lowest (2.59).

Prediction accuracy: ~50% (structural predictions mixed).

B.7 Phase 7: Early Anchoring and Navigational Mechanics

Runs: ~2,360 new + 920 reused

Key question: Does pre-filling a waypoint causally affect the path?

Design: Part A: 8 pairs x 3 pre-fill conditions (incongruent, congruent, neutral) x 4 models. Part B: 8 curvature triangles. Part C: 10 too-central pairs.

Primary findings: - Pre-filling causally displaces bridges: mean 0.515, CI excludes zero [O15]. - Mechanism is associative primacy, not directional heading [G15]. - Bridge survival is bimodal: sadness/dog survive; harmony/germination collapse. - Taxonomic bridges resist displacement (0.140).

- Triangle inequality replicates at 90.6%. - Cross-model distance fails (r = 0.170) [O18, G17]. - Gradient > causal-chain bridges (0.730 vs 0.496).

Prediction accuracy: ~50% (structural replications succeed, mechanistic novel predictions fail).

B.8 Phase 8: Bridge Fragility and Gemini Deficit

Runs: ~2,690 new + 2,960 reused

Key question: Can bridge fragility be predicted from single variables?

Design: Part A: 14 fragility pairs testing route exclusivity (competitor count). Part B: 20 gradient/causal pairs testing Gemini gradient blindness. Part C: 16 distance pairs testing gait normalization.

Primary findings: - Route exclusivity fails: $\rho = 0.116$, opposite direction [G20]. - Gemini gradient blindness fails: interaction 0.046, null [G21]. - Gait normalization: zero improvement (r = 0.212 before and after) [G22]. - O17 replicates: gradient 0.770 vs causal 0.578. - New hypotheses proposed: H9 (dominance ratio), H10 (transformation blindness), H11 (facilitation).

Prediction accuracy: 24% (6/25). The benchmark's lowest — all novel mechanistic predictions fail.

B.9 Phase 9: Dominance, Transformation, and Facilitation

Runs: ~3,037 new + ~5,270 reused

Key question: Can bridge survival be explained by dominance ratio, transformation type, or facilitation crossover?

Design: Part A: 14 pairs testing dominance ratio as fragility predictor. Part B: 20 transformation/gradient pairs retesting Gemini. Part C: 14 facilitation pairs testing the crossover regression.

Primary findings: - Dominance ratio fails: $\rho = 0.157$ [G23]. - Transformation blindness fails with reversal: Gemini transformation mean 0.667 vs gradient 0.293 [G24]. - Crossover regression: slope correct direction but CI includes zero [G25]. - Pre-fill content is secondary modulator; 5/8 Phase 7A pairs replicate. - Marginal bridges show massive facilitation: mean 3.761x [O22].

Prediction accuracy: 20% (5/25). Mechanism ceiling confirmed.

B.10 Phase 10: Model Generality and Relation Classes

Runs: ~1,680 new + 778 reused

Key question: Do structural findings generalize to new models?

Design: Part A: 5 new models (Qwen, MiniMax, Kimi, GLM, Llama 8B) on 12 pairs with reliability probing. Part B: 4 core models on 8 pairs with 3 pre-fill relation classes.

Primary findings: - R1 replicates: new model gaits 0.298–0.508 [R1 extended]. - R2 replicates: all new models exceed 0.60 asymmetry threshold [R2 extended]. - Bridge frequency CI includes zero (0.721 vs 0.817): bridge structure generalizes. - Llama 8B sole outlier (bridge freq 0.200): scale effect [O25]. - Relation class matters (Friedman p = 0.034): unrelated < related [O26]. - Predicted relation class ordering wrong [G27]. G26 resurrected.

Prediction accuracy: 50% (9/18). Recovery from mechanism floor — more replication/structural predictions.

B.11 Phase 11: Expanded Generality, Control Revision, and Robustness

Runs: ~2,040 new + ~340 reused

Key question: Do findings survive more models and protocol variation?

Design: Part A: 4 new models (DeepSeek, Mistral, Cohere, Llama 4 Maverick) on 12 pairs. Part B: 4 control pair candidates across 6 models. Part C: 3 models x 2x2 waypoint/temperature grid.

Primary findings: - R1/R2 universal across all 12 models. Mistral record gait (0.747) [O28]. - Llama scale confirmed: Maverick 0.724 vs 8B 0.200. - All 4 control candidates fail (0/24 cells pass). Stapler-monsoon fails for all 12 models [G28]. - ANOVA: model identity $\eta^2 = 0.242$; protocol ≈ 0 [O30]. - Bridge frequency most robust (> 0.97 all conditions) [O31]. - Asymmetry resolution-dependent: 5wp 0.594, 9wp 0.669 [O32, G29].

Prediction accuracy: 44% (8/18). Intermediate — mix of replication and structural predictions.

B.12 Cumulative Statistics

Phase	New Runs	Prediction Accuracy	Dominant Prediction Type
1–3	~4,700	N/A (exploratory)	—
4	~1,600	81.3%	Characterization
5	~3,720	42.9%	Characterization + structure
6	~2,080	~50%	Structure
7	~2,360	~50%	Structure + mechanism
8	~2,690	24%	Mechanism
9	~3,037	20%	Mechanism
10	~1,680	50%	Replication + structure
11	~2,040	44%	Replication + robustness
Total	**~21,540**		

The prediction accuracy trajectory — 81.3% → 42.9% → ~50% → ~50% → 24% → 20% → 50% → 44% — tracks the shift between prediction types. Replication predictions succeed at ~80%; structural predictions at ~50%; mechanistic predictions at 0–15%.

C The Graveyard — Dead Ends, Failed Hypotheses, and Deprecated Approaches

This appendix documents 29 dead ends across 11 experimental phases. These entries are ordered chronologically and represent an honest accounting of the benchmark's learning curve. One entry (G26) was subsequently resurrected with additional data.

C.1 Summary Table

ID	Phase	Hypothesis	Predicted	Observed	Type
G1	1	Perfect polysemy differentiation	Jaccard = 0.000	Artifact (empty data)	Data error
G2	2	Monotonic convergence	Increasing overlap	U-shape (dual-anchor)	Model too simple
G3	2	Category symmetry (4 of 8)	High symmetry for antonyms, near-synonyms, controls	Asymmetry 0.596–0.986	Asymmetry is universal
G4	3	"Energy" as hot–cold bridge	Bridge frequency > 0	Near-zero (0.036 transitivity)	Association ≠ navigation
G5	4	"Metaphor" as language–thought bridge	Frequency 0.50–1.00	0.00 all models	Intuition ≠ navigation
G6	4	Concrete/abstract predicts Gemini	tree–forest–ecosystem > 0.40	Gemini 0.10	Category too coarse
G7	5	Gemini cue-strength threshold	Threshold highest	Threshold lowest (1.79)	Quantitative model fails
G8	5	Plant > germination (intuitive cue)	Plant near 100%	Germination wins (process > object)	Human intuition fails
G9	5	Dimensional independence of polysemy	Same-axis > cross-axis by 0.40	Cross-axis higher (forced crossings)	Constraint > association
G10	5	Fire as reliable control bridge	Reliable same-axis bridging	Near-zero all models	Too-central failure mode
G11	5	W-shape aggregate prediction	Bridge-present > bridge-absent	No aggregate signal	Aggregation kills signal
G12	6	Forced-crossing reduces asymmetry	0.50–0.70 asymmetry	0.817 (= baseline)	Asymmetry is structural
G13	6	Semantic distance predicts position	r significant	r = 0.239, p = 0.486	Early-anchoring overrides geometry
G14	6	Forced-crossing positional stability	SD < 0.8	SD = 1.71	Obligatory ≠ stable
G15	7	Directional-heading theory (strong form)	Incongruent > congruent displacement	Indistinguishable (0.515 vs 0.436)	Associative primacy dominates

ID	Phase	Hypothesis	Predicted	Observed	Type
G16	7	Polysemous curvature	Higher triangle excess	Difference 0.053, CI includes zero	Distance metric invalid
G17	7	Cross-model distance validity	r > 0.50	r = 0.170	Distances are model-dependent
G18	7	Too-central as binary category	Clean separation	CI includes zero; only fire/water qualify	Category based on intuition
G19	7	Forced-crossing resists pre-fill	Survival > 0.90	Survival 0.267	Obligatory ≠ robust
G20	8	Route exclusivity (competitor count)	$\rho < -0.70$	$\rho = 0.116$	Single variable fails
G21	8	Gemini gradient blindness	Gradient zeros ≥ 5/10	Causal zeros 6/10 (reversed)	Type-based approach wrong
G22	8	Gait normalization rescues distance	Normalized r > 0.50	r = 0.212 (0.000 improvement)	Ordinal disagreement
G23	9	Dominance ratio predicts fragility	$\rho > 0.50$	$\rho = 0.157$	Retrospective overfitting
G24	9	Gemini transformation blindness	Transformation deficit	Transformation advantage (reversed)	Third type-based failure
G25	9	Pre-fill crossover regression	Significant slope	CI includes zero (partial)	Too heterogeneous
G26	10	Bridge generalization fails	CI excludes zero	**Resurrected**: CI includes zero	Infrastructure artifact
G27	10	Relation class ordering	On-axis < unrelated < same-domain	Unrelated < on-axis ≈ same-domain	Related vs. unrelated is binary
G28	11	Control pair revision	≥ 2/4 candidates pass	0/24 cells pass	LLMs navigate between anything
G29	11	Asymmetry universality across protocol	Mean > 0.60 all conditions	5wp: 0.594; 9wp: 0.669	Resolution-dependent

C.2 Detailed Entries

C.2.1 G1 — Phase 1: Polysemy Cross-Pair Jaccard of 0.000 (Artifact)

Hypothesis: Perfect sense differentiation with cross-pair Jaccard of 0.000 for all three polysemy groups.

What happened: Holdout pairs had no pilot data — the zero Jaccard was comparing real data against empty sets.

Resolution: Phase 2 supplementary runs (120 forward runs) corrected values to 0.011–0.062. Sense differentiation is genuine and strong, but not perfect.

Lesson: Always verify the denominator. Zero is suspicious.

C.2.2 G2 — Phase 2: Monotonic Convergence Prediction

Hypothesis: Mirror-matching forward position i against reverse position *(5-i)* would show monotonically increasing overlap, based on the starting-point hypothesis.

What happened: U-shape instead — positions 1 and 5 both elevated (0.102, 0.129), valley in the middle (0.057–0.085). Both endpoints act as anchors.

Resolution: Replaced starting-point hypothesis with dual-anchor hypothesis (O2).

Lesson: Simple unidirectional models underestimate the influence of the target concept.

C.2.3 G3 — Phase 2: Category Symmetry Predictions

Hypothesis: Several categories would show high path symmetry: antonyms, near-synonyms, random controls, nonsense controls.

What happened: Observed asymmetry of 0.596 (antonyms), 0.665 (near-synonyms), 0.908 (random), 0.986 (nonsense). Four of eight category symmetry predictions failed.

Resolution: Established that asymmetry is pervasive and structural (R2). The 0.811 mean was far higher than predicted.

Lesson: Asymmetry is the default. Symmetry is the exception, produced only by strong semantic bridges.

C.2.4 G4 — Phase 3: "Energy" as Hot–Cold Bridge

Hypothesis: "Energy," associated with both "hot" and "cold," would appear as a bridge concept.

What happened: Near-zero transitivity (0.036). Hot–cold paths traverse the temperature gradient (warm, tepid, cool); hot–energy paths traverse physics terminology.

Resolution: Established that the temperature axis and physics-of-energy axis are parallel non-intersecting dimensions. Association is not navigation.

Lesson: Semantic association does not guarantee navigational intermediacy.

C.2.5 G5 — Phase 4: "Metaphor" as Language–Thought Bridge

Hypothesis: Bridge frequency of 0.50–1.00 for metaphor on language–thought. Metaphor is arguably the quintessential bridge between language and thought.

What happened: 0.00 for all four models. Zero appearances out of 80 runs. Yet transitivity was high (Claude 0.437) through "communication," "meaning," "cognition."

Resolution: Sharpened the bridge definition: bridges must name the primary axis of connection, not merely be associated with both endpoints.

Lesson: The biggest single prediction miss in the benchmark. Intuitive semantic importance has zero predictive power for navigational bridge behavior.

C.2.6 G6 — Phase 4: Concrete/Abstract as Gemini's Fragmentation Boundary

Hypothesis: The concrete/abstract axis predicts Gemini's bridge failures. Predicted Gemini success on tree–forest–ecosystem (concrete, hierarchical).

What happened: Gemini observed at 0.10. Bank–deposit–savings (abstract/financial) succeeds at 1.00.

Resolution: Replaced with frame-crossing hypothesis. Forest-to-ecosystem requires a scale transition (botanical to ecological), not an abstractness gradient.

Lesson: Concrete/abstract is too coarse. Frame membership is the relevant variable.

C.2.7 G7 — Phase 5: Gemini Cue-Strength Threshold

Hypothesis: Gemini has a measurably higher cue-strength threshold than other models.

What happened: Gemini's logistic threshold (1.79) is the lowest, not highest. CI on difference includes zero.

Resolution: The quantitative threshold model is falsified. The qualitative frame-crossing model survives.

Lesson: A correct qualitative model can generate incorrect quantitative predictions.

C.2.8 G8 — Phase 5: Plant > Germination (Intuitive Cue Strength)

Hypothesis: "Plant" (rated very-high cue strength) would bridge seed–garden at near-100%.

What happened: Complete inversion. Germination outperforms plant in all 4 models. Claude: 1.00 vs 0.00.

Resolution: Discovered the distinction between associative strength and navigational salience. Process-naming ("germination") outperforms object-naming ("plant").

Lesson: Human intuition about "most related" does not predict navigational routing. Calibrate empirically.

C.2.9 G9 — Phase 5: Dimensional Independence of Polysemy

Hypothesis: Same-axis bridge frequency would exceed cross-axis by at least 0.40.

What happened: Cross-axis mean (0.291) *higher* than same-axis (0.220). Driven by loan–bank–shore at 0.95–1.00.

Resolution: Polysemy creates forced crossing points, not dimensional walls. Cross-axis paths have no alternative to the polysemous bridge.

Lesson: The most reliable bridges are bottlenecks, not associations. Constraint produces reliability.

C.2.10 G10 — Phase 5: "Fire" as Control Bridge

Hypothesis: "Fire" would reliably bridge spark–ash as a non-polysemous control.

What happened: Near-zero frequency across all models. Fire is too central — both endpoints already imply it.

Resolution: Established "too-central" as a new bridge failure mode (O6), distinct from off-axis (G5) and fragmentation.

Lesson: When both endpoints directly imply the candidate bridge, naming it adds no navigational information.

C.2.11 G11 — Phase 5: W-Shape Aggregate Prediction

Hypothesis: Bridge-present pairs show higher W-shape contrast than bridge-absent pairs in aggregate.

What happened: Bridge-present mean 0.0027, bridge-absent 0.0050. No aggregate signal.

Resolution: Effect is model-dependent and position-variable. Phase 6C redesigned the test as peak-detection.

Lesson: Aggregation can kill real signals when the effect is interaction-dependent.

C.2.12 G12 — Phase 6: Forced-Crossing Reduces Asymmetry

Hypothesis: Forced crossings would reduce asymmetry below the 0.811 baseline to 0.50–0.70.

What happened: Forced-crossing asymmetry 0.817, indistinguishable from baseline.

Resolution: Asymmetry is a global structural property, not modulated by local bottlenecks.

Lesson: A shared waypoint does not produce shared paths.

C.2.13 G13 — Phase 6: Semantic Distance Predicts Bridge Position

Hypothesis: Bridge position correlates with semantic distance ratio d(A,bridge)/d(A,C).

What happened: $r = 0.239$, $p = 0.486$. Irrelevant because bridges overwhelmingly anchor at positions 1–2.

Resolution: Position is determined by heading selection, not geometry.

Lesson: Navigational mechanics override geometric intuition.

C.2.14 G14 — Phase 6: Forced-Crossing Positional Stability

Hypothesis: Forced-crossing bridges have low positional variance (SD < 0.8).

What happened: SD = 1.71 vs 0.52 for non-forced. Forced crossings are the most positionally unstable.

Resolution: Frame-crossing timing is model-dependent. Obligatory bridges are mandatory but not positionally fixed.

Lesson: Obligatory $\neq$ stable.

C.2.15 G15 — Phase 7: Directional-Heading Theory (Strong Form)

Hypothesis: Incongruent pre-fills would produce more displacement than congruent pre-fills.

What happened: Displacement magnitudes indistinguishable: incongruent 0.515, congruent 0.436, neutral 0.536. CIs overlap.

Resolution: The mechanism is primarily associative primacy (whatever occupies position 1 anchors the trajectory) with secondary content modulation.

Lesson: Position matters more than content for displacement magnitude.

C.2.16 G16 — Phase 7: Polysemous Curvature

Hypothesis: Polysemous-vertex triangles show higher curvature (triangle excess) than non-polysemous triangles.

What happened: Difference 0.053, CI includes zero. Categories indistinguishable.

Resolution: Polysemy affects *which* concepts appear, not *how far apart* they are. Also, the cross-model distance metric failed validity (G17).

Lesson: Always validate the instrument before interpreting measurements.

C.2.17 G17 — Phase 7: Cross-Model Distance Validity

Hypothesis: Cross-model navigational distance correlation would exceed r = 0.50.

What happened: r = 0.170. Models assign wildly different distances to the same pairs. Some anti-correlate (Grok–Gemini r = –0.580).

Resolution: Navigational distances are fundamentally model-dependent. Blocks cross-model curvature comparison.

Lesson: Gait differences (R1) contaminate all attempts at shared distance metrics.

C.2.18 G18 — Phase 7: Too-Central as Binary Category

Hypothesis: "Too-central" bridges (fire, tree, dough) would show frequency < 0.15, cleanly separated from "obvious-useful" bridges.

What happened: Too-central mean 0.496. "Tree" has frequency 1.000 for 3/4 models.

Resolution: Only fire/water meet the strict operational definition. The original categorization was based on intuition.

Lesson: Human intuition about "too obvious" does not match model navigation.

C.2.19 G19 — Phase 7: Forced-Crossing Resists Pre-Fill

Hypothesis: "Bank" on loan–shore would survive pre-filling at > 0.90 frequency.

What happened: Survival 0.267, positional shift 0.791.

Resolution: Forced crossings are obligatory only under unconstrained navigation. Pre-filling displaces even "obligatory" bottlenecks.

Lesson: Obligatory $\neq$ robust under perturbation.

C.2.20 G20 — Phase 8: Route Exclusivity (Competitor Count)

Hypothesis: Bridges with more competitors show lower pre-fill survival ($\rho < -0.70$).

What happened: $\rho = 0.116$ (opposite direction). Sadness has 8 competitors yet highest survival.

Resolution: Competitor count ignores the distribution of traffic. A bridge with 8 weak competitors is more robust than one with 3 strong competitors.

Lesson: Quantity $\neq$ quality in the competitive landscape.

C.2.21 G21 — Phase 8: Gemini Gradient Blindness

Hypothesis: Gemini selectively fails on gradient-midpoint bridges (interaction ≥ 0.20).

What happened: Interaction 0.046 (null). Gemini's zeros concentrate on causal-chain pairs (6/10), not gradient pairs (1/10).

Resolution: Proposed transformation-chain blindness (G24).

Lesson: When the result is opposite to prediction, the categorization itself may be wrong.

C.2.22 G22 — Phase 8: Gait Normalization Rescues Distance

Hypothesis: Normalizing by model-specific baselines would raise cross-model distance correlation to r > 0.50.

What happened: Raw r = 0.212, normalized r = 0.212. Zero improvement. Pearson correlation is invariant to scalar division.

Resolution: Cross-model disagreement is ordinal, not scalar. Model-independent geometry is blocked.

Lesson: Always check whether a proposed fix is mathematically capable of addressing the pattern.

C.2.23 G23 — Phase 9: Dominance Ratio Predicts Fragility

Hypothesis: Bridge-to-competitor frequency ratio predicts survival ($\rho > 0.50$).

What happened: $\rho = 0.157$. Warm (ratio 1.00) is destroyed; fermentation (ratio 1.07) is bulletproof.

Resolution: Pre-fill content matters more than dominance ratio. Both single-variable structural predictors (G20, G23) have failed.

Lesson: Retrospective signals on small samples (N = 8) do not generalize.

C.2.24 G24 — Phase 9: Gemini Transformation-Chain Blindness

Hypothesis: Gemini shows a transformation deficit (mean < 0.30).

What happened: Gemini transformation mean 0.667 (vs predicted < 0.30). Result reversed from prediction.

Resolution: Third type-based Gemini characterization fails. Whatever pair type is designated as "hard," Gemini does relatively better on it.

Lesson: When a hypothesis fails backward in two successive tests, the classification itself is wrong. Abandon the type-based approach.

C.2.25 G25 (partial) — Phase 9: Pre-Fill Crossover Regression

Hypothesis: Significant negative slope predicting survival from unconstrained frequency; crossover at 0.40–0.50.

What happened: Slope –3.355 (correct direction) but CI includes zero. Crossover at 0.790 (far from predicted).

Resolution: Qualitative threshold survives (marginal facilitation, dominant displacement). Quantitative regression fails.

Lesson: Qualitative predictions succeed; quantitative predictions fail. The benchmark's defining pattern.

C.2.26 G26 — Phase 10: Bridge Generalization Fails (RESURRECTED)

Hypothesis: Bridge frequency CI would include zero for new models.

What happened initially: With Llama 8B only (1 model), CI excluded zero. Entered graveyard.

What happened after: Timeout relaxation (60s → 300s) recovered 3 additional models. With 4-model cohort, CI includes zero. Bridge structure generalizes.

Resolution: The original finding was an infrastructure artifact. Llama 8B's low bridge frequency (0.200) is a scale effect, not a generalization failure.

Lesson: Infrastructure limitations can masquerade as substantive findings. The sole resurrection in the graveyard.

C.2.27 G27 — Phase 10: Relation Class Ordering

Hypothesis: On-axis < unrelated < same-domain in $\geq 5/8$ pairs.

What happened: Unrelated (0.388) < on-axis (0.643) $\approx$ same-domain (0.708). Only 1/8 pairs showed predicted ordering.

Resolution: The distinction is binary (related vs. unrelated), not three-way. Related pre-fills maintain navigational context.

Lesson: Coarse categorical distinctions capture phenomena; fine-grained orderings fail.

C.2.28 G28 — Phase 11: Control Pair Revision

Hypothesis: At least 2/4 new control candidates would pass screening.

What happened: 0/24 cells pass. Top frequencies 0.60–1.00. Retrospective: stapler–monsoon fails for all 12 models.

Resolution: R5 single-pair strict-threshold validation is fundamentally inadequate. LLMs navigate between any two concepts.

Lesson: LLMs have richer associative connectivity than human intuition predicts.

C.2.29 G29 — Phase 11: Asymmetry Universality Across Protocol

Hypothesis: Mean asymmetry > 0.60 under all protocol conditions.

What happened: 5-waypoint conditions: 0.594 (below threshold). 9-waypoint conditions: 0.669–0.684 (above).

Resolution: Asymmetry is resolution-dependent. R2 qualified: robust at ≥ 7 waypoints.

Lesson: Path length affects measurement sensitivity. Threshold calibration is resolution-dependent.

D Prompt Templates

This appendix documents the exact prompt templates used for waypoint elicitation across the benchmark's 11 phases.

D.1 D.1 Standard Elicitation Prompts

Two prompt formats were developed and evaluated during Phase 1 prompt selection. The semantic format was selected as the primary template based on higher extraction accuracy and more interpretable waypoints on the holdout set.

D.1.1 Semantic Format (Primary)

```
Imagine walking through conceptual space from "{A}" to "{B}".
What {N} landmarks do you pass along the way?
List only the landmark concepts, one per line, numbered 1 through {N}.
Do not include the starting or ending concepts.
```

D.1.2 Direct Format (Fallback)

```
List exactly {N} intermediate concepts that form a path from "{A}" to "{B}".
Respond with only the concepts, one per line, numbered 1 through {N}.
Do not include the starting or ending concepts.
```

The direct format was used as a fallback for models that produced unparseable outputs under the semantic prompt, notably during Phase 10A reliability probing of new models.

Default parameters. Unless otherwise noted: N = 7 waypoints, temperature = 0.7, 10–20 independent runs per (model, pair, direction) condition. Phases 1–2 used 20 runs; later phases standardized to 10 runs.

D.2 D.2 Pre-Fill Elicitation Prompt

Introduced in Phase 7A for causal intervention experiments. The first waypoint is pre-specified, and the model generates the remaining N-1 waypoints.

```
The first intermediate concept between "{A}" and "{B}" is "{pre-fill}".
List exactly {N-1} more intermediate concepts that continue the path
from "{pre-fill}" to "{B}".
Respond with only the concepts, one per line, numbered 1 through {N-1}.
Do not include "{A}", "{pre-fill}", or "{B}" in your list.
```

With default parameters, N-1 = 6 (the pre-fill occupies position 1, and the model generates positions 2–7).

D.3 D.3 System Prompt

All elicitations used the following system prompt:

```
You are a helpful assistant that provides single-word or short-phrase
responses representing conceptual landmarks.
```

This system prompt was consistent across all models and phases. No chain-of-thought or reasoning instructions were included; the waypoint task is presented as a direct generation task.

D.4 D.4 API Configuration

All models were accessed via the OpenRouter API (`https://openrouter.ai/api/v1/chat/completions`) with the following parameters:

- `max_tokens`: 256
- `temperature`: 0.7 (default; varied in Phase 11C robustness analysis)
- `top_p`: 1.0
- `frequency_penalty`: 0
- `presence_penalty`: 0

Phase 10A introduced a `--patient` CLI flag that extended the request timeout from 60 seconds to 300 seconds, recovering 3 of 5 new models that failed the default timeout on OpenRouter latency.

D.5 D.5 Response Parsing

Model responses were parsed using a multi-strategy extraction pipeline attempting, in order:

1. JSON array extraction
2. Numbered list parsing (e.g., "1. concept")
3. Bullet list parsing (e.g., "- concept")
4. Comma-separated value extraction
5. Arrow-separated extraction (recognizing `->`, `=>`, `-->`, `==>`)
6. Line-by-line fallback

After parsing, responses passed through the canonicalization pipeline (Section 2.3 of the main text): lowercase, article stripping, lemmatization via the compromise NLP library, and phrase normalization.

D.6 D.6 Prompt Variations Across Phases

Phase	Variation	Reason
1	Both formats tested on holdout pairs	Prompt format selection
1–6	Semantic format only	Standard elicitation
7A–10B	Pre-fill variant added	Causal intervention
10A	Direct format fallback for some models	Parse reliability
11C	N varied (5, 7, 9); temperature varied (0.5, 0.7, 0.9)	Robustness analysis

No other prompt variations were introduced. The consistency of the prompt template across 11 phases is a deliberate design choice supporting cross-phase comparability.

E Statistical Methods

This appendix documents the statistical procedures used throughout the benchmark. All analyses were implemented in TypeScript (Bun runtime) unless otherwise noted.

E.1 E.1 Bootstrap Confidence Intervals

Confidence intervals for Jaccard-based metrics (gait consistency, path asymmetry, bridge frequency, transitivity) are 95% percentile bootstrap intervals computed from 1,000 resamples with a seeded PRNG (seed = 42) for reproducibility.

Procedure. For a metric computed over n independent runs:

1. Resample n runs with replacement (1,000 times).
2. Recompute the metric on each resample.
3. Report the 2.5th and 97.5th percentiles as the 95% CI.

Limitation. Efron and Tibshirani (1993) recommend 5,000–10,000 resamples for publication-quality confidence intervals. The benchmark uses 1,000 for computational efficiency during the exploration-first workflow. CIs should be interpreted as approximate; the qualitative conclusions (whether a CI includes zero) are robust to resample count, but exact CI bounds may shift slightly with more resamples.

E.2 E.2 Permutation Tests for Asymmetry

Statistical significance of path asymmetry is assessed via permutation test. For a given (model, pair) combination with n_f forward runs and n_r reverse runs:

1. Pool all $n_f + n_r$ runs.
2. Randomly assign n_f to "forward" and n_r to "reverse" (1,000 permutations).
3. Compute the mean cross-direction Jaccard for each permutation.
4. The p-value is the fraction of permutations producing a cross-direction Jaccard as low or lower than the observed value, using the Phipson and Smyth (2010) correction to avoid exact-zero p-values:

$$p = \frac{b+1}{m+1}$$

where b is the count of permutations with test statistic as extreme as observed and m is the total number of permutations.

Limitation. With 1,000 permutations, the minimum achievable p-value is approximately 0.001 (after the Phipson–Smyth correction). This limits resolution for Bonferroni-corrected thresholds. The 87% significance rate reported in Section 5.1 (73 of 84 combinations at $p < 0.05$) was computed without multiple-comparison correction. With Bonferroni correction (threshold $0.05/84 \approx 0.0006$), the percentage would be lower, and the permutation test's resolution is insufficient to distinguish near this corrected threshold. The qualitative conclusion — that the vast majority of combinations show genuine asymmetry — is robust, but the specific percentage should be interpreted conservatively.

E.3 E.3 Friedman Test and Post-Hoc Comparisons

The effect of pre-fill relation class on bridge survival (Section 6.3) was tested using the Friedman test, a non-parametric repeated-measures alternative to one-way ANOVA. The 8 concept pairs serve as blocks, and the 3 relation classes (on-axis, same-domain, unrelated) serve as treatments.

Procedure:

1. For each pair, rank the 3 survival rates.
2. Compute the Friedman chi-squared statistic from the rank sums.
3. Assess significance against the chi-squared distribution with k - 1 = 2 degrees of freedom.

Result: $\chi^2 = 6.750$, $df = 2$, $p = 0.034$.

Post-hoc comparisons. Pairwise Wilcoxon signed-rank tests:

Comparison	p-value	Survives Bonferroni ($0.05/3 = 0.017$)?
On-axis vs. unrelated	0.025	No
Same-domain vs. unrelated	0.036	No
On-axis vs. same-domain	0.889	N/A

Neither pairwise comparison survives Bonferroni correction for 3 comparisons. The qualitative conclusion — unrelated pre-fills are more disruptive than related ones — is supported by the omnibus Friedman test but the specific pairwise p-values are suggestive rather than confirmatory.

E.4 E.4 ANOVA for Multiverse Robustness

The Phase 11C robustness analysis used a factorial ANOVA decomposing gait variance across model identity, waypoint count, and temperature.

Design: 3 models (Claude, GPT, DeepSeek) × 2 waypoint counts (5, 9) × 2 temperatures (0.5, 0.9), plus the 7-waypoint / 0.7-temperature baseline assembled from prior-phase data.

Procedure:

1. Compute cell means: mean Jaccard for each (model, waypoint count, temperature, pair) combination.
2. Compute sum of squares for each factor and interaction.
3. Express as eta-squared ($\eta^2 = \mathrm{SS}_{\mathrm{factor}} / \mathrm{SS}_{\mathrm{total}}$).
4. Approximate p-values using the chi-squared distribution (ignoring denominator degrees of freedom).

Results:

Factor	η^2	Approx. p
Model identity	0.242	≈ 0.001
Waypoint count	0.008	≈ 0.520
Temperature	0.002	≈ 0.743
Waypoint × Temperature	≈ 0.000	≈ 0.886

Limitations:

1. P-value approximation.* The chi-squared approximation ignores denominator degrees of freedom. Exact F-test p-values would require a proper mean squares decomposition. The eta-squared effect sizes are more reliable than the exact p-values.
2. *Independence assumption.* Per-pair cells are treated as independent observations without repeated-measures modeling. A mixed-effects model with pair as a random effect would be more appropriate for confirmatory inference.
3. *Coverage.* Only 3 models were tested. The conclusion that model identity dominates protocol variation is supported within this subset but should be extended with caution.
4. *Baseline sparsity.* DeepSeek has no prior-phase data; its 7wp-t0.7 baseline cells are empty. This affects the baseline condition but not the four new conditions.

The analysis should be understood as descriptive/exploratory rather than confirmatory. The qualitative conclusion — that model identity accounts for far more variance than protocol parameters — is robust to these approximations, as the effect size gap (0.242 vs. 0.008 and 0.002) is large.

E.5 E.5 Chi-Squared Goodness-of-Fit Test

Departure from uniform waypoint distributions (Section 5.6) is assessed via chi-squared goodness-of-fit test.

Procedure. For each (model, pair) condition:

1. Count unique waypoints across all runs, treating each waypoint as appearing once per run.
2. Compute expected frequencies under the null hypothesis that all waypoints are equally likely.

3. Compute the chi-squared statistic: $\chi^2 = \sum(O_i - E_i)^2/E_i$.
4. Compute p-value using the Wilson–Hilferty normal approximation.
5. Apply Bonferroni correction at the pair level (dividing the significance threshold by the number of pairs tested).

Note on naming. The codebase function implementing this test is named `ksTestUniform` (after Kolmogorov–Smirnov), but the implementation is chi-squared goodness-of-fit, which is more appropriate for categorical frequency data. The paper uses the correct name (chi-squared).

E.6 E.6 Kendall's W (Coefficient of Concordance)

Gait rank stability across protocol conditions (Section 9.3) is assessed via Kendall's W:

$$W = \frac{12 \sum_{i=1}^{n} (R_i - \bar{R})^2}{k^2(n^3 - n)}$$

where R_i is the rank sum for model i across k conditions, and n is the number of models. W = 1 indicates perfect agreement across conditions; W = 0 indicates no agreement. The observed W = 0.840 indicates strong but imperfect rank stability.

E.7 E.7 Bridge Frequency Comparison (Confidence Interval on Difference)

The structure/content/scale hierarchy (Section 8.2) compares aggregate bridge frequency between the original cohort and expanded cohorts using bootstrap CIs on the difference:

1. For each cohort, compute mean bridge frequency across all (model, pair) conditions.
2. Bootstrap the difference (original mean − new mean) with 1,000 resamples.
3. If the 95% CI includes zero, the cohorts are statistically indistinguishable.

Phase 10A: diff = −0.096, CI [−0.241, 0.064] (includes zero). Phase 11A combined: diff = −0.100, CI [−0.286, 0.089] (includes zero).

E.8 E.8 Canonicalization and Its Statistical Implications

All metrics operate on canonicalized waypoint tokens. The canonicalization pipeline (Section 2.3) handles morphological variants but not synonyms. This has systematic statistical implications:

- **Jaccard similarity** is computed on token identity after canonicalization. Semantically equivalent but lexically distinct waypoints ("melody" vs. "tune," "cemetery" vs. "graveyard") are treated as completely different. All Jaccard-based metrics therefore *underestimate* true conceptual overlap.
- **Gait consistency** is a lower bound on true navigational consistency.
- **Path asymmetry** is an upper bound on true conceptual asymmetry.
- **Bridge frequency** may miss bridges expressed through synonyms rather than exact (or morphologically related) tokens.

The magnitude of this bias is bounded by the canonicalization (which collapses morphological variants) but is otherwise unknown. Embedding-based semantic similarity would help bound this effect but was deferred due to API cost.

E.9 E.9 Multiple Comparison Corrections

The benchmark applies Bonferroni correction in two contexts:

1. **Uniformity tests** (Section 5.6): 8 pairs tested per model; threshold adjusted to $0.05/8 = 0.00625$.
2. **Asymmetry significance** (Section 5.1): 84 pair/model combinations; the 87% significance rate is reported without correction. With Bonferroni ($0.05/84 \approx 0.0006$), the percentage would be lower.

No correction is applied to the phase-level prediction accuracy tracking (Sections 7.1, 7.2), which is descriptive rather than inferential.

For the Friedman post-hoc comparisons (Section 6.3, Appendix E.3), Bonferroni correction for 3 pairwise comparisons (threshold 0.017) is documented but results in neither surviving comparison reaching significance. The paper reports both corrected and uncorrected results for transparency.

F Robustness Analysis Details

This appendix provides the full per-model, per-pair, per-condition breakdown from the Phase 11C multiverse robustness experiment. Section 9 of the main text presents aggregated results; this appendix provides the underlying data.

F.1 F.1 Experimental Design

Models: Claude Sonnet 4.6, GPT-5.2, DeepSeek V3.2

Pairs (forward + reverse): - light → color / color → light (bridge: spectrum) - hot → cold / cold → hot (bridge: warm) - emotion → melancholy / melancholy → emotion (bridge: sadness)

Conditions (2×2 grid + baseline):

Condition	Waypoints	Temperature
Baseline (7wp-t0.7)	7	0.7
5wp-t0.5	5	0.5
5wp-t0.9	5	0.9
9wp-t0.5	9	0.5
9wp-t0.9	9	0.9

Runs: 1,080 new (4 conditions × 3 models × 6 pairs × 15 reps) + ∼340 reused baseline runs from prior phases.

Baseline data limitation: The 7wp-t0.7 baseline is assembled from prior-phase data (Phases 1, 2, 6A). DeepSeek, added in Phase 11A, has no prior-phase data — its baseline cells are empty. Additionally, reverse-direction data for some pairs is missing from prior phases. Baseline values for DeepSeek should be treated as missing, not zero.

F.2 F.2 Gait (Intra-Model Jaccard) by Model and Condition

Model	7wp-t0.7	5wp-t0.5	5wp-t0.9	9wp-t0.5	9wp-t0.9
Claude	0.676	0.794	0.772	0.699	0.624
GPT	0.378	0.413	0.477	0.382	0.365
DeepSeek	0.000*	0.582	0.492	0.592	0.597

*DeepSeek baseline is empty (no prior-phase data); reported as 0.000 by the analysis script.

Observations: - Claude shows the highest gait in every condition (0.624–0.794). - GPT shows the lowest gait in all new conditions (0.365–0.477). - DeepSeek occupies the middle position in all new conditions (0.492–0.597). - 5wp-t0.5 produces the highest gait for both Claude (0.794) and DeepSeek (0.582) — fewer waypoints and lower temperature reduce navigational variability. - 9wp-t0.9 produces the lowest gait for Claude (0.624) — more waypoints and higher temperature maximize variability.

F.2.1 Gait Rank-Order Stability

Condition	Rank 1	Rank 2	Rank 3
7wp-t0.7 (baseline)	Claude (0.676)	GPT (0.378)	DeepSeek (0.000*)
5wp-t0.5	Claude (0.794)	DeepSeek (0.582)	GPT (0.413)
5wp-t0.9	Claude (0.772)	DeepSeek (0.492)	GPT (0.477)
9wp-t0.5	Claude (0.699)	DeepSeek (0.592)	GPT (0.382)
9wp-t0.9	Claude (0.624)	DeepSeek (0.597)	GPT (0.365)

Kendall's W = 0.840. Claude is always first. The baseline anomaly (GPT > DeepSeek) reflects DeepSeek's missing baseline data, not genuinely lower consistency.

F.3 F.3 Asymmetry by Model, Pair, and Condition

F.3.1 Full Breakdown

Model	Condition	Pair	Asymmetry	95% CI
Claude	7wp-t0.7	hot–cold	0.726	[0.697, 0.750]
Claude	7wp-t0.7	emotion–melancholy	1.000	[1.000, 1.000]
GPT	7wp-t0.7	hot–cold	0.471	[0.414, 0.522]
GPT	7wp-t0.7	emotion–melancholy	1.000	[1.000, 1.000]
DeepSeek	7wp-t0.7	hot–cold	1.000	[1.000, 1.000]
DeepSeek	7wp-t0.7	emotion–melancholy	1.000	[1.000, 1.000]
Claude	5wp-t0.5	hot–cold	0.750	[0.750, 0.750]
Claude	5wp-t0.5	emotion–melancholy	0.512	[0.466, 0.550]
GPT	5wp-t0.5	hot–cold	0.434	[0.372, 0.503]
GPT	5wp-t0.5	emotion–melancholy	0.520	[0.461, 0.568]
DeepSeek	5wp-t0.5	hot–cold	0.665	[0.641, 0.689]
DeepSeek	5wp-t0.5	emotion–melancholy	0.684	[0.644, 0.719]
Claude	5wp-t0.9	hot–cold	0.726	[0.690, 0.750]

Model	Condition	Pair	Asymmetry	95% CI
Claude	5wp-t0.9	emotion–melancholy	0.519	[0.473, 0.562]
GPT	5wp-t0.9	hot–cold	0.392	[0.354, 0.435]
GPT	5wp-t0.9	emotion–melancholy	0.555	[0.445, 0.654]
DeepSeek	5wp-t0.9	hot–cold	0.653	[0.621, 0.684]
DeepSeek	5wp-t0.9	emotion–melancholy	0.712	[0.667, 0.760]
Claude	9wp-t0.5	hot–cold	0.671	[0.647, 0.696]
Claude	9wp-t0.5	emotion–melancholy	0.625	[0.583, 0.662]
GPT	9wp-t0.5	hot–cold	0.514	[0.460, 0.562]
GPT	9wp-t0.5	emotion–melancholy	0.682	[0.645, 0.717]
DeepSeek	9wp-t0.5	hot–cold	0.783	[0.760, 0.804]
DeepSeek	9wp-t0.5	emotion–melancholy	0.740	[0.708, 0.766]
Claude	9wp-t0.9	hot–cold	0.655	[0.620, 0.690]
Claude	9wp-t0.9	emotion–melancholy	0.612	[0.565, 0.656]
GPT	9wp-t0.9	hot–cold	0.596	[0.555, 0.638]
GPT	9wp-t0.9	emotion–melancholy	0.725	[0.684, 0.769]
DeepSeek	9wp-t0.9	hot–cold	0.760	[0.732, 0.785]
DeepSeek	9wp-t0.9	emotion–melancholy	0.754	[0.720, 0.784]

F.3.2 Condition Means

Condition	Mean Asymmetry	95% CI	> 0.60?
7wp-t0.7 (baseline)*	0.599	[0.471, 0.726]	no
5wp-t0.5	0.594	[0.512, 0.675]	no
5wp-t0.9	0.593	[0.502, 0.688]	no
9wp-t0.5	0.669	[0.597, 0.733]	**yes**
9wp-t0.9	0.684	[0.631, 0.736]	**yes**

*Baseline from 2 valid cells only (Claude and GPT on hot–cold).

Key pattern: The waypoint-count effect is consistent across all three models. At 9 waypoints, every model×pair combination shows asymmetry ≥ 0.514; at 5 waypoints, GPT hot–cold drops to 0.392–0.434. Temperature has negligible effect: within each waypoint count, 5wp-t0.5 (0.594) $\approx$ 5wp-t0.9 (0.593) and 9wp-t0.5 (0.669) $\approx$ 9wp-t0.9 (0.684).

F.4 F.4 Bridge Frequency by Pair, Model, and Condition

F.4.1 Full Breakdown

light → color (bridge: spectrum)

Model	7wp-t0.7	5wp-t0.5	5wp-t0.9	9wp-t0.5	9wp-t0.9
Claude	1.000	1.000	1.000	1.000	1.000
GPT	0.975	1.000	1.000	1.000	1.000
DeepSeek	0.000*	0.933	0.800	1.000	1.000

Model	7wp-t0.7	5wp-t0.5	5wp-t0.9	9wp-t0.5	9wp-t0.9

hot → cold (bridge: warm)

Model	7wp-t0.7	5wp-t0.5	5wp-t0.9	9wp-t0.5	9wp-t0.9
Claude	1.000	1.000	1.000	1.000	1.000
GPT	1.000	1.000	1.000	1.000	1.000
DeepSeek	0.000*	1.000	1.000	1.000	1.000

emotion → melancholy (bridge: sadness)

Model	7wp-t0.7	5wp-t0.5	5wp-t0.9	9wp-t0.5	9wp-t0.9
Claude	1.000	1.000	1.000	0.933	1.000
GPT	0.975	1.000	1.000	1.000	0.867
DeepSeek	0.000*	1.000	1.000	1.000	1.000

*DeepSeek 7wp-t0.7 entries are 0.000 due to missing baseline data, not actual zero bridge frequency.

F.4.2 Condition Means (Excluding DeepSeek Baseline)

Condition	Mean Bridge Freq	95% CI
7wp-t0.7 (baseline)	0.992	[0.983, 1.000]
5wp-t0.5	0.993	[0.978, 1.000]
5wp-t0.9	0.978	[0.933, 1.000]
9wp-t0.5	0.993	[0.978, 1.000]
9wp-t0.9	0.985	[0.956, 1.000]

All conditions exceed 0.97. The two lowest individual cells are DeepSeek's spectrum at 5wp-t0.9 (0.800) and GPT's sadness at 9wp-t0.9 (0.867). Even these represent only minor dips from ceiling.

F.5 F.5 ANOVA Interaction Test

Factorial analysis testing gait (intra-model Jaccard) against model identity, waypoint count, temperature, and their interaction. Data from the four new conditions only (baseline excluded due to DeepSeek sparsity).

Factor	η^2 (SS Proportion)	p-value (approx.)
Model identity	0.242	≈ 0.001
Waypoint count	0.008	≈ 0.520
Temperature	0.002	≈ 0.743

Factor	η^2 (SS Proportion)	p-value (approx.)
Waypoint × Temperature	≈ 0.000	≈ 0.886

Methodological note: p-values use a chi-squared approximation to the F-distribution (ignoring denominator degrees of freedom) and treat per-pair cells as independent without repeated-measures modeling. Only 3 models were tested. The effect sizes (η^2) are more reliable than the exact p-values. The qualitative conclusion — that model identity dominates while protocol parameters are null — is robust to the approximation, but the analysis should be understood as descriptive/exploratory rather than confirmatory.

F.6 F.6 Waypoint Scaling

Fraction of waypoints shared across different path lengths, computed as mean Jaccard similarity between the waypoint sets of matched pairs at different waypoint counts.

Comparison	Shared Fraction
5wp vs 7wp	0.761
7wp vs 9wp	0.504
5wp vs 9wp	0.481
Reference: O10 (5wp vs 10wp, Phase 1)	0.705

The 5-to-7 scaling (0.761) is consistent with O10 (Phase 1's 5-to-10 at 0.705): shorter paths are genuine coarse-grainings of longer paths. The 7-to-9 scaling (0.504) is lower, suggesting that additional waypoints from 7 to 9 introduce genuinely new content rather than interpolating between existing waypoints.

F.7 F.7 Control Pair Screening (Phase 11B)

F.7.1 Candidate Results

4 candidates screened across 6 models (Claude, GPT, Grok, Gemini, DeepSeek, Mistral), 10 runs each. Frequency gate: top-waypoint frequency < 0.15. Entropy gate: waypoint entropy > 5.0.

Candidate	Max Top Freq	Min Entropy	Best Model Entropy	Passes?
accordion–stalactite	1.000	3.29	Gemini (4.92)	fail
turmeric–trigonometry	1.000	3.44	DeepSeek (4.77)	fail
barnacle–sonnet	1.000	3.44	Gemini (4.57)	fail
magnesium–ballet	1.000	3.65	Gemini (4.92)	fail

0 of 24 model×candidate cells pass either gate.

F.7.2 Retrospective: Stapler–Monsoon Across 12 Models

Model	Cohort	Top Waypoint	Top Frequency
Claude	original	office	0.775
GPT	original	paperwork	0.650
Grok	original	flood	0.700
Gemini	original	humidity	0.775
MiniMax	phase10a	paper	0.933
Kimi	phase10a	paper	0.933
Qwen	phase10a	cloud	0.867
Llama 8B	phase10a	office	0.800
DeepSeek	phase11a	paperclip	0.600
Mistral	phase11a	workplace	1.000
Cohere	phase11a	office supply	1.000
Llama 4	phase11a	paperweight	1.000

All 12 models fail R5 under strict criteria (top frequency < 0.15). The original 4 models show frequencies of 0.650–0.775; the Phase 10A and 11A models show 0.600–1.000. Under these criteria, stapler–monsoon has never been a valid control for any model.

F.8 F.8 Phase 11C Predictions Summary

#	Prediction	Result	Value	Verdict
1	Gait rank-order survives ($W > 0.70$)	$W = 0.840$	above threshold	confirmed
2	Asymmetry > 0.60 all conditions	5wp: 0.594, 0.593	below threshold	not confirmed
3	Bridge freq survives (spectrum > 0.50, warm > 0.30)	all > 0.97	far above threshold	confirmed
4	Temperature 0.5 increases Jaccard by 0.05–0.15	$\Delta = 0.226$	exceeds range	not confirmed
5	Temperature 0.9 decreases Jaccard, all > 0.10	$\Delta = 0.203$	magnitude wrong	not confirmed
6	5wp bridge freq $>$ 9wp	5wp $= 0.985$, 9wp $= 0.989$	no difference	not confirmed
7	Control pair unstructured all conditions	structured	—	not confirmed